

Enhancing a VR Bicycle Simulator by Realistically Integrating Tilt Dynamics into Steering And Balance Control

Master's Thesis

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Abstract

Virtual reality (VR) bicycle simulators are promising approaches for traffic safety research and training, yet achieving realistic cycling dynamics remains a challenge. Most existing systems treat tilt as a calculated output rather than an input. This often requires expensive actuated platforms. This thesis investigates an alternative approach: integrating user-induced tilt as a direct input into steering control through a passive adjustable resistance mechanism.

Building on previous work at the UniTyLab of Heilbronn University (HHN), this research addressed reliability issues before implementing a passive tilt resistance system with three adjustable stiffness levels. In addition, multiple steering models were developed to integrate handlebar and tilt inputs, including a velocity-based approach that weights inputs according to the speed of the bicycle. This emphasizes tilt at low speeds where balance dominates, and handlebar input at higher speeds where kinematic steering prevails.

A two-phase evaluation with 20 participants revealed unexpected findings. Rather than identifying a universally optimal configuration, the study showed that individually selected settings significantly outperformed the baseline configuration. The improvement was substantial, amounting to 26 % (Cohen's $d = 1.51$, $p = 0.0001$). The body weight of the participants was strongly correlated with the stiffness of the preferred configuration ($r = 0.630$, $p = 0.003$). This suggests a practical prediction approach for an initial configuration recommendation.

The comparison with a previous study using the same simulator validated the enhancements through substantial improvements in stress management. Participants showed significantly higher coping resources, representing a 24 % improvement (Cohen's $d = 2.37$, $p < 0.001$). This indicates increased confidence in handling challenging scenarios. However, presence measures did not show improvement, suggesting that passive tilt sensing enhances control without necessarily increasing immersion.

These findings challenge the assumption that a physically accurate bicycle simulation requires a complex actuated platform. The passive adjustable approach provides a viable middle ground, offering meaningful control enhancements through relatively simple mechanical modifications. More importantly, the results highlight that the effectiveness of the simulator depends on matching configurations with individual users rather than pursuing universal optimal settings. This has fundamental implications for the design of the simulator, suggesting that personalization may be more valuable than improvements in physical fidelity.

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1 Introduction

Virtual reality (VR) bicycle simulators have become an increasingly relevant research area, combining immersive technologies with cycling mechanics to create an authentic VR riding experience. These simulators integrate components such as motion platforms, sensors, and visual environments to approximate real-world cycling conditions [54]. Recent validation studies further demonstrated that such simulators can be used effectively in both scientific research and practical training contexts [19].

Research Use Case: VR bicycle simulators enable investigations of traffic safety under controlled and reproducible conditions. They allow researchers to study scenarios that would be difficult or unsafe to create on the road, such as intersection conflicts, cyclist–driver interactions, or the influence of environmental factors. For example, simulators have been used to examine cyclist perception of safety in dynamic traffic environments [37], to assess hazardous perception skills [51], and to analyze interactions with motor vehicles at intersections [35]. These studies provide insights that help improve urban cycling safety.

Training Use Case: Beyond research, bicycle simulators are also used as safe training tools [21, 40]. They can be used for people with low confidence in cycling or anxiety in real traffic situations. Users can be confronted with scenarios such as urban traffic, busy intersections, or adverse weather. This enables them to gradually build confidence and practice skills in a safe environment. Previous studies have also highlighted their contribution to sustainable mobility and urban planning [49].

1.1 Motivation

This work is part of the continuous development of the VR bicycle simulator at Heilbronn University’s UniTyLab. The foundational setup was implemented by Flaig et al. [13], establishing the hardware architecture and the sensor-actuator pipeline that connects the physical components to the Unity simulation. Subsequent projects extended the simulator, for example, by enhancing the Elite Rizer gradient simulator [48] and creating realistic German urban scenarios.

Building on this, Baumann et al. [3] improved the steering sensor and conducted the first user study, which demonstrated the potential of the simulator for training and research purposes. Their evaluation also revealed some shortcomings, including limited braking realism, overall system reliability, and a riding experience that was often perceived as unstable and unconvincing. Later, Hessenauer et al. [23] addressed further limitations by introducing pedal and back wheel sensors to improve response time and braking behavior.

In summary, these projects have created a strong foundation for the VR bicycle simulator but also highlighted existing issues. The most promising direction for further development was to investigate approaches that create more realistic cycling dynamics and to identify and resolve the

technical issues that currently limit reliability. This thesis therefore contributes to the ongoing refinement of the simulator with the aim of improving both system stability and cycling experience.

1.2 Objectives

The focus of this thesis is to enhance the realism and reliability of the simulator through targeted improvements. Before investigating the effectiveness of tilt-steering, fundamental system limitations required resolution. The work therefore concentrated on three main objectives:

- **Establish Reliable Foundation:** Increase robustness by addressing technical weaknesses in sensor reliability and overall system behavior that would otherwise confound experimental results and general riding experiences.
- **Improve Dynamics:** Improve cycling dynamics, which are often perceived as unstable and unconvincing, by developing and implementing enhanced tilt dynamics and steering control.
- **Evaluate Enhanced System:** Conduct a two-part evaluation to assess the effects of improved bicycle dynamics and to examine the potential of the simulator for training and research. This evaluation tested specific hypotheses (Section 7.1) and allows comparison with the findings of the previous study by Baumann et al. [3].

Objectives 1 and 2 establish the experimental setup required for an evaluation. Objective 3 builds on this foundation by conducting empirical research. Together, these objectives ensured that the simulator is not only technically more reliable but also capable of providing a more realistic cycling experience that can be systematically validated.

1.3 Structure of the Thesis

This thesis is organized into 13 chapters that address the objectives of improving the VR bicycle simulator through enhanced reliability, improved dynamics, and a systematic evaluation.

Chapter 1: Introduction to the motivation, objectives, and overall research context.

Chapter 2: Review of existing research on realism and human perception, bicycle dynamics models, and bicycle simulator implementations.

Chapter 3: Description of the existing system setup, including hardware components, data flow architecture, and identified issues.

Chapter 4: Gap analysis comparing the current setup with related work, leading to the approach for integrating tilt-steering through passive resistance systems and proposed steering models.

Chapter 5: System reliability and hardware improvements, including brake sensors, power supply, and cable management.

Chapter 6: Improved tilt-steering dynamics, including a passive adjustable tilt mechanism and Unity-based steering model implementation.

Chapter 7: Study methodology, including research hypotheses, participant criteria, VR scenarios, and study design.

Chapter 8: Measurement approaches and statistical analysis plan.

Chapter 9: Results of both study phases, including analyses of personalization effects, weight-based configuration, learning effects, and stress management results.

Chapter 10: Discussion and interpretation of results, highlighting individual differences, absence of universal configurations, and links to existing research.

Chapter 11: Limitations of the study, including sample constraints, technical limitations, and experimental design issues.

Chapter 12: Future work, outlining directions for technical enhancements, methodological extensions, and application domains.

Chapter 13: Conclusion summarizing technical achievements, research findings, and methodological contributions, with emphasis on personalization approaches in VR bicycle simulator design.

2 Related Work

This chapter reviews previous research relevant to this thesis. Although the thesis has three primary objectives: (1) improving the reliability of the simulator setup, (2) improving steering and balance to achieve a more realistic and stable cycling experience, and (3) evaluating these improvements through a two-part user study. The first objective primarily involved technical refinements and engineering adjustments; therefore, no dedicated related work section is required. In contrast, the second objective was built directly on existing research. Consequently, this chapter addressed three key areas: realism and human perception, bicycle dynamics with an emphasis on tilt control, and related implementations of VR bicycle simulators. The approaches to evaluating simulators and user studies were discussed separately in Chapters 7 and 8, where they are directly related to the experimental design of this thesis.

2.1 Realism and Human Perception

Human perception and realism are central to this thesis. Therefore, the following section discusses key concepts from these areas.

2.1.1 Objective vs Perceived Realism in Bicycle Simulators

When evaluating the performance of a simulator, it is useful to distinguish between objective realism and perceived realism.

- **Objective realism** describes how precisely the simulator reproduces real-world dynamics in measurable terms (e.g. steering response, roll motion, braking behavior).
- **Perceived realism** describes how realistic the experience feels to the user. It depends on psychological factors such as immersion, presence, and coherent multisensory feedback (visual, vestibular, tactile).

Hammami et al. [19] make this distinction explicit with the **Degree of Realism (DR)** and the **Perceived Level of Realism (PLR)**. Their results show that users sometimes do not perceive measurable differences between simulators and real-world conditions, even when objective data reveal them. This illustrates that technical accuracy alone is not enough for a convincing user experience.

Other studies underscore this point. Matviienko et al. [33] found that higher hardware fidelity, such as tilt mechanisms, can improve realism, but only if users maintain good control and the sensory cues are consistent. Similarly, Van Gisbergen et al. [50] demonstrated that adding more visual detail does not automatically improve engagement or natural behavior when other signals do not match.

Overall, these findings suggest that realism in bicycle simulators is a balance: systems must not only be technically accurate but also provide coherent and consistent cues that feel convincing to the user.

2.1.2 Importance of Visual–Vestibular Integration

In VR, the eyes and the vestibular system (inner ear) should tell a consistent story about motion and orientation. Visual motion alone can create a sense of self-motion (vection), but if not matched by body cues, symptoms of motion sickness can occur [25, 38]. Sensory conflict theory provides an explanation: When expected and sensed motions disagree, discomfort and sickness are likely [39].

For VR simulator design, this has two practical implications. First, adding physical motion (for example, tilt) can reinforce visual cues and improve comfort and control [54]. Second, the timing and amplitude of the signals matter: roll and steering should be coordinated so that the user experiences a coherent pattern of forces and visuals [31]. In short, good visual–vestibular integration increases immersion and supports the use of natural balance.

2.1.3 Non-linear Perception of Tilt

The human perception of tilt is not linear. For tilting around the roll axis, small angles are often perceived to be larger than they are (E-effect), whereas larger angles are perceived to be smaller (A effect) [15]. These systematic biases must be taken into account when integrating realistic tilt into VR cycling dynamics.

Such nonlinearities are consistent with general psychophysical principles, specifically the **Weber–Fechner law**. According to this law, the perceived intensity of a stimulus increases proportionally to the logarithm of its physical magnitude [10]. In the context of tilt perception, this implies that sensitivity decreases with increasing tilt angle: small deviations from the upright position are exaggerated, whereas large tilts are perceptually compressed. This scaling principle provides a theoretical foundation for the A- and E-effects observed in tilt perception and highlights why simple linear mappings between physical roll angles and perceived tilt might be insufficient in simulator design.

2.2 Bicycle Dynamic Models

Bicycle dynamics models describe how steering, roll (tilt), and speed interact to keep a bicycle balanced and on a defined path. Because this work aimed to reproduce a realistic feeling bicycle dynamic, it was essential to review different modeling approaches.

2.2.1 Carvallo–Whipple (Benchmark) Model

The Carvallo–Whipple model works in the following way: It describes bicycle mechanics by treating the bicycle as four rigid bodies: the rear frame with the cyclist, the front frame (fork + handlebar) and the two thin wheels, connected by ideal hinges on the steer axis. The wheels roll without slip on flat, level ground. Other effects such as tire flexibility, suspension, and aerodynamics are neglected [34, 26].

The **main essence** is that leaning (roll) and steering are coupled, with the coupling changing at different speeds. A small handlebar input can create lean (roll), and a small lean can induce steering through geometry and gyroscopic effects. At low speeds, the bicycle is usually unstable; at medium speeds, it can be self-stable; at higher speeds, a slow capsize tendency appears [34, 26].

Common 2-DOF form For small roll ϕ and handlebar angle δ , a standard linearized form is [34]:

$$\mathbf{M} \ddot{\mathbf{q}} + v \mathbf{C}_1 \dot{\mathbf{q}} + (g \mathbf{K}_0 + v^2 \mathbf{K}_2) \mathbf{q} = \mathbf{f}, \quad \mathbf{q} = \begin{bmatrix} \phi \\ \delta \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} T_\phi \\ T_\delta \end{bmatrix}. \quad (2.1)$$

Here, $\mathbf{M}, \mathbf{C}_1, \mathbf{K}_0, \mathbf{K}_2 \in \mathbb{R}^{2 \times 2}$ are constants set by geometry and mass, v is forward speed, g is gravitational acceleration, and T_δ is the steering torque (often the only applied input, so $\mathbf{f} = \mathbf{B} \tau_\delta$ with $\mathbf{B} = [0 \ 1]^\top$). For unforced analyses, set $\mathbf{f} = \mathbf{0}$ as in [36].

Integration of roll and handlebar angles In (2.1), the state $\mathbf{q} = [\phi \ \delta]^\top$ contains the roll angle (tilt) ϕ and the handlebar angle δ . The off-diagonal entries couple these two motions, which means that a steering torque T_δ not only turns the handlebar; it also affects the roll angle (lean), and a small lean feeds back into the steering. The speed-dependent terms determine which effect dominates: at very low v , the $g \mathbf{K}_0$ term (gravity + basic geometry) dominates; as v increases, the $v \mathbf{C}_1$ term (mainly gyroscopic and kinematic) becomes more dominant; at higher v , the $v^2 \mathbf{K}_2$ term grows, producing the typical progression from low-speed weave, through a self-stable range, to a high-speed capsize tendency.

2.2.2 Other Bicycle Models

As discussed in Section 2.1.1, perceived realism and physical precision do not always align. Therefore, it also makes sense to consider different modeling approaches that trade physical detail and parameter accuracy for robustness, easier tuning, and lower computational cost, which can be advantageous for real-time VR applications.

Kinematic single-track (“bicycle–car”) model. This model ignores balance and tire slip. The steering directly sets the path curvature, which then sets the yaw rate. It is useful for routing and traffic logic, but does not explain how a bicycle balances [41]. From the kinematic model (Ch. 2, Eq. 2.8; Ch. 3, Eqs. 3.15–3.16) [41], we have $\dot{\psi} = v/R$ and $\tan \delta = \ell/R$; therefore $\kappa = 1/R = \tan \delta / \ell$ and $\dot{\psi} = v \kappa$.

$$\kappa = \frac{\tan \delta}{\ell}, \quad \dot{\psi} = v \kappa, \quad (2.2)$$

with wheelbase ℓ , steering angle δ , curvature κ , yaw rate $\dot{\psi}$, and speed v . With this model, the required steering angle (δ) can be calculated given a desired curvature (κ) or radius (R), or the yaw rate ($\dot{\psi}$) can be determined given the speed of the bike (v) and the steering angle (δ) [41].

Quasi-static lean relation (steady turns). In a constant turn, the bicycle leans so that gravity balances the lateral acceleration from the roll angle (lean). Resulting in a simple rule for mapping speed and turn radius to a platform roll target [2]:

$$\phi \approx \arctan\left(\frac{v^2}{g R}\right) \approx \frac{v^2}{g R} \quad (\text{small angles}). \quad (2.3)$$

In (2.4), R is the turn radius, and g is the gravitational acceleration. For small steering angles, one may approximate the relationship between radius and steering by $R \approx \ell/\delta$ [41]. This relation can be used for small, steady corners and slow roll angle changes; it does not take into account the transient balance dynamics or tire slip.

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2.3 Overview of Different Bicycle Simulators

The implementations of various bicycle simulator designs vary significantly in terms of degrees of freedom, control mechanisms, and display technologies. Table 2.1 provides a comparison of different bicycle simulator setups, highlighting their capabilities in various features.

Table 2.1: Comparison of VR Bicycle Simulators by Feature Set and Degrees of Freedom. DoF codes: F = Free (passive), A = Actuated, S = Sensed, X = Exists/Implemented (unspecified). Combinations such as A+S or F+S indicate multiple capabilities.

Bicycle Simulator Setup	Surge DoF (Forward/Backward)	Roll DoF (Tilt/Lean)	Pitch DoF (Elevation)	Handlebar Steering	Brake	Rear Bike Trainer	Pedal Motion Sensor	Back Wheel	Reverse Motion Input	Headwind	Head Mounted Display	External Display
HHN UniTyLab [13, 3, 23]	F	F+S	A	S	S	A+S	S	S	S	A	X	-
TU Wien [54, 52, 53]	-	A+S	A+S	A+S	S	A+S	-	-	-	A	X	-
TU Delft (BIKE) [18]	-	A ¹ +S	-	S	A	-	S	S	-	-	-	X
Doshisha [55]	-	A+S	-	S	S	-	-	S	-	-	-	X
GEU [45]	-	F	-	A+S	S	-	-	A+S	-	A	-	X
Győr [19]	-	-	-	S	S	-	-	A+S	-	-	X	-
FIVIS [22, 21]	-	A	A	A+S	S	-	-	A+S	-	-	-	X
KAIST [28, 29]	A	A	A	A+S	A+S	-	-	A+S	-	-	X	X
DLR/Wuppertal [11, 17, 32]	-	A	A	A+S	S	A+S	-	-	-	A	X	X
TUM [24]	-	-	-	S	S	A	-	S	-	-	X	X
THN [49]	-	-	-	S	-	A+S	-	S	-	-	X	-
MUARC [40]	-	-	-	S	S	A+S	-	-	-	-	X	-
UniBw (BikeCG) [20]	-	-	-	-	-	A+S	-	-	-	A	X	-
Chalmers/Aalborg [35]	-	-	-	S	S	-	S	-	-	-	X	-

When comparing features, several common design patterns can be identified in the simulator setups. The choice between head-mounted displays and external screens varies between systems, with some relying on VR headsets and others on projection-based environments. Most systems measure speed input through bicycle trainers or rear wheel sensing, and the majority include actuated resistance control to simulate variable pedaling conditions. Almost all systems implement steering input via sensed handlebars, ranging from simple sensing to actuated force-feedback approaches. Brake inputs are also common, though not universal, and range from basic sensing to actuated brakes. Some simulators incorporate pedal motion sensors to capture pedaling intensity,

¹Roll is achieved through the rider’s natural balancing actions (passive/free) while lateral (sway and yaw) motion is actively actuated by linear motors beneath the contact points.

while headwind simulation remains relatively rare. The HHN UniTyLab setup [23, 13, 3] is unique in offering reverse motion input capability. Comprehensive motion feedback across roll, pitch, and surge axes is rare; most systems provide only partial motion cues. In the following analysis, particular attention was paid to roll implementations, since roll is central to bicycle balance and steering behavior.

2.3.1 Challenges Without Tilt Integration

Simulators that lack tilt feedback exhibit several limitations:

- **Reduced balance realism and impaired control:** Simulators without physical tilt capabilities suppress vestibular feedback, which is essential for natural balance control. Haasnoot et al. [18] show that realistic roll and lateral motion significantly improve both objective alignment with real-world behavior and subjective feeling of realism. Dialynas et al. [8] also emphasize that vestibular input, linked to bicycle roll, is a crucial component of rider feedback control and cannot be fully replaced by visual cues alone.
- **Lowered subjective immersion:** Wintersberger et al. [54] report that static, non-tilting bicycle simulators reduce spatial presence and perceived realism. Their study shows that even small roll movements noticeably enhance immersion, while the absence of motion feedback creates a mismatch between visual and bodily sensations, decreasing realism and comfort.
- **Increased simulator sickness risk:** The same study found that visual–vestibular conflicts arise when strong visual motion is not accompanied by corresponding physical tilt. These mismatches were shown to increase simulator sickness, particularly in conditions with abrupt or absent tilt feedback.
- **Training fidelity gaps:** Martinez Garcia et al. [32] emphasize that accurate leaning behavior and motion perception are essential to evoke realistic cycling behavior and to ensure that simulator results can be meaningfully transferred to real-world scenarios. Likewise, Sporrel et al. [46] argue that many simulators lack proper balance control and natural lean–steer coupling, which undermines behavioral validity, particularly for safety research and fall prevention training.

In summary, the exclusion of tilt dynamics reduces realism and may compromise both the usability and the validity of bicycle simulators, especially in contexts where authentic user behavior is required.

2.3.2 Roll Integration Approaches

Existing simulator implementations demonstrate different approaches to roll integration, each with distinct advantages and limitations. These approaches could be categorized by their mechanical complexity, control, and intended applications.

TU Wien Motion-Based Implementation The implementation of TU Wien employs a commercially available *Next Level Racing Motion Platform V3*, originally designed for racing games and flight simulators [54, 53, 52]. The platform provides up to 10 ° tilting in both lateral and longitudinal axes with maximum angular velocities of 20°/s and accelerations of 360°/s². The team compared three tilt conditions: strong, weak, and static baseline.

Initial attempts to implement a physically accurate tilt proved problematic. The complete implementation of the quasi-static lean equation $\phi = \arctan(v^2/gR)$ resulted in uncomfortably aggressive platform movements. For example, the transition from a left curve to a right curve at 15 km/h with a radius of 10 m would immediately command the platform from -10 ° to +10 °, resulting in severe motion sickness. The researchers fixed this by progressive scaling: first mapping a theoretical maximum of 40 ° lean to the platform's 10 ° range, then further reducing to ± 5 °, and finally implementing a weak heuristic tilt function. This conservative approach, limiting platform motion to approximately ± 2.3 °, provided the optimal balance between immersion and comfort. The weak tilt condition improved perceived realism without inducing motion sickness, demonstrating that modest motion cues can be highly effective when properly tuned.

TU Delft Bicycle Simulator The TU Delft setup uses lateral actuation at the contact points of the wheel beneath the frame of the bicycle to generate realistic roll dynamics without tilting the entire base [18]. In this design, the natural weight shifts of the rider and the input of the steering produce a lean ride, while the base remains horizontally stable. Control is informed by the Carvallo-Whipple model (see Section 2.2.1), ensuring that lean–steer coupling follows established bicycle dynamics rather than pre-programmed tilt cues.

The evaluation results reported by Haasnoot et al. [18] indicate improved alignment with real cycling behavior, as well as increased perceived realism. The system emphasizes natural rider control over automated motion cueing and avoids the complexity of a full six-degree-of-freedom platform. In particular, the setup uses an external display instead of a head-mounted display and is therefore not classified as a VR application.

Doshisha University Tilt Control System The simulator developed at Doshisha University implements direct roll angle (tilt) control coupled to steering, using an industrial AC servomotor installed on the rear wheel to tilt the bicycle frame [55]. Based on the detected steering angle δ and the rear wheel speed v , the controller computes a target roll angle from the balance of centrifugal and gravitational moments, consistent with the quasi-static lean relation in Eq. (2.4) (see Section 2.2.2) and aligned with the lean–steer coupling principles of the Carvallo–Whipple

model in Eq. (2.1). The servomotor adjusts the tilt within a range of approximately $\pm 15^\circ$, allowing systematic investigation of how actively actuated roll influences balance and steering dynamics.

The evaluation results reported in [55] focus mainly on tracking the trajectory in tasks such as slalom and right turns; subjective evaluations of user experience were limited. The virtual environment was rendered in Unity and projected on three surrounding screens, rather than displayed through a head-mounted display, to minimize simulator sickness [55].

Gustave Eiffel University (GEU) Passive System The simulator developed at Gustave Eiffel University provides passive roll through a mechanical lateral suspension that allows the bicycle to tilt slightly when the rider leans left or right [45]. Importantly, this approach does not sense or measure the tilt angle for feedback into the simulation dynamics: the platform simply allows free rolling motion without any sensors to detect the actual tilt position. The virtual environment is displayed on a multi-screen setup that provides a wide field of view, rather than through a head-mounted display. This approach is robust and cost-effective, but since vestibular signals are weak at low speed or standstill and not coordinated with the visual scene, perceived realism may be reduced compared to actively actuated tilt systems that incorporate tilt sensing and feedback [45].

FIVIS Hydraulic Hexapod Platform The FIVIS simulator mounts a real bicycle on a hydraulic Stewart platform (hexapod) that provides six actuated degrees of freedom [21, 22]. The virtual environment is projected onto a multiscreen display system rather than presented through a head-mounted display. The steering input is measured with a potentiometer and the forward speed with an electro-optical rear-wheel sensor. An eddy-current brake on the rear wheel provides controllable pedal resistance and support force-feedback evaluation [22]. During turns, the simulator computes a target roll angle from the cycling speed v and the turn radius R and applies torque to the bicycle frame to achieve it, consistent with the quasi-static relation in Eq. (2.4) [21]. The motion platform can also overlay short vibrations to mimic surface irregularities, and forces or accelerations recorded during outdoor rides can be replayed to drive the platform [21]. The system has been used in contexts such as traffic education, stress induction, and biomechanics, including integration with the CUELA measurement system [22].

KAIST Stewart-Platform Implementation The first KAIST bicycle simulator mounted a real bicycle on a Stewart six-degree-of-freedom platform actuated by electric motors via ball screws, with a load capacity of approximately 130 kg and a tracking bandwidth of roughly 7 Hz [28]. Visuals were provided on a projection screen or through an optional head mounted display. Haptic feedback was implemented on two channels: the handlebar used a brushless DC motor in series with a magneto-rheological clutch (MR) to generate both active (motor-driven) and passive (variable-damping) torques, while the pedal unit combined an AC servo motor with an MR brake to emulate downhill assist and uphill resistance [28]. The roll angle (tilt) ϕ was calculated with

a non-linear bicycle dynamics solver based on Newton–Euler equations, including tire slip and camber forces, and the resulting body attitude was assigned to the platform motion using classical washout filtering [28].

A second version introduced multi-user networking, 3D audio, and a redesigned motion system that reduced the platform to four active degrees of freedom (heave, roll, pitch, and yaw via turntable). The lever drive was simplified to direct motor actuation for a faster response, and the effects of the road condition were mapped into the background vibration and rolling resistance in the haptic channels [29]. Overall, the KAIST system demonstrated an ambitious and early integration of motion cueing and haptics.

DLR/Wuppertal Two-Degree-of-Freedom System The bicycle simulator developed at DLR/University of Wuppertal mounts a real bicycle on a two-degree-of-freedom motion platform (roll and pitch) with position control on both axes [17]. The early versions focused on the integration of steering, braking, and resistance modules, with plans to implement model-based lean control. Subsequent iterations introduced a lateral dynamics block that computes a time-varying roll reference from steer input δ and speed v , using a linearized Whipple model of bicycle dynamics [11, 32]. Thus, the roll axis follows a calculated $\phi_{\text{ref}}(t)$ rather than a fixed mapping $\phi(v, R)$, while the pitch axis provides cues for braking and road inclination. Evaluations combined objective measures (lean, steer, speed) with subjective instruments (SSQ, presence, acceptance). The results showed improved lateral dynamics compared to previous versions, although perceived realism and steering controllability remain challenging [11]. In general, focusing on roll and pitch offers meaningful motion cues with less complexity than full six-DoF platforms [17].

Concluding summary Across the reviewed implementations, roll integration can be grouped into two main categories:

1. **Quasi-static approaches**, where the roll angle ϕ is calculated from speed v and turn radius R using Eq. (2.4).
2. **Model-based approaches**, where lean is derived dynamically from bicycle equations of motion, such as the linearized Carvallo–Whipple model (see Section 2.2.1) or full Newton–Euler formulations.
3. **Passive tilt approaches**, where the platform allows free rolling motion but does not sense or measure the tilt angle to input the simulation.

Display technology also varies: Although some sets rely on multiscreen projection, others use head-mounted displays, increasing the importance of coherent visual–vestibular cueing (see Section 2.1.2). In most cases, roll is not an independent input but emerges from the steering angle δ , the speed v , and the path curvature κ , so ϕ follows these states. Only mechanically coupled designs allow rider-induced weight shifts to directly generate lean, which then feeds back into steering dynamics.

3 Existing System Setup

This chapter describes the current state of the VR bicycle simulator setup. It is divided into three parts. The first part describes the hardware components used, based on the existing setup and documentation. The second part focuses on the software architecture, including how the components are connected and communicate. The third part summarizes the main issues and limitations of the current system.

The setup is based on the foundational setup by Flaig et al. [13], who implemented the hardware architecture and the initial sensor-actuator pipeline. Building on this, student projects developed the virtual environment of typical german Urban Traffic. Baumann et al. [3] then improved the steering and braking subsystems and conducted the first user study, while Hessenauer et al. [23] added enhanced pedal (and wheel) sensing for more responsive input.

Figure 3.1 illustrates the original configuration documented by Flaig et al. [13]. The later updates introduced by Baumann et al. [3] and Hessenauer et al. [23] (improved steering, braking, and pedal sensing) are not visible in this image, but are described in the text below.

3.1 Hardware Components

The simulator consists of a combination of commercially available third-party devices and custom-built components. Each element can serve one or more functions: providing sensor input, providing actuator feedback, or supporting the overall structural setup. Table 3.1 gives an overview of the main components, their roles and origin (third party vs. custom), while Figure 3.1 shows their physical arrangement within the simulator.

3.1.1 Smart Trainer

In addition to serving as the rear wheel mount of the bicycle, the Direto XR-T Smart Trainer [47] is used to dynamically adjust the resistance of the pedaling according to the simulated terrain. It supports gradients up to 24 % and considers the current simulated velocity when setting resistance, thereby contributing to a realistic cycling experience.

The Direto XR-T communicates via Bluetooth Low Energy (BLE), a standard interface in indoor cycling systems. A Python-based control script running on the main PC connects to the trainer using the Bleak library [4]. The script continuously receives simulation data and updates the resistance of the trainer in real time.

Originally, the smart trainer also provided speed measurement. However, due to the latency and low sampling rate of its built-in sensor, this role was reassigned to a dedicated sensor system developed in a student project (see Section 3.1.10). Since then, the trainer has been used primarily for resistance control, while external Hall-effect sensors with an ESP32 microcontroller supply more

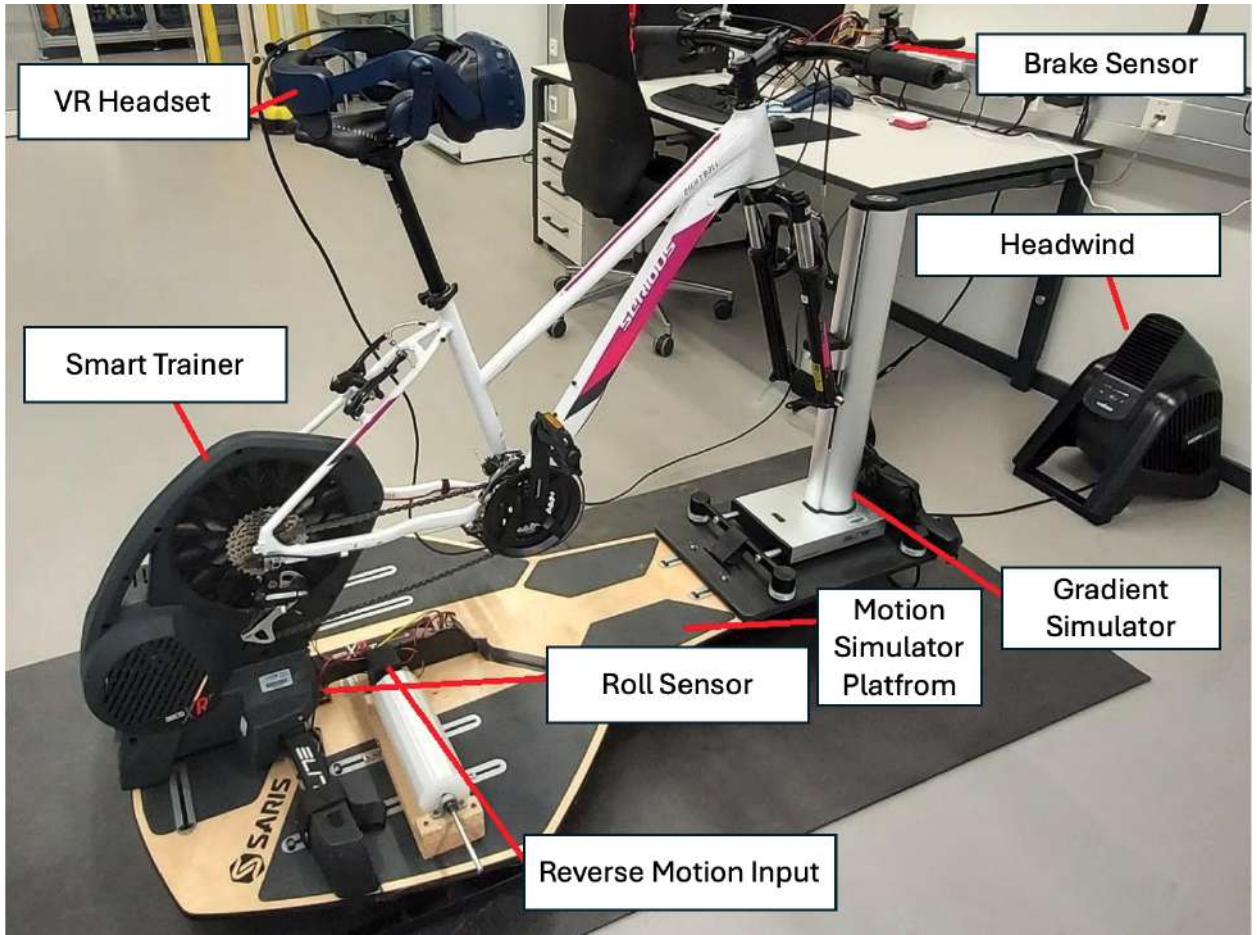


Figure 3.1: Original simulator configuration as documented by Flaig et al. [13].

accurate speed and pedal data. This separation of responsibilities improved both responsiveness and realism.

The Python control script now only receives gradient and velocity data from the Unity simulation over the User Datagram Protocol (UDP) and writes the corresponding resistance values to the trainer via BLE.

3.1.2 Gradient Simulator

To further enhance the realism of the riding experience, an ELITE Rizer gradient simulator [48] is used as a front wheel mount. This device adjusts the physical incline of the bicycle to simulate gradients up to $\pm 15^\circ$. An additional feature is the ability to rotate the front fork, allowing handlebar movement within a range of approximately $\pm 17^\circ$.

Although the Rizer can detect three distinct handlebar positions (left, center, and right), this resolution is insufficient for realistic steering input. Therefore, an alternative steering solution is used, as described in Section 3.1.3.

The Rizer communicates through BLE, and a similar approach to the Smart Trainer (see Section 3.1.1) is used: a Python-based script built on the Bleak library [4].

Table 3.1: Overview of Components and Their Categories: Sensors, Actuators, and Third-Party Devices

Component	Sensor	Actuator	Third Party
Smart Trainer	Pedaling Speed	Pedaling Resistance	ELITE DIRETO XR-T [47]
Gradient Simulator	Directional Steering	Elevation Gradient	ELITE RIZER [48]
Headwind	-	Air Resistance	Wahoo KICKR HEAD-WIND [12]
Virtual Reality Headset	Head Tracking	Head-Mounted Display	HTC VIVE Pro [7]
Motion Simulator Platform	-	-	Saris MP1 Platform [42]
Roll Angle Sensor	Platform Roll Angle	-	-
Brake Lever	Brake Lever Angle	-	-
Reverse Motion Input	Rotation of Roller	-	-
Wheel & Pedal Sensor	Wheel Speed & Cadence	-	-

The script performs two core functions:

- Continuously receive simulated gradient values from the main PC over UDP and translates these into physical incline adjustments on the Rizer.
- Establish and maintain a BLE connection with the Rizer, sending the appropriate commands to raise or lower the incline incrementally, based on changes in the target gradient.

3.1.3 Steering Sensor

The steering angle sensor was developed as part of the research project described in [3], which focused on improving the steering input for the VR bicycle simulator. The sensor aimed to address the limitations of earlier setups that used the built-in input of the gradient simulator (see Section 3.1.2).

Initially, the ELITE Rizer was used to detect three discrete steering positions: left, straight, and right. Although functional, this approach did not reflect the continuous nature of real-world bicycle steering, which reduced the sense of realism in the simulation.

To address this, a custom sensor setup was designed using an optical photoelectric encoder. The encoder allows for precise steering angle measurements, with a resolution of approximately 0.6° per step. It is mechanically connected to the bicycle fork, while the housing is mounted to the frame using a 3D printed bracket.

An ESP32 microcontroller processes encoder data using interrupt-based detection on two signal phases, which also enables it to determine the direction of rotation. To reduce the influence

of mechanical noise from the gradient simulator, a software-based high-pass filter was applied. The ESP32 communicates steering data to the Control PC via UDP, where it is integrated into the simulation in real time. This setup allows for smoother and more continuous steering input, contributing to a more natural interaction between the rider and the system.

3.1.4 Brake Sensor

As braking is an integral part of cycling, sensory input from the brake lever was essential for the VR bike simulator. To meet this requirement, a custom brake lever with a mounted rotary Hall sensor was designed. The custom solution extends the pivot bolt of the brake lever to allow connection to the Hall sensor rotary input axis. The sensor housing is mounted via a 3D printed bracket, allowing precise measurement of the angular position of the brake lever.

The Hall sensor is interfaced with an ESP32 microcontroller, which transmits angle data to the main PC via UDP over a wireless network. The microcontroller continuously samples the sensor at a fixed interval, formats the readings into a JSON structure, and sends them over the network, enabling real-time monitoring of the brake lever position.

3.1.5 Motion Simulator Platform

The complete setup of the VR bicycle simulator is installed on a Saris MP1 Infinity Motion Platform [42], which enables natural cycling motion by allowing side-to-side (roll) movement and fore / aft translation, i.e. movement along the axis parallel to the roll direction.

3.1.6 Roll Angle Sensor

As the bicycle is mounted on the motion simulator platform described in Section 3.1.5, it can tilt sideways (roll). To reflect this motion in the simulation by tilting the virtual bicycle accordingly, the roll angle is measured using a BNO055 sensor [44]. The sensor is mounted directly on the motion simulator platform and operates in nine degree of freedom (9DoF) mode. It is connected to an ESP32 microcontroller, which transmits the roll angle to the main PC via UDP.

3.1.7 Headwind

To simulate headwind, the Wahoo KICKR headwind [12] is integrated into the setup. The Headwind is a BLE device that dynamically adjusts airflow based on the speed of the bicycle in the simulation. A Python script, developed using the Bleak library [4], establishes a BLE connection to the fan and controls its speed in real time. The script listens for speed data sent via UDP from the simulation. Upon receiving the data, it decodes the JSON-formatted message, extracts the fan speed value, and sends a corresponding command to the Headwind device via BLE. Depending on



Figure 3.2: Reverse Motion Input

the value, the fan is activated, adjusted, or turned off, creating a dynamic airflow effect that reflects the simulated riding conditions.

3.1.8 Virtual Reality Headset

To visualize the virtual Unity environment for the cyclist, the HTC VIVE Pro head-mounted display is used [7]. The system accurately tracks the user's position and orientation, dynamically adjusting the view in real time. This enables a realistic and immersive 360° experience.

3.1.9 Reverse Motion Input

As illustrated in Figure 3.2, this component consists of a bearing-mounted roller, a magnetic AS5600 encoder, and an ESP32 microcontroller. It is positioned just behind the right pedal of the bicycle, making it easily accessible with the user foot. When the user rotates the roller by foot, the magnetic encoder detects the rotation. The ESP32 continuously transmits this input via UDP to the main PC, allowing the simulation to trigger reverse movement of the virtual bicycle. This feature is especially useful in situations where the user encounters obstacles, offering a practical way to maneuver out of otherwise inaccessible or blocked areas within the virtual environment.

3.1.10 Back Wheel and Pedal Sensors

The back wheel and pedal sensors were developed as part of a student project to improve the responsiveness and realism of the bicycle simulator [23]. This development focused on replacing the delayed speed data from the Direto XR Trainer with faster and more precise measurements.

The custom setup consists of two Hall-Effect sensors and an ESP32 microcontroller. The Back Wheel Sensor uses five magnets mounted on the rear wheel. As the wheel rotates, the Hall effect sensor detects each passing magnet and triggers an interrupt on the ESP32. This setup allows the system to calculate the speed of the wheel based on the timing between the detected pulses. Similarly, the Pedal Sensor uses four magnets placed on the crank arm to measure pedaling activity. Each sensor is connected to its own interrupt pin, allowing independent and parallel data collection.

During the project, two software versions were implemented. Version 1 uses a sliding average over a 1-second window, which smooths out the values but introduces some delay. Version 2 improves upon this by calculating the speed directly from the time difference between interrupts, which makes it more responsive to changes in motion. Both sensors communicate with the main PC via UDP over a wireless network, achieving data update rates below 100 ms.

The collected data is integrated into the Unity simulation to control acceleration and visual pedal rotation. By combining the data from both sensors, the simulator can now better represent the real cycling dynamics, especially during rapid starts and stops.

To support easy installation, a 3D printed mounting system was also designed for the sensors and ESP32 module. This housing simplifies access to the microcontroller and provides a stable attachment to the bicycle frame. In general, the new sensor system is a major improvement to the simulator, providing low latency and high-accuracy input that aligns closely with the real-world actions of the rider.

3.2 Data Flow

The data flow between the components of the VR bicycle simulator is illustrated in Figure 3.3. The main PC is responsible for collecting, distributing and processing sensor data, as well as running the Unity-based simulation. Communication between components is managed through the BLE or UDP protocol, depending on the specific sensor or device. A set of Python-based scripts acts as an intermediary, forwarding sensor and control data to the Unity simulation via UDP. The VR headset communicates directly with Unity via the SteamVR plugin.

Compared to the original implementation described in [3], the data flow diagram has been updated to reflect the integration of the custom back-wheel and pedal sensors described in Section 3.1.10. These sensors transmit high-frequency speed and pedaling data via UDP to the main PC, replacing the slower and less accurate measurements previously provided by the smart trainer. This update improves system responsiveness without altering the overall communication architecture.

3.3 Identified Issues and Limitations

Although the VR bicycle simulator provides a promising foundation for immersive urban cycling experiences, several limitations were identified through user evaluation, system analysis, and hands-on experimentation. These issues can be grouped into three main categories: user-study feedback, observed hardware and software instability, and limitations discovered during development.

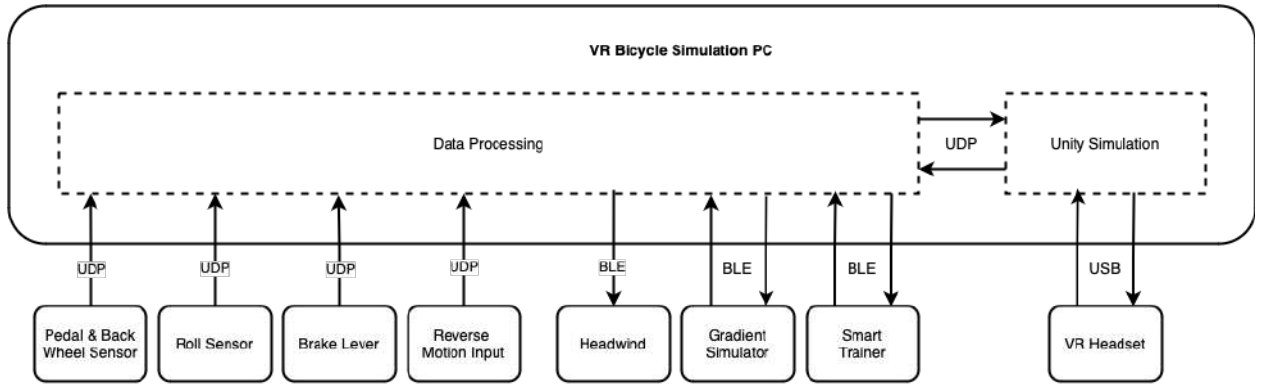


Figure 3.3: System-level data flow of the VR bicycle simulator. Updated to include the Back Wheel and Pedal Sensors. Based on original version from [3].

3.3.1 Study-Based Observations

The simulator was evaluated in a user study with 24 participants with different VR and cycling experiences, as reported in [3]. Although several improvements were well received, particularly the updated steering sensor, the study also highlighted a number of challenges.

- **Immersion and Motion Sickness:** Participants reported that immersion increased with time, suggesting that familiarity with the system reduces initial discomfort. However, motion sickness remained a significant problem, with high ratings for nausea and disorientation during the first sessions.
- **Acceleration and Braking Realism:** Feedback on the braking system was generally positive, but impressions of acceleration and deceleration realism were mixed. The braking dynamics were considered improved, but further refinement of the deceleration behavior was suggested. These aspects were later enhanced by Hessenauer et al. [23] through improved back-wheel and pedal sensors (see Section 3.1.10).
- **Sensor Evaluation:** Steering input was rated as clearly more realistic after hardware updates, while braking and acceleration realism received only moderate scores. This indicates that while progress has been made, consistency across all control channels remains an area for improvement.
- **Roll–Steer Decoupling:** A key limitation identified was the lack of a consistent link between the roll of the motion platform and the virtual steering model. This decoupling limited the sense of natural balance and sometimes caused unrealistic ‘wiggling’ or instability, particularly during turns or low-speed maneuvers.

3.3.2 Hardware and Power Supply Limitations

During system integration and testing, several technical limitations were identified, primarily related to the power supply and sensor stability.

- **Unstable Shared Power Supply:** Multiple ESP32 microcontrollers that are responsible for reading roll, reverse motion input, and brake input were initially powered via a single 5 V rail. This setup often resulted in inconsistent startup behavior, where certain components did not initialize properly and required manual rebooting. These problems were likely caused by voltage drops and current spikes across the shared line. Partial mitigation was achieved by switching some sensors (for example, brakes and steering) to individual USB power sources, significantly improving their stability.
- **Limited USB Access for Debugging:** Shared 5 V wiring also restricts physical access to ESP32 USB ports. This hinders firmware updates, code debugging, and sensor calibration, which complicates maintenance and iterative development.
- **Brake Sensor 'Sticking' behavior:** The current implementation of the brake sensor exhibits occasional faults in which the reported brake value does not return to zero after release. This creates an undesired state in which braking remains active within the simulation, leading to confusing or frustrating behavior for the user.

3.3.3 Mechanical and Mounting Challenges

Although Hall-Effect sensors for the back wheel and pedal system significantly improved measurement latency, the current mechanical setup introduces reliability concerns.

- **Temporary Sensor Mounting:** Both sensors are mounted using tape or provisional supports, allowing for physical displacement or misalignment over time. Even slight changes can cause magnets to move outside the detection range of the sensor, resulting in data loss or inaccurate readings. This undermines the intended responsiveness of the speed and cadence system.
- **Need for Robust Enclosures:** Although 3D-printable housings have been designed for several components, some have not yet been printed or permanently installed. Proper mechanical integration would reduce vibration artifacts and improve long-term robustness.

3.3.4 Summary and Outlook

The analysis reveals two main categories of identified issues that align directly with the objectives established in Section 1.2. First, hardware and power supply limitations along with the reliability concerns of the brake sensor detailed in Section 3.3 confirm the need for immediate technical

improvements to enhance the robustness of the system and braking realism. Second, the general feeling of "wobbly" and unstable dynamics, particularly the lack of coherent tilt–steering integration identified through user feedback, substantiates the objective of improving bicycle dynamics.

The latter issue, concerning bicycle dynamics and tilt integration, will be systematically investigated in the next chapter through a comprehensive gap analysis (Chapter 4) comparing the existing setup (Chapter 3) with related work (Chapter 2). The former issues related to reliability and braking behavior will be addressed through direct engineering improvements without requiring further investigation into alternative approaches.

4 Gap Analysis and Approach Development

This chapter examines the limitations of the current simulator setup through a systematic comparison with related work, leading to the development of a new approach of the integration of tilt dynamics into steering control that addresses the current issue while regarding existing hardware constraints.

4.1 Gap Analysis: Current Setup vs. Related Work

4.1.1 Project Constraints and Analysis Focus

This analysis considers specific constraints: The current setup must be improved through modifications and additions rather than complete redesign, reflecting the practical limitations of time, budget, and development resources. Furthermore, consistent with the discussion in Section 2.1.1, this work emphasizes perceived realism over objective physical accuracy.

4.1.2 Evaluation Framework for Comparison

When comparing potential approaches, the following evaluation criteria are critical to the success of this project. The complexity of implementing any solution within existing hardware constraints is a major consideration: solutions that require extensive mechanical modifications would exceed the available resources. Cost-effectiveness and development time further limit the viable solution space, particularly given the timeline of this project.

It is not evident that perceived realism necessarily increases with platform complexity or strong actuation, which supports considering alternative approaches. As shown by Wintersberger et al. [54], even small tilt amplitudes ($\approx \pm 2.3^\circ$) proved to be more effective than aggressive motion signals, suggesting that subtle signals may be sufficient to improve perceived realism.

The final constraint is technical feasibility within the existing setup. The current infrastructure, specifically the passive roll capability of the Saris MP1 platform [42] and the BNO055 roll angle sensor, provide a foundation for tilt integration without requiring complex mechanical modifications.

4.1.3 Tilt Integration Approaches in Related Work

The review of related bicycle simulator implementations, described in Chapter 2, shows three primary categories of tilt integration: actuated motion platforms, quasi-static tilt mappings, and model-based dynamic approaches. Each approach has specific trade-offs between implementation complexity, hardware requirements, and user experience.

Actuated systems such as those developed by TU Wien [54] and TU Delft [18] require substantial mechanical engineering and custom hardware. Although technically sophisticated, they require extensive resources. The KAIST Stewart platform [28] and the FIVIS hexapod system [21] go further, actuating all six degrees of freedom and therefore requiring advanced control algorithms and significant mechanical infrastructure.

Quasi-static approaches, exemplified by Doshisha University [55], calculate the tilt as a function of speed and path/steering using relations derived from Eq. (2.4). Although conceptually straightforward, these implementations treat tilt as a consequence rather than an input to the cycling dynamics. The DLR / Wuppertal system [17, 11] extends this concept with model-based lean calculations using linearized Carvallo–Whipple dynamics, but still positions tilt as output rather than control input.

4.1.4 Current HHN Setup Analysis

The existing HHN UniTyLab setup has capabilities that distinguish it from other implementations. The Saris MP1 platform provides natural passive roll capability, allowing users to tilt the bicycle through a weight change without requiring actuated motion. The BNO055 sensor accurately measures the roll angle of the platform, providing real-time feedback about the user-induced tilt. The custom steering sensor provides continuous measurement of the angle of the handlebar with a resolution of 0.6° [3].

Although both the roll angle and the handlebar angle are measured independently, no coupling mechanism connects these inputs to produce unified steering behavior. The current approach treats steering as purely handlebar driven, ignoring the roll angle (tilt) available from the sensed motion platform and potentially contributing to the "wobbly" dynamics reported in user studies [3].

4.1.5 Identified Research Gap and Approach Rationale

The comparison reveals a significant gap: existing implementations predominantly use tilt as an output calculated from other parameters (speed, steering angle, path curvature), whereas the potential for using tilt as an input to influence steering dynamics remains largely unexplored. This represents an inversion of the conventional approach that may offer new insights into visual–vestibular integration.

Most reviewed systems implement the relationship $\phi = f(\delta, v, R)$, where the roll angle ϕ follows from steering δ , speed v , and turn radius R . The proposed approach investigates the inverse problem: $\delta_{\text{effective}} = g(\delta_{\text{handlebar}}, \phi, v)$, where effective steering emerges from the combination of handlebar input and user-induced tilt.

Given hardware constraints and uncertainty about whether expensive actuated systems provide proportional benefits for perceived realism, a passive sensing approach is both pragmatic and scientifically valuable. This approach leverages the existing Saris MP1 platform's natural roll capability

while adding sensor-based feedback to create interlinked tilt–steering coupling without requiring complex mechanical modifications.

4.2 Design for Enhanced Tilt–Steering Integration

4.2.1 Incremental Development Strategy

The proposed enhancement follows a systematic and incremental approach that allows for empirical evaluation and iterative refinement. This strategy addresses two aspects:

1. Variable resistance control for the passive tilt mechanism
2. Multiple mathematical models for combining tilt and handlebar inputs into unified steering commands

The incremental approach enables controlled experimentation with different parameter settings and model configurations. This supports the development of evidence-based design decisions.

4.2.2 Passive Tilt Resistance System

The goal of this component is to design an additional or mechanical modification for the existing Saris MP1 motion platform [42]. The system should offer adjustable resistance on the roll axis, allowing quick and simple changes in the stiffness levels during testing. Adjustments must be fast enough to allow a single participant to experience multiple settings within one user study session.

The purpose of this design is two-fold. First, it supports the search for a generally suitable stiffness configuration that could enhance balance and realism for most riders. Second, it enables the investigation of rider-specific preferences, for example, whether optimal resistance levels are linked to body mass, cycling experience, or individual balance strategies. Identifying such relationships could help define a universal setting or adaptive configurations for different user groups.

The details of the implementation of this resistance mechanism will be presented in Chapter 6, where practical solutions are described for modifying the Saris MP1 platform.

4.2.3 Tilt–Steering Integration Models

These three mathematical models are proposed to combine the handlebar angle θ_h and the roll angle θ_r into a unified steering command θ_s . Each model represents a different hypothesis on how tilt and steering inputs should be integrated to achieve a perceived realistic bicycle dynamic.

Baseline Model. The baseline configuration maintains the current approach, where the steering output directly reflects the handlebar input:

$$\theta_s = \theta_h \quad (4.1)$$

This serves as a baseline for a comparative evaluation, allowing the assessment of whether tilt integration provides measurable benefits over existing implementation.

Nonlinear Model. The nonlinear approach incorporates tilt influence through a sinusoidal relationship:

$$\theta_s = \theta_h + \alpha \cdot \sin(\theta_r) \quad (4.2)$$

where α represents a scaling parameter that controls the magnitude of tilt influence.

The sine function is justifiable through multiple theoretical principles. Fraser et al. [15] show that human tilt perception follows nonlinear patterns, with both E-effect (small angles perceived as larger) and A-effect (large angles perceived as smaller). The sinusoidal relationship captures this perceptual non-linearity while providing natural compression of large tilt angles and amplification of subtle movements.

More generally, the **Weber–Fechner law** describes sensory perception as following a logarithmic rather than a linear function of stimulus intensity [10]. In the context of tilt, this implies that the sensitivity is greater near the upright position and decreases as the angles increase. Although the sine function is not a direct implementation of Weber–Fechner scaling, it reflects the same principle of *relative sensitivity*: small tilts are perceived as more salient than large ones.

Velocity-Weighted Model. The velocity-weighted approach implements speed-dependent blending between handlebar and tilt inputs:

$$\theta_s = w(v) \cdot \theta_h + (1 - w(v)) \cdot \theta_r \quad (4.3)$$

where $w(v) = \frac{v^2}{v^2 + v_0^2}$ represents a speed-dependent weighting function with characteristic velocity v_0 .

This model is derived from the established literature on bicycle dynamics. The speed-dependent weighting reflects the principle of the Carvallo-Whipple model [34] that the characteristics of the bicycle dynamics are coupled to velocity. At low speeds, manual control (tilt input) dominates, while at higher speeds, kinematic steering becomes more important [34].

This approach inverts the quasi-static relationship $\phi \approx v^2/(gR)$ from Eq. (2.4), using tilt to infer the intended curvature rather than to calculate the tilt from the curvature. This positioning echoes Åström et al.’s [1] speed-dependent coupling gains but applies them to input fusion rather than the dynamics themselves.

At low speeds, the weighting function $w(v) \approx 0$, causing the steering output to be dominated by

the tilt input ($\theta_s \approx \theta_r$). This reflects the reality that low-speed bicycle control is heavily dependent on balance adjustments. At high speeds, $w(v) \approx 1$, and the handlebar input dominates ($\theta_s \approx \theta_h$), consistent with the kinematic steering behavior described in Eq. (2.2).

5 Implementation: System Reliability and Hardware Improvements

This chapter addresses existing technical problems and introduces hardware updates that aim to improve system reliability and functionality. Key issues included unstable power supply, unreliable sensor mounting, and restricted access to critical components. Without resolving these issues, reliability would negatively affect general usability and the conduct of evaluation studies.

5.1 Brake Sensor Improvements

5.1.1 Problem

The original brake sensor occasionally caused a 'sticking' effect, where the brake remained active in the simulation even after the user released the brake lever. This impaired user control and reduced the realism of the cycling experience. This issue can be observed in the previous system setup as shown in Figure 3.1.

5.1.2 Solution

To address this, both hardware and software aspects were refined:

- An external button connected to the ESP32 was repurposed to manually reinitialize the angular offset of the brake sensor during runtime, allowing quick recalibration when needed.
- The system now calculates the angle of the brake instead of sending the raw Hall sensor values. During initialization or reset, an offset is applied to establish a stable zero point, preventing sticking.
- A threshold buffer ignores angles smaller than 5 ° to reduce noise and accidental input. Only angles between 5 ° and 32 ° are considered valid.
- In Unity, this valid range (5 ° to 32 °) is linearly assigned to a normalized value between 0 and 1; angles below 5 ° are treated as zero input.

5.1.3 Benefit

These improvements resulted in smoother braking control, improved realism, and reduced user frustration during simulation. The updated brake sensor setup, including the microcontroller in a protective case with a reset button, is shown in Figure 5.1.

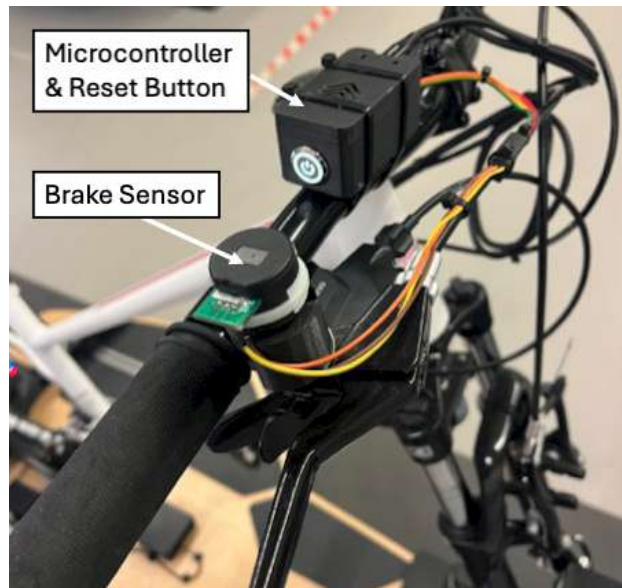


Figure 5.1: Updated brake sensor and microcontroller in case with reset button

5.2 Power Supply and Cable Management

5.2.1 Problem

Originally, all ESP32 microcontrollers shared a single 5 V power rail. Some sensors were connected via a USB hub, but this setup was temporary and incomplete. Loose USB cables led to frequent disconnections and unstable signals, particularly for sensors on moving parts, such as the steering and brake systems. In addition, mechanical strain on cables caused by handlebar movements contributed to unreliable sensor behavior, as illustrated in Figure 3.1.

5.2.2 Solution

To improve stability:

- Each ESP32 microcontroller was assigned an individual USB power connection, replacing the shared 5 V line, thereby reducing voltage drops and electrical interference.
- USB cables were routed and secured along the frame of the bicycle to minimize mechanical stress and prevent accidental disconnections.

5.2.3 Benefit

These updates improved overall system stability, particularly during startup, extended test runs, and firmware updates. The improved cable management and dedicated power supply connections are shown in Figure 5.2.

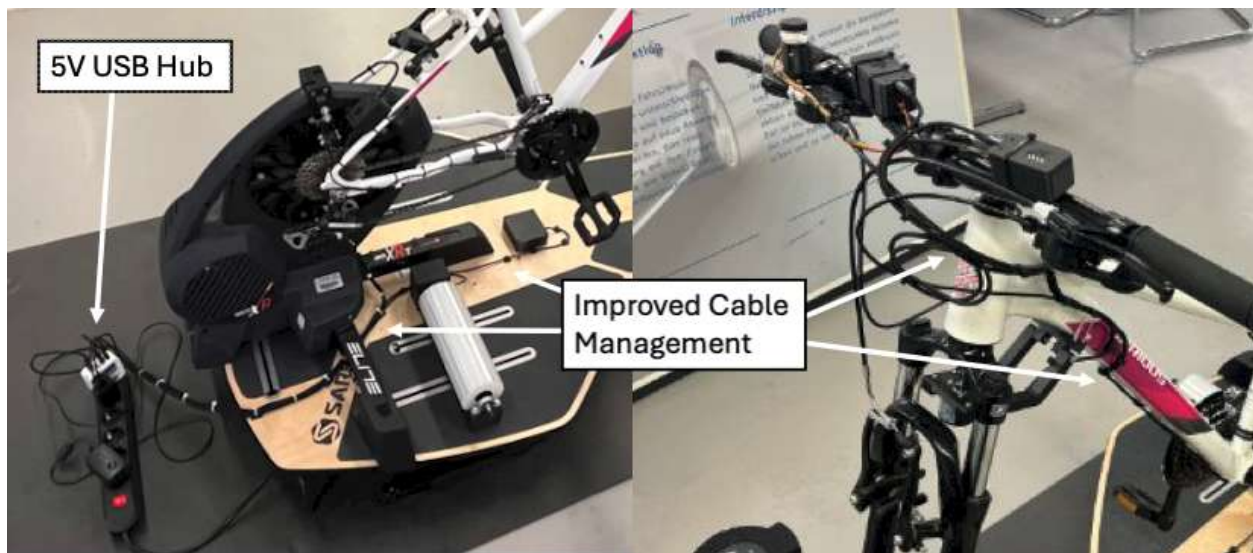


Figure 5.2: Updated power supply and improved cable management

5.3 Reverse Motion Input

5.3.1 Problem

The original reverse motion input was only provisionally mounted on a wooden base, which made the setup mechanically unstable and unreliable. In addition, the electronics were connected via two fixed cables for a 5 V DC power supply, without direct access to a micro-USB interface. This limited both usability and maintainability: the component could not be debugged, reprogrammed, or powered independently without disassembling the entire part. A comparison of the previous solution (left) and the updated implementation (right) is shown in Figure 5.3.



Figure 5.3: Comparison of reverse-motion input: previous provisional setup (left) and updated implementation (right)



Figure 5.4: Updated reverse-motion input: (left) opened enclosure with electronics; (right) integrated into the simulator setup

5.3.2 Solution

To address these issues, the component was redesigned.

- The enclosure was updated to make the USB port accessible from the front. The previous layout did not provide enough clearance for the connector due to interference with the middle foot of the rear bike trainer.
- The entire frame that holds the roller (white cylinder) and the electronics casing was redesigned in CAD, then manufactured as modular 3D printed parts.
- New 3D printed clamps were developed to fix the roller on its axis. These clamps are glued to the ends of the cylinder using double-sided tape, ensuring reliable positioning.
- The casing now uses standard screws 3 mm, allowing it to be easily opened, closed, and disassembled while remaining mechanically robust.

5.3.3 Benefit

The component now boots more consistently, as the microcontroller can be powered directly via USB, independently of the rest of the system. Debugging and firmware updates are easily possible without disassembly, and mechanical parts can be reprinted and replaced if needed. The updated design is shown in Figure 5.4, with the opened enclosure and electronics on the left, and the fully integrated component on the right.

5.4 Roll Angle (Tilt) Sensor

5.4.1 Problem

In the previous setup, the roll angle sensor and its associated circuit board (including the microcontroller) were directly taped to the Saris MP1 platform [42]. This posed a significant risk, as the electronics were neither securely mounted nor adequately protected against accidental contact, dust, or liquids, which could easily cause damage. Furthermore, the power supply relied on two fixed cables for a direct 5 V DC connection. This caused problems with voltage drops and current spikes and also prevented direct access to the microcontroller via USB for software updates or debugging.

5.4.2 Solution

To address these issues, both the hardware design and the firmware were improved:

- A new enclosure was designed and mounted flat on the platform. The lid is fixed with four standard 3 mm screws, allowing the casing to easily open and close while ensuring stable protection. The design provides access to the USB port of the microcontroller without disassembly.
- The electronics were reviewed and the power supply was switched to a single 5 V USB connection. This reduced instability and enabled direct access to the microcontroller.
- The firmware was updated to include a more robust boot sequence. Additional delays were added to ensure the microcontroller was fully initialized before attempting to read data from the BNO055 9DoF sensor. On startup, the current roll angle is now determined and used as an offset, resulting in a more reliable reference.

5.4.3 Benefit

The redesign improved reliability by ensuring a stable power supply, protecting the electronics, and enabling easy debugging and software updates. The updated boot sequence further increased robustness, providing consistent initialization and accurate roll angle measurements. The new roll angle sensor unit is shown in Figure 5.5.

5.5 Back Wheel and Pedal Sensors

5.5.1 Problem

The initial implementation of the back wheel and pedal sensors by Hessenauer et al. [23] was functionally valid, but the Hall sensors themselves lacked a strong mounting on the bicycle frame



Figure 5.5: Updated roll angle (tilt) sensor in protective enclosure

and the rear wheel trainer. As a result, precise positioning of the sensors relative to the magnets was difficult to maintain, which occasionally reduced reliability.

5.5.2 Solution

To improve robustness, two new sensor mounts were developed:

- For the back wheel of the bicycle trainer, a custom mounting bracket was designed and 3D printed. This bracket allows for precise adjustment of the Hall sensor relative to the magnets on the flywheel, ensuring stable signal detection (see Figure 5.6, left).
- The Hall sensor for the magnets on the sprocket was mounted using a repurposed piece of the back wheel mount. This provided a simple and stable solution while reusing existing design elements (see Figure 5.6, right).

5.5.3 Benefit

The new mounts improved reliability by keeping both sensors in stable alignment with their respective magnets. They also make future adjustments more convenient, as the mounts allow for fine repositioning and easy replacement if required. The updated designs are shown in Figure 5.6.

5.6 Summary

The hardware updates in this chapter substantially increased overall system reliability and reduced the failure modes related to power delivery, loose cabling, and insecure sensor mounting. By enclosing electronics, switching to dedicated USB power, and adding robust mounts, components are now easier to access, debug, and improve maintainability during development. Firmware refinements (e.g. stable boot sequences and calibrated offsets) further enhanced the consistency of

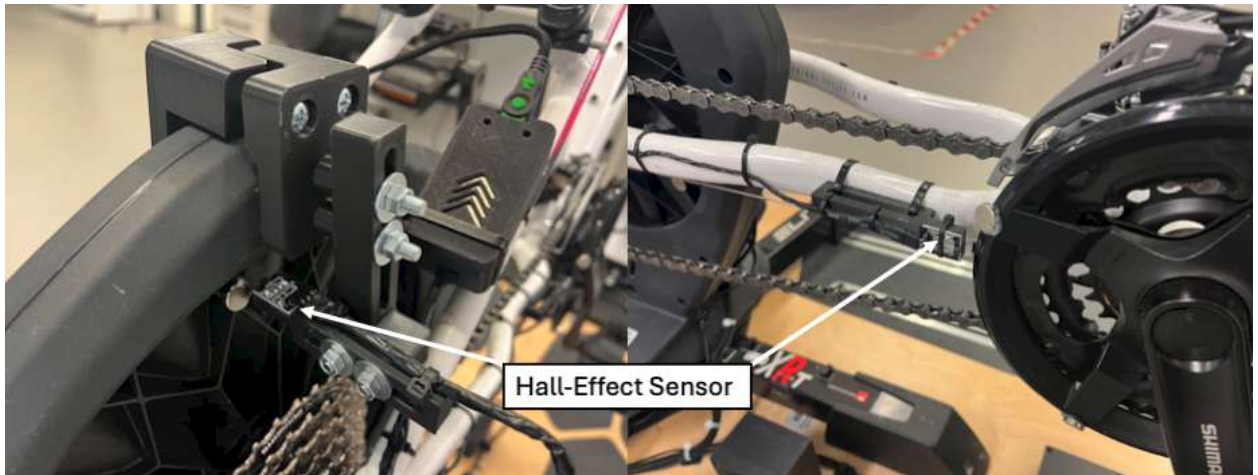


Figure 5.6: Updated mounts for back wheel sensor (left) and pedal sensor (right)

sensor readings and startup behavior. These changes provide a stable baseline for future work, including additional firmware tuning, alternative sensing options, and iterative mechanical refinements. Importantly, the improved robustness benefits user studies: Sessions can proceed with fewer interruptions and less technician overhead, allowing participants and researchers to focus on experimental tasks rather than reliability issues.

6 Implementation: Improved Steering and Balance Dynamics

This chapter describes the implementation of improved steering and balance dynamics for the simulator. The work follows the strategy outlined in Section 4.2, which consists of two main elements:

- the implementation of variable resistance for the passive tilt mechanism (see Subsection 4.2.2), and
- the integration of multiple mathematical models that combine tilt and handlebar inputs into unified steering commands (see Subsection 4.2.3).

However, before these integration steps, the general stiffness of the setup was improved to provide a stable foundation for tilt-based dynamics. Therefore, the following sections begin with modifications to increase structural rigidity, followed by the implementation of the enhanced tilt–steering interaction.

6.1 Setup Stiffness

During development of the adjustable tilt mechanics, the setup revealed noticeable play. The cause was the insufficiently rigid mounting of the rear bicycle trainer and the front gradient simulator on the Motion platform. In the previous setup, both devices were secured with Velcro straps (gradient simulator on the left, trainer on the right in Figure 6.1), which introduced mechanical play. This reduced stiffness, hindered reproducible and consistent tilt dynamics, and degraded the accuracy of roll angle sensing.

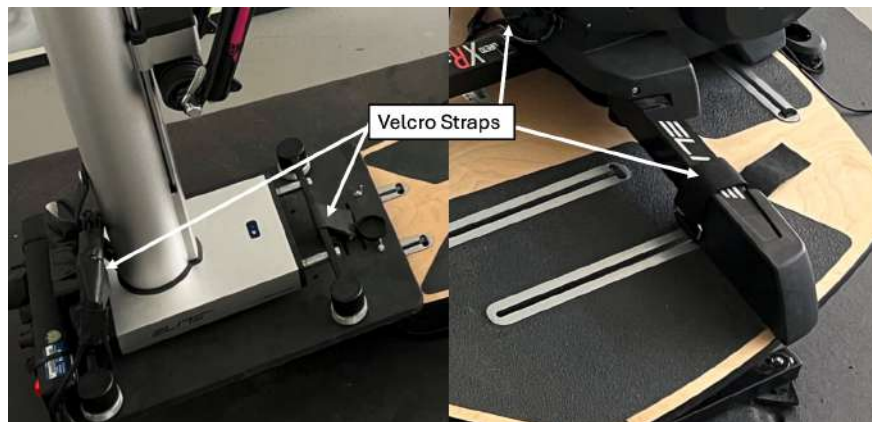


Figure 6.1: Velcro-strap mounting of the gradient simulator (left) and bicycle trainer (right) on the motion platform

To increase stiffness, both mounting points were converted to bolted connections:

- **Rear bicycle trainer:** The Velcro straps were removed and two 11 mm holes were drilled into the platform. The trainer is now fastened with two standard M10 screws and washers, creating a play-free fit (Figure 6.2, right).
- **Gradient simulator:** In the improved configuration, the simulator is mounted directly to the adapter plate with four flat countersunk M8 screws (Figure 6.2, left). The adapter plate itself is then attached to the motion platform using the existing anchor system (Figure 6.2, middle).

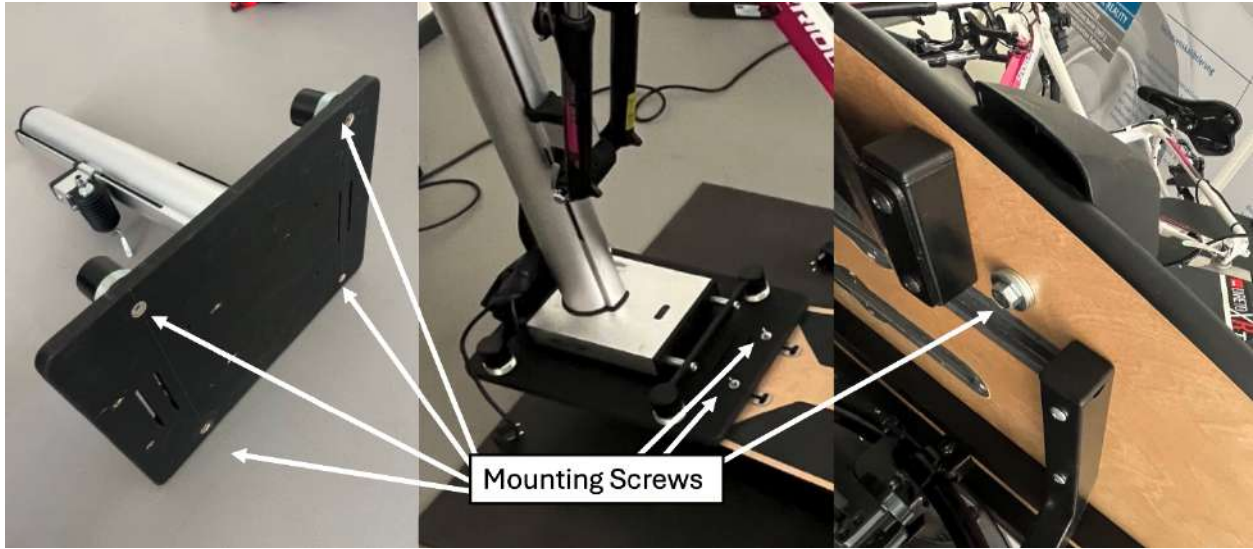


Figure 6.2: Rigid mounting of the gradient simulator (left and middle) and bicycle trainer (right)

The updated bolted setup eliminates mechanical play between the components and the platform, resulting in a stiffer assembly. This improves the repeatability of the tilt dynamics, increases the fidelity of roll-angle measurements, and enhances overall safety and user confidence during operation.

6.2 Tilt-Steering Bicycle Dynamics

6.2.1 Working Principle of the Saris MP1 Tilt Mechanism

The Saris MP1 Motion Platform [42] provides a passive roll degree of freedom, allowing the bicycle to tilt laterally in response to the shifts in weight of the cyclist. The platform and its frame are connected by a central pivot joint aligned parallel to the longitudinal axis of the bicycle. This joint enables the platform to roll to the left or right. The resistance to tilt is provided by a flat and round steel spring that is clamped in the center of the frame. As the rider leans, the spring is deflected, generating a restoring torque that increases with the lean angle. This creates a natural but

relatively soft tilt response, which, as already reported in Chapter 3, can feel unstable or 'wobbly' in simulator use.

In addition to roll, the MP1 platform also allows a limited passive fore–aft motion. This rules out simple solutions such as adding dampers or springs directly between the bicycle frame and the platform, because these would interfere with longitudinal motion or require extensive modifications. Therefore, the chosen approach builds on existing roll mechanics and adds adjustable resistance in a way that does not require irreversible changes to the current platform. The principle of this extension is shown in Figure 6.3: the original spring setup is combined with additional supports to allow the adjustment of the stiffness depending on the configuration.

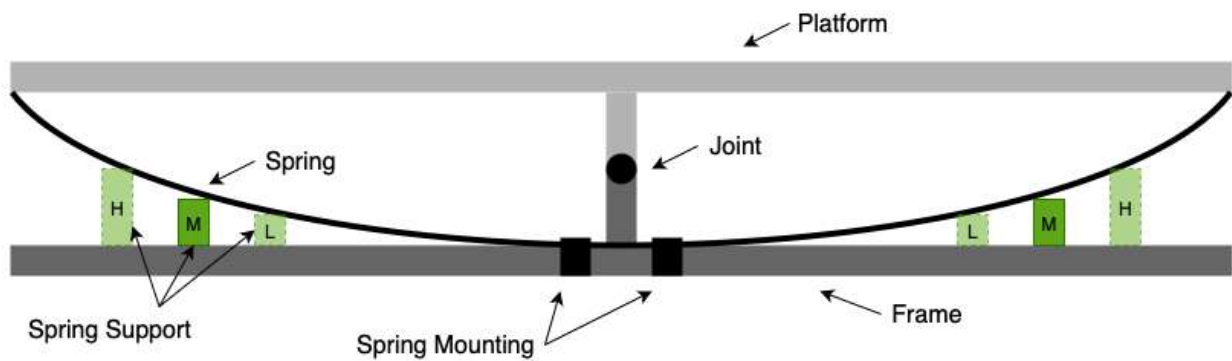


Figure 6.3: Concept for extending the Saris MP1 tilt mechanism with passive adjustable resistance

6.2.2 Concept for Passive Adjustable Tilt Resistance

The goal of this work is to extend the Saris MP1 platform with an additional mechanism that allows an adjustable resistance for passive tilt. This addresses the unstable riding experience reported in earlier evaluations by allowing the stiffness to be increased or decreased depending on experimental needs. The concept in Figure 6.3 introduces additional spring supports between the flat steel spring and the frame.

The effective stiffness depends on both the distance of the support from the center of the spring and on the height of the padding (as illustrated in Figure 6.3, supports labeled L, M, and H for the low, medium and high positions). Supports positioned further away from the center and with higher pads result in greater resistance.

To make this adjustment practical, the supports must be variable in height and fixed linearly to prevent lateral movement. A rail and carriage design was chosen to meet these requirements while keeping the modification reversible and fully compatible with the existing MP1 platform.

6.2.3 Passive Adjustable Tilt Rail

The rail system is based on two 25×25 mm T-slot aluminum profiles mounted to the existing Saris MP1 platform frame. These profiles guide the carriages and allow for linear positioning using T-nuts. The aluminum profiles were cut to the correct length and fastened in place using custom 3D printed adapters, which connect the two rails without interfering with the spring or other mechanical parts. The adapters were printed with PLA at 50 % infill, which provides sufficient stability for this use case. An overview of the frame with the integrated rail extensions is shown in Figure 6.4.

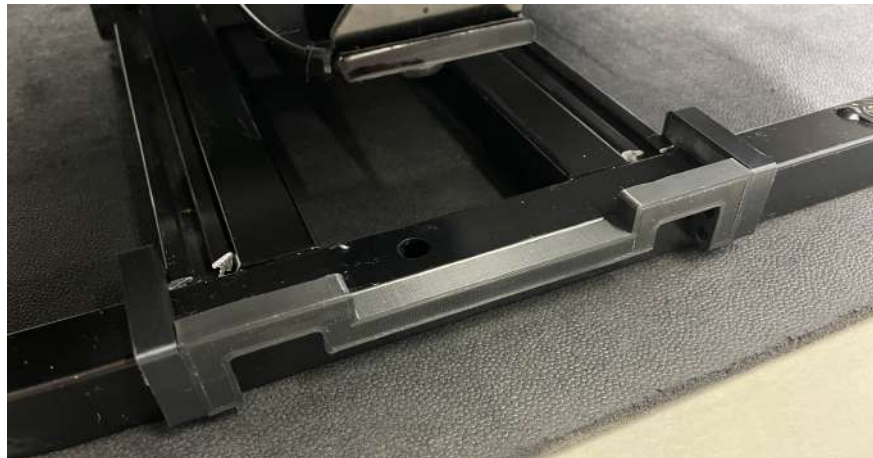


Figure 6.4: Integration of the rail extensions into the MP1 frame

6.2.4 Passive Adjustable Tilt Carriage

The carriages are placed below the flat spring between the two existing steel rods in the frame. They can be fixed at arbitrary positions along the aluminum profiles, enabling different stiffness settings. As illustrated in Figure 6.5, the spring does not rest directly on the carriage. Instead, interchangeable pads of different heights fill the gap between the spring and the carriage base, allowing the resistance to be adjusted.

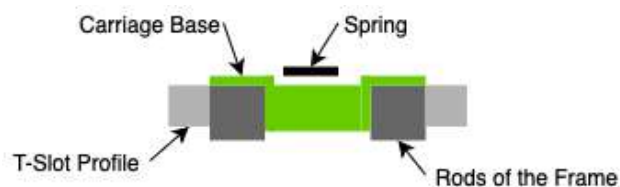


Figure 6.5: Concept sketch of the carriage base supporting the spring via pads

Because the carriage must withstand high loads, a stainless steel base was manufactured (Figure 6.6). It consists of welded L- and U-profiles with a flat steel plate that provides strength and

corrosion resistance. The choice of stainless steel and welding was necessary due to space constraints: in low-resistance settings near the center of the spring, the underside of the spring aligns with the upper rods of the frame. This creates a vertical offset that the carriage must bridge while still carrying the spring load. Therefore, a robust welded steel solution was chosen to provide both the required strength and the compact geometry needed in the limited available space.



Figure 6.6: Carriage base manufactured from welded stainless steel

The carriage housing is constructed from 3D printed PLA parts (infill 50 %) and is fixed with M3 screws. It fulfills two main purposes: first, to hold and guide the pads so they can be mounted securely on the carriage base; and second, to keep the entire carriage fixed onto the aluminum rails while allowing easy repositioning. Adjusting the pad height and repositioning the carriage must be able to be done in a short time. The PLA material is sufficiently strong for the housing in this application, while the pads themselves were printed with 100 % infill to withstand compressive forces.

Figure 6.7 shows the assembled carriage in different configurations and the three implemented pad thicknesses. Subfigure (a) shows the empty carriage frame with open bracket, subfigure (b) the closed carriage with base and pad, and subfigure (c) the three pad heights (7.5 mm, 10 mm, and 12.5 mm).

For pad changes, a hinged clamping bracket with hand screws allows quick replacement without tools. The three discrete pad heights provide low, medium, and high stiffness. These settings were deliberately chosen as coarse steps rather than fine increments, because it is not yet clear how riders will perceive different stiffness levels. Large, clearly noticeable differences help to identify the general direction of suitable settings. With limited time and a small number of study participants, more than three options would not be practical. The selected levels are clearly distinguishable from each other and also from the baseline (without pad), creating a solid foundation for experimentation.

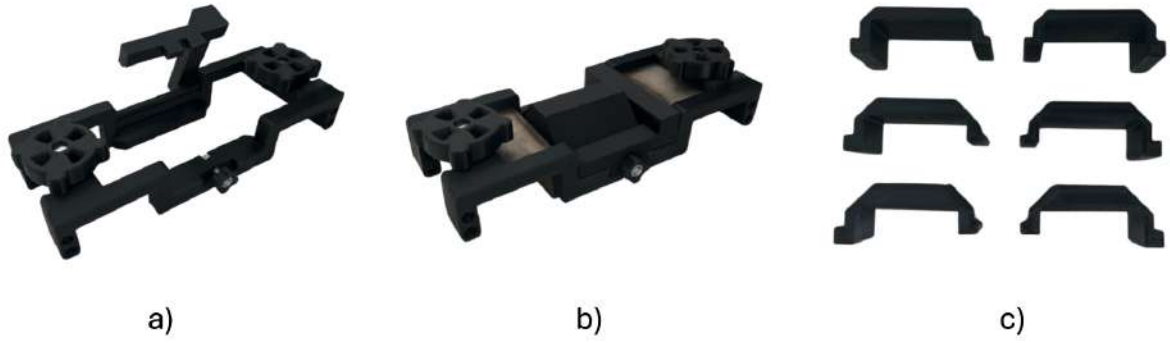


Figure 6.7: Carriage design with interchangeable pads: (a) empty frame with open bracket, (b) closed carriage with base and pad, (c) pad heights 7.5 mm, 10 mm, and 12.5 mm

6.2.5 Final Assembly

Two mirrored carriages were built and mounted on the left and right sides of the Saris MP1 platform spring. The final assembly of the complete system is shown in Figure 6.8.

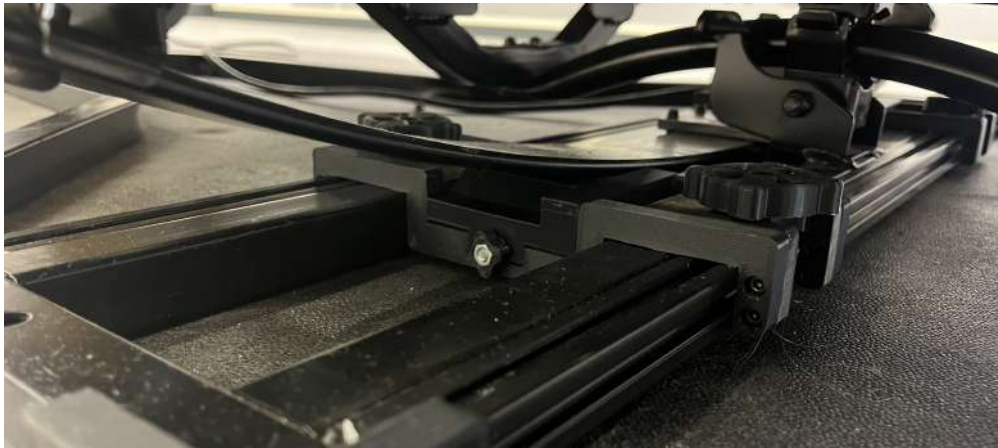


Figure 6.8: Final assembly of the two mirrored carriages integrated into the Saris MP1 platform

6.3 Tilt-Steering Models in Unity Implementation

The three integration models described in Section 4.2.3 were implemented in Unity within the physics script `BicycleController.cs`. All relevant inputs (handlebar angle θ_h , roll angle θ_r , and velocity v) are available in real time, allowing the steering command θ_s to be calculated directly in the simulation loop and updated seamlessly during runtime.

6.3.1 Implementation Notes

- **Baseline Model:** $\theta_s = \theta_h$ serves as reference case.
- **Nonlinear Model:** $\theta_s = \theta_h + \alpha \cdot \sin(\theta_r)$ with α tuned experimentally for natural behavior.
- **Velocity-Weighted Model:** $\theta_s = w(v) \cdot \theta_h + (1 - w(v)) \cdot \theta_r$ with $w(v) = \frac{v^2}{v^2 + v_0^2}$ the parameter v_0 was adjusted for a smooth transition between the low and high speed regimes.

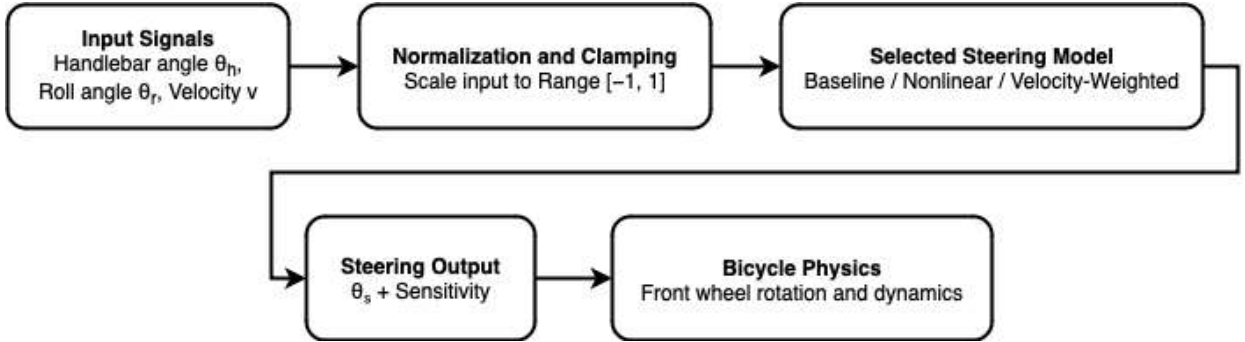


Figure 6.9: Processing pipeline for steering integration in Unity: input signals are normalized, processed by the selected model, and mapped to the bicycle physics.

6.3.2 Steering Logic in Unity

The steering logic follows a structured five-step pipeline (see Figure 6.9):

1. **Input acquisition:** The raw input signals are read: handlebar angle θ_h , roll angle θ_r and forward velocity v .
2. **Normalization and clamping:** All inputs are normalized and clamped to the range $[-1, 1]$ to ensure comparability and prevent outliers from destabilizing the simulation.
3. **Model selection:** Based on the Unity Inspector setting (see Figure 6.10), the chosen steering model (baseline, non-linear or velocity weighted) combines the inputs according to its equation, using constant parameters defined in the script (see Table 6.1).
4. **Steering output:** The model produces a unified steering command θ_s and an associated sensitivity factor.
5. **Physics application:** The steering command is passed to the bicycle physics, directly controlling the rotation of the front wheel and thus the simulated steering geometry.

This modular pipeline ensures that all steering models are tested under identical conditions while keeping the implementation transparent and easily adjustable.

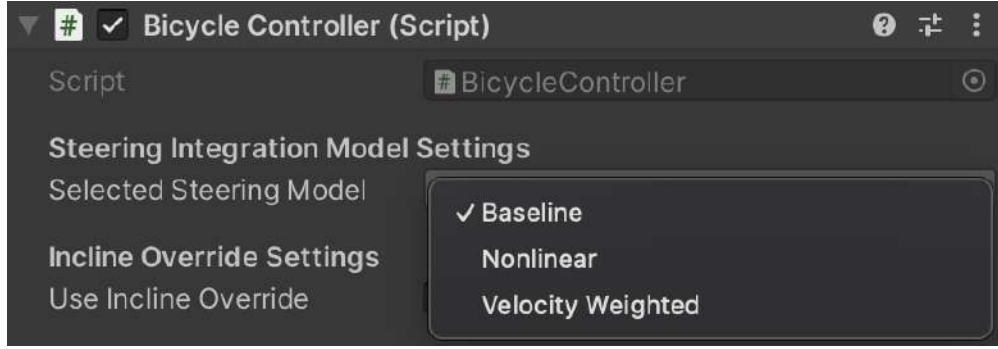


Figure 6.10: Unity Inspector interface of the bicycle object with drop-down menu for selecting the steering model.

6.3.3 Practical Integration

Within Unity, the three models are exposed through an enumeration, allowing the researcher to switch between them directly in the Inspector without editing or recompiling code (see Figure 6.10). This flexibility allows the simulator to be reconfigure instantly during experiments, which is particularly valuable for structured comparisons and user studies.

Table 6.1: Predefined constants for the three steering models in Unity implementation.

Model	Parameter	Value
Baseline	Sensitivity	1.0
Nonlinear	Gain α	0.25
	Sensitivity	0.9
Velocity-Weighted	Characteristic velocity v_0	1.5
	Sensitivity	0.85

The sensitivity parameters for each model were calibrated by exploratory testing with three participants: one with previous VR bicycle simulator experience and two without. Due to limited time and resources, systematic parameter derivation was not feasible. Instead, practical tuning was applied to ensure a stable and plausible steering behavior across the models. The chosen constant values are summarized in Table 6.1. Although this leaves open questions regarding optimal parameterization, it provided a workable basis for validating the feasibility of tilt-steering integration within the scope of this thesis.

7 Study Design and Methodology

To evaluate the performance of the improvements made and to provide a general assessment of the VR bicycle simulator, a two-part study was conducted.

The first part focuses on verifying whether changes in bicycle dynamics regarding tilt resistance and steering models lead to measurable improvements. This includes examining which of the possible combinations offers the best overall performance (if there is any improvement) and exploring whether the optimal configuration may be related to the rider's body weight.

The second part addresses a broader evaluation of the simulator as a training tool. Here, the goal is to determine whether the current setup can already produce measurable training effects, especially in challenging urban traffic scenarios. Finally, the findings are compared to the recent study by Baumann et al. [3], to further validate the progress of the simulator.

7.1 Research Hypotheses

This study investigates the effects of enhanced steering and tilt dynamics on the performance of VR bicycle simulators through two evaluation phases.

7.1.1 Phase 1: Tilt-Steering Dynamics Investigation

Phase 1 examines the effectiveness of improved tilt-steering integration through a systematic evaluation of different configuration combinations.

Hypothesis 1: Enhanced Configuration vs. Baseline Performance

H₁: *The best performing enhanced tilt-steering configuration outperforms the baseline configuration (N-B) in terms of composite user experience scores.*

- **H_{1,0}:** There are no significant differences in composite performance scores (comfort + realism + control) between the baseline configuration (N-B) and any enhanced configuration.
- **H_{1,1}:** The best performing enhanced tilt-steering configuration yields significantly higher composite performance scores than the baseline configuration (N-B).

This hypothesis investigates whether any of the new tilt-steering configurations provides measurable improvement over the existing baseline system.

Hypothesis 2: Personalized Configuration vs. Baseline Performance

H₂: *Individually selected personalized configurations outperform the baseline configuration.*

- **H_{2,0}:** There is no significant difference in performance between the personally chosen configurations of the participants and the baseline configuration (N-B).
- **H_{2,1}:** Individually chosen configurations of the participants significantly outperform baseline configuration (N-B) in composite performance scores.

This hypothesis examines whether allowing individuals to select their preferred configuration provides measurable benefits over the baseline system.

Hypothesis 3: Weight-Configuration Correlation

H₃: *Participants' body weight correlates with their preferred stiffness settings.*

- **H_{3,0}:** There is no significant correlation between the body weight of the participants and their preferred stiffness level (based on the highest rated configuration).
- **H_{3,1}:** There is a significant positive correlation between participants' body weight and their preferred stiffness level (heavier participants prefer stiffer settings).

This hypothesis explores whether physical characteristics can predict configuration preferences, providing a basis for initial configuration recommendations. The stiffness level of each participant's highest-rated configuration during trials is used as preference.

Hypothesis 4: Steering Model Main Effect

H₄: *Configurations with enhanced steering models (NL or VB) yield higher composite user experience scores than configurations with the baseline steering model (B).*

- **H_{4,0}:** The average composite scores of the configurations with the enhanced steering models (NL or VB) do not differ significantly from the average composite scores of the configurations with the baseline steering model (B).
- **H_{4,1}:** The average composite scores of the configurations with enhanced steering models (NL or VB) are significantly higher than the average composite scores of the configurations with the baseline steering model (B).

This hypothesis tests whether enhanced steering dynamics provides a general performance benefit, regardless of stiffness settings.

7.1.2 Phase 2: Learning Effects and Training Applications

Phase 2 evaluates the effectiveness of the simulator for training applications and compares the performance with the previous study [3].

Hypothesis 5: Danger Exposure Benefit

H₅: *Prior danger scenario exposure enhances performance in subsequent danger scenarios compared to safe-to-danger transitions.*

- **H_{5,0}:** Participants who experience safe-to-danger transitions show no significant differences compared to those who experience danger-to-danger transitions in terms of stress levels, cybersickness symptoms, presence scores, immersion, and safety perception.
- **H_{5,1}:** Participants who experience danger-to-danger transitions demonstrate significantly better performance compared to safe-to-danger transitions across multiple measures: lower stress levels, reduced symptoms of cybersickness, increased presence and improved perception of environmental realism and safety.

This hypothesis tests whether progressive difficulty training (first exposure to danger) provides superior learning outcomes by comparing the differences between safe-danger versus danger-danger exposure conditions on 10 dependent measures.

Hypothesis 6: System Improvement Validation

H₆: *The current enhanced VR bicycle simulator shows improved performance compared to the previous study by Baumann et al. [3].*

- **H_{6,0}:** Across both rides of the danger–danger exposure condition, the current system shows no significant improvement in stress management and immersion compared to the previous study.
- **H_{6,1}:** Across both rides of the danger–danger exposure condition, the current system demonstrates significantly better stress management and higher immersion scores compared to the results of the previous study.

This hypothesis validates the effectiveness of tilt steering enhancements through a direct comparison with the established baseline performance from the previous study. The comparison will examine the general results of the exposure condition to danger and danger, measuring stress levels and presence scores on seven dependent measures.

7.2 Participants

7.2.1 Inclusion and Exclusion Criteria

To ensure comparability with the recent study [3], the same criteria were applied to the participants.

Inclusion Criteria

- Participants had to be able to ride a bicycle in the real world. This was important because the simulator used in this work was not designed to teach basic cycling skills, but rather as a training tool for traffic scenarios and as a research platform to evaluate cycling behavior.
- Participants had to be able to assess the realism, comfort, and usability of different simulator configurations (e.g., steering models and tilt resistance). Such evaluations would not be meaningful if they did not have prior cycling ability.

Exclusion Criteria

- Participants with medical conditions that would prevent safe participation were excluded from the study.
- Participants with previous negative experiences in simulators or virtual reality were also excluded. This served as both an ethical precaution, avoiding unnecessary discomfort, and a methodological one, since prior negative experiences could bias the results (e.g., increasing susceptibility to simulator sickness).

7.2.2 Demographic and Background Measures

Before the experimental phases, participants completed a short demographic questionnaire to collect variables relevant to the research hypotheses. The following information was collected using categorical response options:

- **Basic Demographics:**
 - Age: 6 categories (18-24, 25-34, 35-44, 45-54, 55-64, 65+)
 - Gender: 4 options (Male, Female, Other, Prefer not to say)
- **VR Experience:**
 - Self-reported familiarity: 3 levels (Not familiar, <10h, 10h+)

This measure allows to investigate potential familiarity effects, as previous VR exposure could influence adaptation rates and motion sickness.

- **Cycling Background:**

- Weekly cycling frequency: 3 categories (<1x, 1x, Several times)
- Urban cycling frequency: 3 categories (<1x, 1x, Several times)

These measures assessed the real-world cycling experience and exposure to urban traffic to put simulator performance and realistic ratings into context.

- **Physical Characteristics:**

- Body weight: Continuous variable collected in kilograms via text input field

The weight data was considered particularly important given the theoretical relationship between the riders' mass and the optimal tilt resistance setting cycling dynamics. This created the basis for testing Hypothesis 3 with respect to the prediction of the weight-based configuration.

The demographic questionnaire was implemented in German and English, consistent with all other measurement instruments (see Appendix [A.2](#)).

7.2.3 Sample Size Justification

Statistical Power Analysis

In behavioral and HCI research, a significance level of $\alpha = 0.05$ and the desired power of $1 - \beta = 0.80$ are widely adopted as standard thresholds [6, 9]. These criteria balance the risk of Type I errors (false positives, rejecting the null hypothesis when it is in fact true) and Type II errors (false negatives, failing to reject the null hypothesis when a true effect exists), while providing sufficient sensitivity to detect practically meaningful effects within the scope of this thesis.

For the primary hypotheses testing, the following assumptions were used:

- Significance level: $\alpha = 0.05$
- Desired power: $1 - \beta = 0.80$
- Effect sizes guided by preliminary testing and prior VR cycling research

The effect sizes were informed by two sources: (1) preliminary pilot testing on the developed simulator, which suggested that the differences between the steering models were subjectively noticeable but not extreme, and (2) related studies of VR cycling and simulator, which typically reported perceptual and performance effects of medium to large size.

For example, Wintersberger et al. [54] ($n = 31$) found medium to large effects of simulator tilt on presence and realism, as well as negative effects of excessive tilt on performance and simulator sickness. Similarly, Matviienko et al. [33] ($n = 24$) reported large effects on cycling speed, head movements, and safety ratings, with realism outcomes in the medium range. Martinez Garcia et

al. [17] validated a motion-enabled bicycle simulator with $n = 27$ participants and found very large effects in the lean angle and velocity dynamics. Together, these studies confirm that VR bicycle simulator research typically yields medium to large effects, which can be detectable with samples in the range of 20 to 30 participants.

Sample Size Decision

Based on these considerations, the present study was designed for $n = 20$ participants, with an upper limit of $n = 24$. This range balances statistical sensitivity with feasibility under the practical constraints of this thesis. Each participant session required approximately 60 minutes, plus an additional 15 minutes for preparation, including setup, experimental phases, and questionnaires. With only two weeks of data collection time and a throughput of three to four participants per week, expanding the sample size beyond 24 was not achievable.

The chosen range is directly comparable to previous work: Wintersberger et al. [54] used $n = 31$, Matviienko et al. [33] $n = 24$, and Martinez Garcia et al. [17] $n = 27$. Furthermore, our group's earlier work on immersion factors in VR cycling (Baumann et al. [3]) used a sample of $n = 24$, which further supports the practical applicability of this sample size, although no effect sizes were analyzed in that study.

Phase 1 Power Considerations

As described in Section 7.4.2, each participant completed seven trials, covering different steering–stiffness configurations while avoiding excessive fatigue. This within-subject design with seven trials per participant provides substantial statistical power through repeated measurements. With $n = 20$ participants, the effective sample size for configuration comparisons approaches 140 observations (7×20).

Power analysis indicates adequate detection capability for:

- Medium-to-large effect sizes in paired comparisons ($d \geq 0.5$), following Cohen's conventions for standardized mean differences [6].
- Correlations of $r \geq 0.6$, corresponding to strong associations. Although $r = 0.5$ is conventionally considered large, with $n = 20$ only correlations slightly above this threshold can be reliably detected with power $\approx .80$.
- Mean differences of ≥ 0.5 points on the 7-point Likert scales, representing a moderate perceptual shift (half a response category on average). A complete category shift (1.0 point) would indicate a very strong effect, so 0.5 was chosen as a practically meaningful benchmark.
- Medium effect sizes for the *Analysis of Variance* (ANOVA) main effects ($\eta^2 \geq 0.06$), following Cohen's conventions where $\eta^2 = 0.01$ represents a small effect, $\eta^2 = 0.06$ a medium effect and $\eta^2 = 0.14$ a large effect [6].

Phase 2 Power Considerations

The 2×2 repeated-measures design (scenario type × exposure order) provides sufficient power for detecting:

- Differences between groups in adaptation patterns with $d \geq 0.8$. Given the smaller effective group sizes ($n \approx 10$ –12 per group), the study is able to detect only large effects reliably in comparisons between subjects.
- Comparisons with the means of the previous study using one sample t tests. Here, detectable effects are also limited to medium or larger magnitudes, as smaller deviations from reference means would fall below the sensitivity of the current sample size.

7.3 VR Unity Scenarios

The virtual scenarios used in this study were developed in Unity based on a simulation environment created during a student project that continued the work of Flaig et al. [13]. This environment models a typical urban setting in Germany, including roads, bike lanes, traffic lights and signs, buildings, and parks. Vehicles were also implemented, following predefined routes via a way-point system.

This base environment was later adapted in the work of Baumann et al. [3], where two different danger scenarios were created for user studies. For the current work, these scenarios were reused to allow comparability with previous results. In addition, a safe scenario was implemented to provide a neutral baseline. In total, four different scenarios were implemented: one parameter scenario for Phase 1 (steering and stiffness evaluation), and three urban traffic scenarios for Phase 2 (general simulator evaluation).

7.3.1 Parameter Scenario

The parameter scenario was created for configuration testing in Phase 1. It uses the same base as the urban environment but was intentionally kept as simple as possible to minimize distractions. The course is predefined with start and end signs and is bounded by traffic cones (Figure 7.1).

The course was designed to be relatively short to reduce fatigue over seven trials while still containing left and right curves, as well as a straight segment for higher velocities. This allowed for a balanced exposure to different cycling dynamics within a short session duration.

7.3.2 Safe Scenario

The safe scenario (Figure 7.2) represents a calm baseline condition within the urban setting. It is set during the day with clear visibility and no other traffic participants. The task of the participants



Figure 7.1: Parameter scenario used in Phase 1 for configuration testing.



Figure 7.2: Safe scenario: daytime, no traffic participants.

was to follow the bike path on the right side of the street for the length of a block, which takes approximately 30 seconds to ride. This scenario provides a non-threatening reference for comparing performance and perceived stress against the danger (challenging) scenarios.

7.3.3 Danger Scenario 1

Danger Scenario 1 (Figure 7.3) is set at night, with fog and limited visibility. The street lights illuminate the edge of the road, while the cars move in both directions on the street. The cyclist rides on the bike path to the right. Approximately one third of the way down the route, the bike lane is blocked by a construction site, forcing the participant to find a way around it. Given the poor



Figure 7.3: Danger Scenario 1: nighttime with fog, traffic, and a blocked bike lane.

visibility and the presence of traffic in both directions, this situation is challenging and potentially dangerous.

To ensure safety, cars were programmed with simple ray-based collision detection: If a cyclist or another vehicle is detected within 1 to 2 meters, the vehicle stops. However, vehicles do not actively anticipate the cyclist, maintaining the impression of limited foresight typical of real traffic.

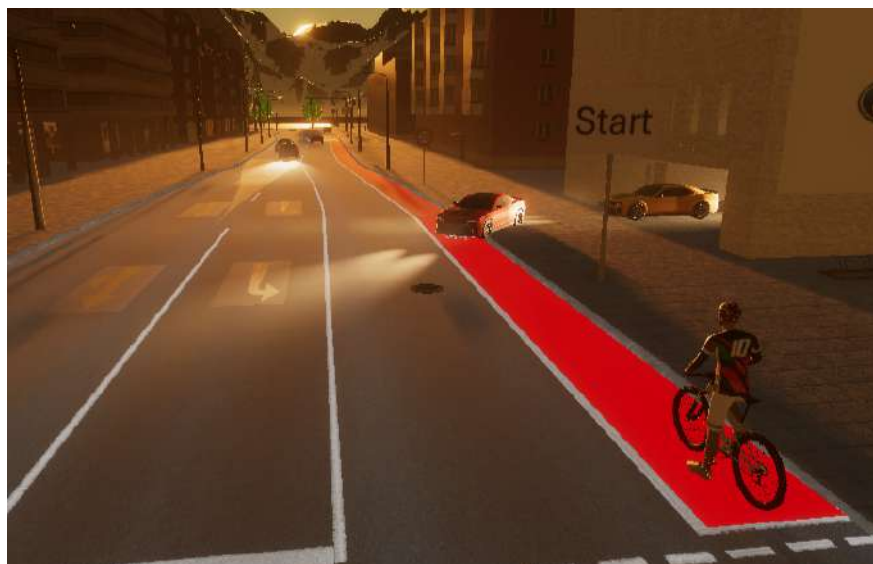


Figure 7.4: Danger Scenario 2: sunset with poor visibility and cars exiting a parking garage across the bike lane.

7.3.4 Danger Scenario 2

Danger Scenario 2 (Figure 7.4) is set at sunset, where glare and low light reduce visibility. The cyclist again follows the bike path on the right, while the cars drive in the adjacent lane. The critical event occurs when cars suddenly exit a parking garage and cross the bike path without checking for approaching cyclists. This forces the participant to react quickly, either by braking or maneuvering around the vehicles. As in Danger Scenario 1, vehicles employ local ray-based collision detection.

Together, these four scenarios enabled systematic evaluation across both study phases: controlled configuration testing in Phase 1 and urban traffic challenges in Phase 2. Including safe and dangerous scenarios allowing us to investigate learning effects, perceived realism, and safety in realistic but ethically responsible VR cycling conditions.

7.4 Study Design

7.4.1 Overall Study Structure

The study used a two-phase design to evaluate the integration of tilt steering and the overall performance of the VR bicycle simulator. The first phase focused on evaluating different tilt-steering integration configurations to identify optimal settings, while the second phase provided a general evaluation of the VR bicycle simulator's performance in various traffic scenarios using the preferred configuration from Phase 1. The complete procedure lasted approximately 60 minutes per participant and was conducted in a controlled laboratory environment at Heilbronn University.

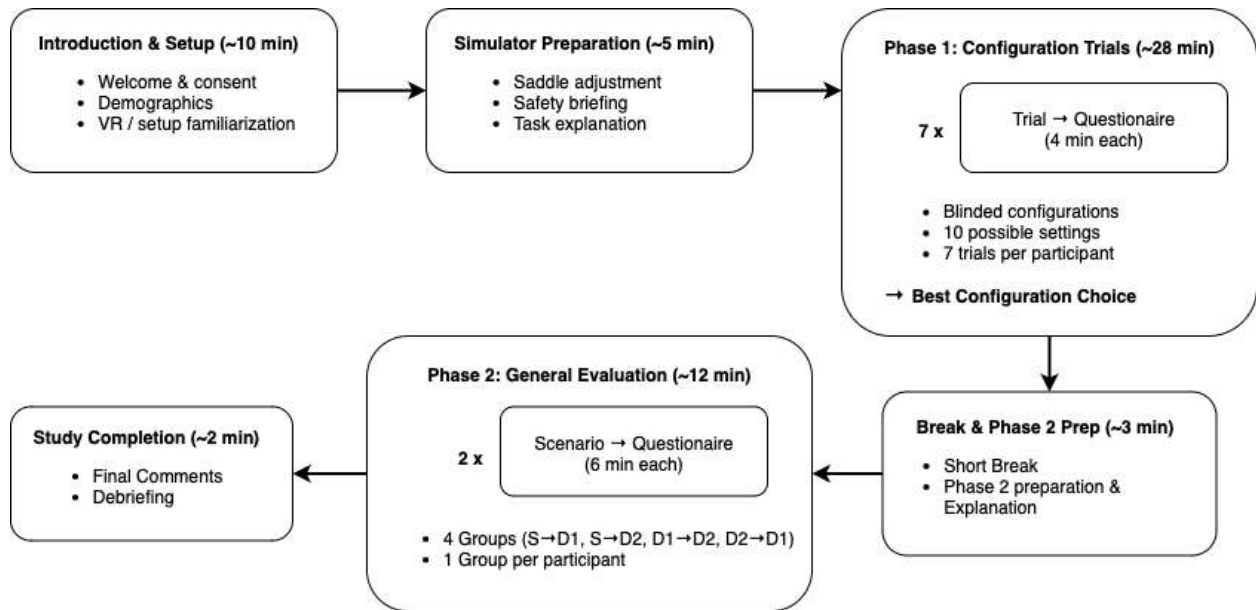


Figure 7.5: Complete study procedure flowchart showing the sequential phases, timing estimates, and decision points for each participant

Study Flow Overview

The procedure followed a sequence designed to minimize bias and ensure participant safety while evaluating different simulator configurations and traffic scenarios (see Figure 7.5):

1. **Introduction and Setup** (≈ 10 minutes)
 - Welcome and brief explanation of study purpose
 - Privacy statement and informed consent
 - Demographics questionnaire completion
 - VR familiarization (SteamVR welcome world exploration)
2. **Simulator Preparation** (≈ 5 minutes)
 - Individual saddle height adjustment for comfort
 - Safety briefing and getting on/off procedures
 - Explanation of Phase 1 task
3. **Phase 1: Configuration Evaluation** (≈ 28 minutes total (4 x 7 min))
 - Seven trials with different stiffness-steering model combinations
 - Post-trial questionnaire after each configuration
 - Best configuration selection for Phase 2
4. **Transition Break & Phase 2 Preparation** (≈ 3 minutes)
 - Short rest
 - Phase 2 explanation
5. **Phase 2: Scenario Evaluation** (≈ 12 minutes total (2 x 6 min))
 - Two traffic scenarios using preferred configuration from Phase 1
 - Post-scenario questionnaire after each scenario
6. **Study Completion** (≈ 2 minutes)
 - Final comments
 - Debriefing and appreciation

7.4.2 Phase 1: Steering and Stiffness Evaluation

The first phase focused on systematically evaluating different combinations of steering models and stiffness levels in the bicycle simulator. Three new stiffness settings were implemented (*Low*, *Medium*, and *High*), and combined with the three steering models (*Baseline*, *Non-Linear*, and *Velocity-Based*). This resulted in a total of nine experimental combinations. In addition, an additional baseline condition was included (*No Stiffness – Baseline Steering*), giving ten unique configurations in total.

Table 7.1: Overview of the 10 tested configurations

Code	Description
N-B	No stiffness – Baseline steering (General Baseline)
L-B	Low stiffness – Baseline steering
M-B	Medium stiffness – Baseline steering
H-B	High stiffness – Baseline steering
L-NL	Low stiffness – Nonlinear steering
M-NL	Medium stiffness – Nonlinear steering
H-NL	High stiffness – Nonlinear steering
L-VB	Low stiffness – Velocity-based steering
M-VB	Medium stiffness – Velocity-based steering
H-VB	High stiffness – Velocity-based steering

Trial Optimization and Balancing

To balance the need for sufficient trials per participant with the risk of exhaustion and VR fatigue, a trial optimization procedure was applied. Each participant completed seven trials, which allowed coverage of all conditions while keeping the total session length feasible. The design was first optimized for $n = 20$ participants (14 repetitions per configuration) to ensure feasibility if only 20 participants could be recruited. Then it was extended to $n = 24$, where each condition appeared 16 to 17 times while still remaining balanced between positions.

The trial order was optimized to ensure that each configuration appeared equally often in all positions and that the baseline condition (*N-B*) was evenly distributed. This was achieved by running a custom Python script that iteratively generated and reordered the trial sets until the imbalance between conditions and positions was minimized.

For complete transparency, the optimized trial sequences for all 24 participants are provided in the Appendix [A.1](#). Each row represents the trial order of one participant with seven configurations per session. This ensured that conditions were not only balanced in general, but also well

Table 7.2: Final distribution matrix for $n = 24$ participants (16–17 appearances each)

Setting	Pos1	Pos2	Pos3	Pos4	Pos5	Pos6	Pos7	Total
N-B	3	2	3	2	2	2	3	17
L-B	2	2	3	3	2	2	2	16
L-NL	2	2	2	3	3	3	2	17
L-VB	2	2	3	3	2	2	3	17
M-B	2	3	3	3	2	2	2	17
M-NL	3	2	2	2	3	2	3	17
M-VB	2	3	2	2	3	3	2	17
H-B	2	2	2	2	3	3	2	16
H-NL	3	3	2	2	2	2	3	17
H-VB	3	3	2	2	2	3	2	17

distributed across the trials.

This optimization ensured that the study design was fair (balanced exposure to conditions) and practical (limited fatigue of participants), providing a reliable basis to compare the different steering and stiffness configurations.

Phase 1 Procedure

Participants were blinded to the specific configuration that was tested in each trial. Before each trial, the researcher adjusted the simulator to the predetermined stiffness setting and steering model. Each trial followed this sequence:

1. **Pre-trial Setup:** Configuration adjustment by researcher
2. **VR Trial:** Cycling session in the virtual environment
3. **Questionnaire:** Completion of post-trial evaluation (comfort, realism, control ratings)

After completing all seven trials, participants selected their preferred configuration for Phase 2, independent of their numerical ratings. This choice represented their subjective preference for phase 2.

7.4.3 Phase 2: Traffic Scenario Evaluation

The second phase focused on evaluating the effect of different traffic scenarios on realism, stress, and perception of safety. Three scenarios were defined:

- **Safe (S)**: Daytime, open road, no traffic
- **Danger 1 (D1)**: Night, fog, dense traffic
- **Danger 2 (D2)**: Sunset, cars crossing bike lane

Group Assignment and Scenario Order

To systematically vary the order of exposure, four groups were created, each combining two scenarios.

- **G1**: Safe (*S*) → Danger 1 (*D1*)
- **G2**: Safe (*S*) → Danger 2 (*D2*)
- **G3**: Danger 1 (*D1*) → Danger 2 (*D2*)
- **G4**: Danger 2 (*D2*) → Danger 1 (*D1*)

Participants were assigned to groups in a rotating pattern based on their ID (1 → G1, 2 → G2, 3 → G3, 4 → G4, 5 → G1, etc.). This ensured a balanced distribution between groups, regardless of whether the final sample size was $n = 20$ or extended to $n = 24$. Effectively, this created two larger meta-groups: one starting with a Safe scenario (G1 and G2), and one starting with a Danger scenario (G3 and G4).

Table 7.3: Group distribution for Phase 2 depending on total sample size

Group	Scenario order	Participants ($n = 20$)	Participants ($n = 24$)
G1	S → D1	5	6
G2	S → D2	5	6
G3	D1 → D2	5	6
G4	D2 → D1	5	6

Phase 2 Procedure

Using the configuration selected in Phase 1, participants experienced two traffic scenarios. Each part lasted approximately 6 minutes and followed this sequence:

1. **Pre-scenario Setup:** Researcher loaded the appropriate traffic scenario
2. **Scenario Brief:** Brief explanation of the upcoming environment without revealing specific hazards
3. **VR Scenario:** 2-minute cycling session in the assigned traffic environment
4. **Comprehensive Questionnaire:** Post-scenario evaluation covering stress appraisal, motion sickness, presence, and perceived realism/safety

7.4.4 Bias Control Measures

Several measures were implemented to minimize researcher bias and maintain the validity of the study.

Standardization Efforts

- Identical introduction and safety briefing for all participants
- Consistent explanation of tasks
- Standardized questionnaire presentation and data collection procedures
- Blinded configuration assignment (participants unaware of specific settings)

Practical Limitations

Despite efforts to maintain strict protocols, some deviations were unavoidable. Participants occasionally asked questions during trials or requested clarification about procedures. The researcher provided the necessary assistance while trying to maintain neutrality and avoid leading responses. During the completion of the questionnaire, participants independently worked, with the researcher remaining available but not providing guidance on responses unless technical difficulties arose.

Interaction Protocols

- Minimal verbal interaction during VR sessions
- Neutral responses to participant questions about their performance
- Encouragement focused on safety rather than performance evaluation

7.4.5 Safety Protocols

The following safety measures were implemented to ensure participant well-being throughout the study: All participants received safety instructions that emphasize their right to terminate participation at any time if they experienced discomfort, motion sickness, or any other adverse effects. The voluntary nature of participation was explicitly reinforced.

8 Measures and Analysis Plan

8.1 Measures

Due to the time constraints of this work, the study relied only on subjective measures for evaluation. This approach was considered sufficient for the research objectives of assessing tilt-steering dynamic configuration preferences and general simulator performance, while staying within the available research time frame.

All questionnaires were implemented in German and English within the study application to accommodate the language preferences of the participants and ensure adequate comprehension.

8.1.1 Phase 1: Configuration Evaluation Measures

Measurement Approach

Phase 1 used a three-question approach to evaluate each tilt-steering dynamic configuration. After each trial, participants rated their experience in three dimensions using 7-point Likert scales ranging from 1 ("not at all") to 7 ("very much"):

- **Comfort:** "How comfortable was this setting?"
- **Realism:** "How realistic did this feel?"
- **Control:** "How much control did you feel?"

Rationale for Scale Selection

The decision to use a three-part evaluation for each trial was driven by practical considerations, aiming to balance participant fatigue from more extensive questionnaires with the need for detailed and meaningful insight into the performance of each configuration.

The selection of comfort, realism, and control as evaluation dimensions was based on their relevance to the main research questions about the effectiveness of tilt-steering dynamic integration. These dimensions capture the essential aspects of user experience that would differentiate between configuration performance: physical comfort during interaction, perceived authenticity of the cycling experience, and subjective sense of agency within the virtual environment.

Final Configuration Selection

After completion of all seven trials, participants selected their preferred configuration that would be used in Phase 2. This selection was made independently of their numerical scores, allowing participants to consider holistic impressions that might not be fully captured by the quantitative

measures. The participants also rated their confidence in this choice on a 7-point scale from "Not confident" to "Very confident."

An option for qualitative feedback on configuration preferences or specific observations that could inform future development was included. The complete questionnaire items, including the German and English versions, are provided in Appendix [A.2](#).

8.1.2 Phase 2: Traffic Scenario Evaluation Measures

Phase 2 used a comprehensive assessment approach using established measures to evaluate the responses of the participants to different traffic scenarios. The measurements covered four primary domains: stress index, motion sickness, presence and immersion, and perceived realism and safety.

Stress Appraisal (PASA)

The responses to stress in virtual traffic scenarios were measured using the *Primary Appraisal Secondary Appraisal* (PASA) scale developed by Gaab [16]. The PASA is a 16-item questionnaire that measures situational cognitive appraisals.

The scale employs a 6-point response ranging from "completely false" to "completely true" and captures four primary dimensions: threat perception, challenge appraisal, self-concept of abilities, and control beliefs. These dimensions are combined to calculate primary appraisal scores (threat + challenge) and secondary appraisal scores (abilities + control), with the overall stress index calculated as the difference between primary and secondary evaluation scores.

The PASA was selected for its validated application in measuring cognitive stress responses in challenging situations, making it particularly suitable for evaluating participant reactions to danger scenarios implemented in this study.

Motion Sickness (CSQ-VR)

Virtual reality motion sickness symptoms were evaluated using the *Cybersickness in Virtual Reality Questionnaire* (CSQ-VR) developed by Kourtesis et al. [27]. The CSQ-VR represents an advancement over traditional motion sickness questionnaires (SSQ, VRSQ) with demonstrated superior internal consistency and VR-specific validation.

The questionnaire assesses three primary symptoms: nausea (stomach discomfort, acid reflux, urge to vomit), vestibular symptoms (dizziness, lightheadedness, spinning sensations) and oculomotor discomfort (eyestrain, blurred vision, headache). Each item is rated on a 7-point scale from "absent feeling" to "extreme feeling."

The CSQ-VR was chosen due to its specific validation in VR environments and superior psychometric properties compared to earlier motion sickness measures, ensuring accurate assessment of cybersickness symptoms in the context of a bicycle simulator.

Presence and Immersion (IPQ)

The sense of presence and immersion in the virtual environment was measured using the *Igroup Presence Questionnaire* (IPQ) developed by Schubert, Friedmann, and Regenbrecht [43]. The IPQ is a widely validated instrument for assessing the presence in virtual environments. It consists of three subscales and one general item. The three subscales are the following.

- **Spatial Presence:** the sense of being physically located in the virtual environment
- **Involvement:** the degree of attention devoted to the virtual environment and the level of engagement experienced
- **Experienced Realism:** the subjective impression of realism within the virtual environment

Items are rated on scales ranging from 5 to 7 points depending on the specific question, with endpoints varying by item (e.g. “not at all” to “very much,” or “very aware” to “not aware”). In addition, the IPQ includes a general item that measures the overall “sense of being there,” that is closely associated with all subscales, particularly in spatial presence.

The IPQ was selected as an established standard for presence measurement in VR research, allowing comparison with existing studies and providing a validated assessment of the quality of presence experienced in the bicycle simulator.

Perceived Realism and Safety

Scenario-specific evaluations were performed using custom items designed to capture aspects particularly relevant to traffic scenarios. Two primary dimensions were assessed:

- **Environmental Realism:** "How realistic did the environment feel?" (7-point scale: "Not realistic" to "Very realistic")
- **Safety Perception:** "How safe or threatened did you feel?" (7-point scale: "Very threatened" to "Very safe")

In addition, optional qualitative feedback was collected after each run. This allowed us to capture additional information on the experiences of the participants that could not be reflected in the quantitative measures.

Questionnaire Subset Selection and Justification

Due to the constraints of this work and the potential fatigue of participants, subsets of the established questionnaires were used in Phase 2. This approach balanced the need for multidimensional measurement (stress, presence, realism, safety) with the practical limits of study time, so that participant engagement could be sustained while still ensuring the results remained meaningful and reliable.

PASA Subset Implementation The original 16-item PASA questionnaire was reduced to four items, with a representative question selected from each of the four primary subscales.

- **Threat Appraisal:** "This situation is very uncomfortable for me."
- **Challenge Appraisal:** "This situation challenges me."
- **Self-concept of Abilities:** "I can best protect myself from failure in this situation through my own actions."
- **Control Beliefs:** "I have no idea what I should do in this situation."

Each item retained the original 6-point Likert scale ranging from "Completely False" to "Completely True," maintaining consistency with the theoretical framework of the transactional stress model.

CSQ-VR Subset Implementation The CSQ-VR implementation maintained six items that covered all three symptom categories to ensure a comprehensive assessment of the symptoms of cybersickness.

- **Nausea symptoms** (2 items): Including stomach discomfort, acid reflux, and tension to vomit
- **Vestibular symptoms** (2 items): Covering dizziness, lightheadedness, and postural instability
- **Oculomotor symptoms** (2 items): Addressing visual fatigue, eyestrain, and discomfort

The original 7-point scale from "Absent" to "Extreme" was maintained to preserve the response structure and enable meaningful comparison with established cybersickness research.

IPQ Subset Implementation The IPQ was condensed to 4 items representing the core presence dimensions and the general presence item:

- **Spatial Presence:** "To what extent did you have a sense of 'being there' in the computer-generated world?"
- **Involvement:** "I had a sense of acting in the virtual space, rather than operating something from outside."
- **Experienced Realism:** "How real did the virtual world seem to you?"
- **Real-world Awareness** (reverse-coded): "How aware were you of the real-world surroundings while navigating in the virtual world?"

Each item maintained a question-specific 5-point scale appropriate to the construct being measured, ensuring the preservation of the original response format and interpretability.

Questionnaire Design Since only subsets of the questionnaires were used, the reliability is lower than in the full versions. However, the chosen elements cover the key parts of each construct, so the main validity is kept. In this way, the study remains realistic for the given time and participant limits, while still allowing us to see meaningful differences. Using subsets also keeps the results comparable to earlier studies, but with less burden on the participants.

8.1.3 Methodological Consistency and Comparison

Alignment with Previous Research

The measurement approach for Phase 2 was deliberately aligned with the methodology employed by Baumann et al. [3] in the evaluation of the same VR bicycle simulator. This consistency was implemented to allow for a direct comparison between the current enhanced system and the previous baseline performance, providing validation of the improvements implemented in this thesis work. Replication of measurement instruments ensures that the observed differences in participant responses can be assigned to system enhancements rather than methodological variations.

Comparison Limitations

Although the measurement approach remained consistent, a direct comparison is possible only for a subset of the data in the current study. Specifically, meaningful comparisons can be drawn between the previous study results and the current study's Groups 3 and 4 (the danger-to-danger meta-group), as these participants experienced the same scenario sequence (Danger 1 followed by Danger 2, or vice versa) used in the previous research.

Groups 1 and 2, who experienced safe-to-danger transitions, provide valuable insights into learning effects and scenario difficulty progression, but cannot be directly compared to the previous study's results due to the different scenario exposure pattern.

Scientific Value of Methodological Replication

The decision to replicate the measurement approach, despite some limitations in direct comparability, serves several scientific purposes. First, it enables validation of the enhanced simulator's performance against established benchmarks. Second, it contributes to the accumulation of comparable data within this specific research context, supporting meta-analytic approaches in future work. Finally, it demonstrates the reliability and applicability of the selected measurement instruments in VR bicycle simulation research.

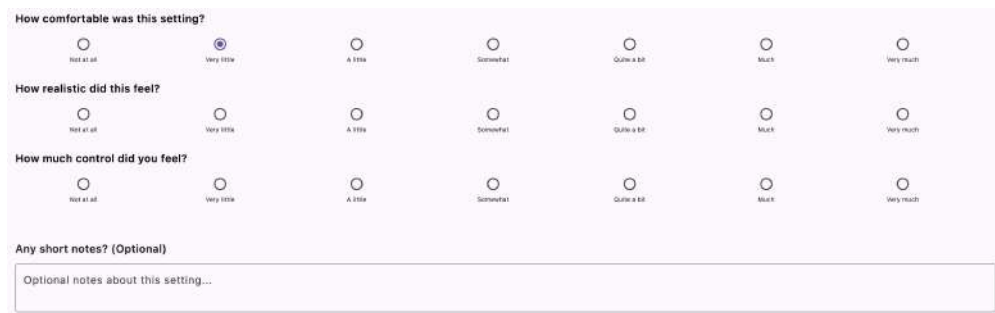
8.1.4 Data Collection Procedures

All questionnaires were conducted digitally through the study application, ensuring standardized presentation and automated data logging. The questionnaires were presented immediately after

each trial in Phase 1 and after each scenario in Phase 2. This timing was chosen to capture the participants' immediate impressions before memory decay or interference from subsequent tasks could occur.

Participants completed all questionnaires on their own. The researcher was available for technical support if needed, but avoided any influence on the content of the responses. To accommodate all participants, the application was bilingual: At the beginning of the session, participants selected German or English and then all questionnaires were presented in the chosen language.

A snapshot of the questionnaire interface is shown in Figure 8.1. The example illustrates the three-question page used at the end of each Phase 1 trial.



The screenshot displays a questionnaire interface with a light purple background. It contains three questions, each followed by a horizontal scale of seven radio buttons. The first question is "How comfortable was this setting?" with options: Not at all, Very little, A little, Somewhat, Quite a bit, Much, and Very much. The second question is "How realistic did this feel?" with the same options. The third question is "How much control did you feel?" with the same options. Below the questions is a section titled "Any short notes? (Optional)" with a text input field labeled "Optional notes about this setting...".

Figure 8.1: User interface of the questionnaire tool (example page from Phase 1 with three items).

8.2 Analysis Plan

8.2.1 Data Exploration and Validation

Before hypothesis testing, data validation was performed to ensure the integrity of the data set and verify the implementation of the experimental design. The analysis covered the following.

- **Dataset completeness:** Verification of participant completion rates, missing data assessment, and response scale boundary validation
- **Sample characterization:** Analysis of demographic distributions (age, gender, VR experience), physical characteristics (weight), and cycling experience patterns (general and urban frequency)
- **Design verification:**
 - Phase 1: Confirmation of balanced configuration testing across trial positions
 - Phase 2: Validation of group assignment and scenario sequence implementation
- **Data quality assessment:** Identification of response patterns, outlier detection, and assessment of response variability

These procedures will establish the foundation for statistical analysis and ensure that observed effects reflect genuine experimental differences rather than methodological issues.

8.2.2 Phase 1: Analysis Planning

Phase 1 analysis uses a systematic approach to evaluate tilt-steering configurations while accounting for within-subject dependencies and learning effects that occur in repeated measures designs. The analysis includes four key steps: data preprocessing, learning effect assessment, hypothesis testing with two analytical approaches, and effect size evaluation.

Data Preprocessing and Composite Score Calculation

The primary outcome measure is a composite score calculated as the arithmetic mean of three dimensions (comfort, realism, and control), each measured on 7-point Likert scales. The composite score formula is as follows:

$$\text{Composite Score} = \frac{\text{Comfort} + \text{Realism} + \text{Control}}{3} \quad (8.1)$$

Configuration data will be parsed into constituent components (stiffness level: N/L/M/H and steering model: B/VB/NL) to enable both configuration-specific and component-level analyses.

Learning Effects Assessment and Detrending

Since each participant completes seven trials in sequence, we need to check for and control learning effects. Within-subject designs are particularly prone to order effects and practice-related improvements [5]. We will use linear regression analysis across all data points, with trial position predicting composite scores, to measure learning trends. If we find significant overall trends ($p < 0.05$), we will apply a standard linear detrending procedure:

1. Fit a linear trend for each participant: $y_i = \beta_0 + \beta_1 \cdot \text{trial}_i + \epsilon_i$
2. Remove the trend component while keeping the mean: $y_{\text{detrended},i} = y_i - \hat{\beta}_1 \cdot \text{trial}_i$
3. Check that detrending worked by confirming no significant trends remain in the adjusted data

In these equations, y_i represents the composite score for trial i , β_0 is the intercept, β_1 is the learning slope, trial_i is the trial number (1-7), and ϵ_i is the error term. This approach follows established practices for controlling order effects in within-subject experimental designs [5].

Dual Analysis Strategy

All hypotheses will be tested using both raw and detrended data to provide comprehensive insight. This dual approach serves multiple purposes:

- **Raw data analysis:** Preserves ecological validity and represents actual user experience
- **Detrended analysis:** Isolates configuration effects from temporal adaptation, providing unbiased estimates of true differences
- **Convergent validation:** When both analyses yield consistent results, confidence in findings is strengthened
- **Divergent insights:** Discrepancies between analyses reveal the influence of learning effects on specific outcomes.

Together, these complementary perspectives provide a more in-depth understanding of the data by distinguishing configuration effects from temporal adaptation.

Statistical Testing Framework

Each hypothesis will be tested using an appropriate statistical method for its data structure and comparison type.

- **H₁:** Independent samples t-test comparing the best-performing enhanced configuration against baseline (N-B) using overall mean scores
- **H₂:** Paired samples t-test comparing each participant's personally chosen configuration against baseline, including only participants who made valid configuration choices
- **H₃:** Pearson correlation between body weight and preferred stiffness level (ranked as: N=0, L=1, M=2, H=3), using each participant's highest-rated configuration
- **H₄:** One-way ANOVA comparing all three steering models, followed by individual t-tests between baseline and any enhanced model showing potential differences.

All tests will use $\alpha = 0.05$ without correction for multiple comparisons, as justified in the power analysis (Section 7.2.3). Each hypothesis tests a distinct research question, making correction unnecessary.

Effect Size Evaluation and Power Thresholds

All results will be evaluated against predefined effect size thresholds established during power analysis (Section 7.2.3):

- Cohen's $d \geq 0.5$ for paired comparisons (medium-to-large effects)
- Pearson's $r \geq 0.6$ for correlations (strong relationships detectable with $n = 20$)
- Eta-squared $\eta^2 \geq 0.06$ for ANOVA effects (medium effects)
- Mean differences ≥ 0.5 points on 7-point scales (meaningful user-perceived changes)

These thresholds were chosen based on what can be reliably detected with our sample size and what represents significant improvements for the development of VR bicycle simulators. Effect sizes will be calculated using standard methods [6] and reported with confidence intervals to show both the size and reliability of any effects found.

8.2.3 Phase 2: Analysis Planning

Phase 2 analysis examines learning effects and system improvements through two hypotheses testing scenario-based training effectiveness (H_5) and comparison with the results of the previous study (H_6).

Group Structure and Meta-Group Definition

Phase 2 used four experimental groups (G1-G4) combined into two meta-groups for analysis:

- **Safe-Start** (G1, G2): Participants who experienced safe-to-danger scenario transitions ($n \approx 10$)
- **Danger-Start** (G3, G4): Participants who experienced danger-to-danger scenario transitions ($n \approx 10$)

Hypothesis 5: Danger Exposure Benefit

H₅: Prior exposure to danger scenarios improves performance in later danger scenarios compared to safe-to-danger transitions.

Dependent Measures (10 total):

- PASA: Primary Appraisal, Secondary Appraisal, Stress Index (3 measures)
- CSQ-VR: Total motion sickness score (1 measure)
- IPQ: General, Spatial, Involvement, Experienced Realism (4 measures)
- Single-item: Perceived realism, Perceived safety (2 measures)

Statistical Approach: Independent samples t-tests compare second-scenario performance between the Safe-Start and Danger-Start groups. This comparison between subjects isolates the effect of prior exposure to danger while controlling for scenario order. Effect sizes are calculated using Cohen's d with pooled standard deviation [6].

The pooled standard deviation combines the variability of both groups, where n_1 and n_2 are the sample sizes, and s_1^2 and s_2^2 are the variances for each group:

$$\sigma_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \quad (8.2)$$

Cohen's d measures the standardized difference between the group means, where $\bar{x}$ represents the mean score for each group:

$$d = \frac{\bar{x}_{\text{Danger-Start}} - \bar{x}_{\text{Safe-Start}}}{\sigma_{\text{pooled}}} \quad (8.3)$$

Improvement is defined directionally as:

- Lower values indicate improvement: stress indices, motion sickness
- Higher values indicate improvement: presence scores, perceived safety

Hypothesis 6: System Improvement Validation

H₆: The current enhanced VR bicycle simulator shows improved performance compared to the previous study [3].

Dependent Measures (7 total):

- PASA: Primary Appraisal, Secondary Appraisal, Stress Index (3 measures)
- IPQ: General, Spatial, Involvement, Experienced Realism (4 measures)

Statistical Approach: Two-sample independent t-tests compare current Danger-Start group means (averaged across both scenarios) against means from Baumann et al. ($n = 24$). Raw participant data from the previous study was obtained and used to calculate the averages per participant on both rides, allowing a proper statistical comparison that accounts for sampling differences in both studies [30].

The effect sizes are calculated using the same pooled standard deviation method described in 8.2.3, adapted to compare the participants in the current study with the participants in the previous study.

This approach provides degrees of freedom $df = n_{\text{current}} + n_{\text{previous}} - 2 = 10 + 24 - 2 = 32$, which gives adequate statistical power to detect meaningful differences between studies.

Effect Size Evaluation and Power Thresholds

Beyond significance testing, all Phase 2 results will be evaluated against predefined effect size thresholds established during power analysis (Section 7.2.3):

- Cohen's $d \geq 0.8$ for between-subjects comparisons (H_5) - large effects detectable with adequate power given balanced group sizes
- Cohen's $d \geq 0.5$ for two-sample comparisons (H_6) - medium-to-large effects reliably detectable with unbalanced samples ($n = 10$ vs. $n = 24$)

These thresholds were selected on the basis of statistical detectability with the available sample sizes and practical significance for the development of VR bicycle simulators. The effect sizes will be calculated using standard formulas [6] and reported to provide complete information about the magnitude of the effect.

Methodological Notes

- **Sample Considerations:** H_6 uses only danger start groups (G3, G4) to match the order of the scenarios in the previous study, avoiding confusion from the different sequences of scenarios.
- **Missing Data:** Participants with incomplete Phase 2 data are excluded separately for each measure by list deletion.
- **Comparative Framework:** The two-hypothesis approach allows both internal validation (training effects) and external validation (system improvements), providing complete evaluation of the enhanced simulator's effectiveness.

9 Results

9.1 Data Exploration and Validation

This section presents the data exploration and validation performed before hypothesis testing. The analysis confirmed the quality of the data set and validated the successful implementation of the experimental design.

9.1.1 Dataset Completeness and Quality

The data set includes complete data from all 20 participants who completed both study phases. No participants dropped out during the study and all sessions were completed successfully, resulting in a completion rate 100%. Data checks confirmed that all participants completed exactly seven trials in Phase 1 and two scenarios in Phase 2, with correct group assignments in all cases.

All responses remained within their expected ranges. Phase 1 ratings fell within the 1-7 Likert scale range, and Phase 2 responses met their respective scale requirements. The data set did not contain missing demographic information or incomplete questionnaire responses, demonstrating that the digital data collection worked as intended.

9.1.2 Sample Characteristics

Demographic Profile

The study sample ($n = 20$) showed various demographic characteristics while maintaining a good representation of key variables relevant to VR cycling simulation research. The age distribution was concentrated in the younger adult categories, with 50.0 % of the participants aged 25-34 years and 35.0 % aged 18-24 years, while the older age groups had fewer participants. The gender distribution showed that the majority of the participants were male (70.0 %, $n = 14$) compared to women (30.0 %, $n = 6$), reflecting common participation patterns in technology and VR research (Figure 9.1). The levels of VR experience were well distributed across categories: 35.0 % reported no VR familiarity, 35.0 % had less than 10 hours of experience and 30.0 % had 10 or more hours of VR exposure (Figure 9.2). This distribution provides enough variation to assess potential effects related to experience on simulator performance and adaptation.

Physical and Cycling Characteristics

The physical characteristics of the participants and their cycling experience showed diverse distributions suitable for assessing different user responses to the bicycle simulator configurations. The weight measurements showed a normal distribution with a mean of 80.2 kg ($SD = 13.8$ kg)

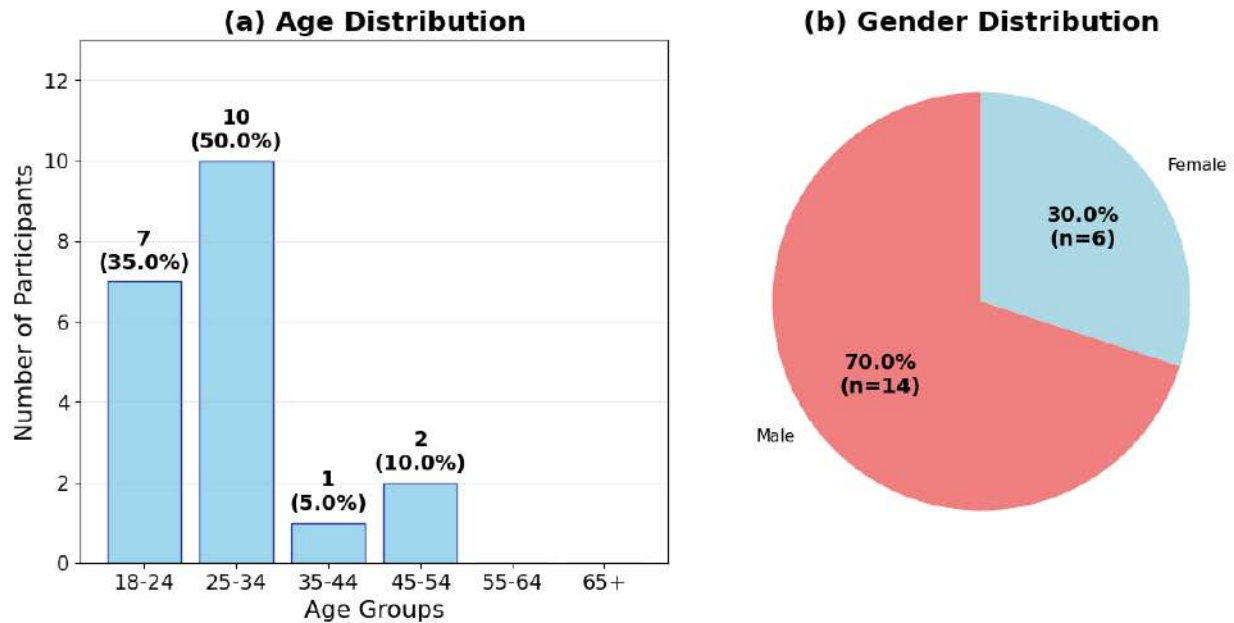


Figure 9.1: Demographic characteristics of study participants showing (a) age distribution and (b) gender distribution (n=20)

and a range of 55.0 to 107.0 kg (Figure 9.2). No statistical outliers were detected using the interquartile range method, indicating a representative sample without extreme values that might disproportionately influence weight-based configuration analyses.

The cycling experience analysis showed that 50.0 % of the participants cycled several times a week, 20.0 % cycled approximately once a week and 30.0 % cycled less than once a week. Urban cycling patterns showed different characteristics: 35.0 % cycled several times a week in urban environments, 25.0 % cycled approximately once a week, and 40.0 % cycled less than once a week in urban settings. Cross-tabulation analysis indicated that participants who cycle frequently in general tend to also cycle frequently in urban environments, although some frequent general cyclists reported lower urban cycling frequencies (Figure 9.3).

9.1.3 Experimental Design Verification

Phase 1 Configuration Balance

The analysis of Phase 1 configuration tests confirmed the correct implementation of the balanced experimental design outlined in Section 7.4.2. The 10 configurations were tested exactly 14 times across the 20 participants, showing perfect frequency balance. The balance matrix between configurations and positions showed a clean distribution between test positions, with each configuration appearing 1-3 times in each of the seven test positions, precisely matching the planned distribution shown in Table 7.2. Balance statistics showed an average standard deviation of 0.29 across positions and an average range of 1.00, confirming that configurations were evenly distributed across trial positions and effectively controlling for order effects.

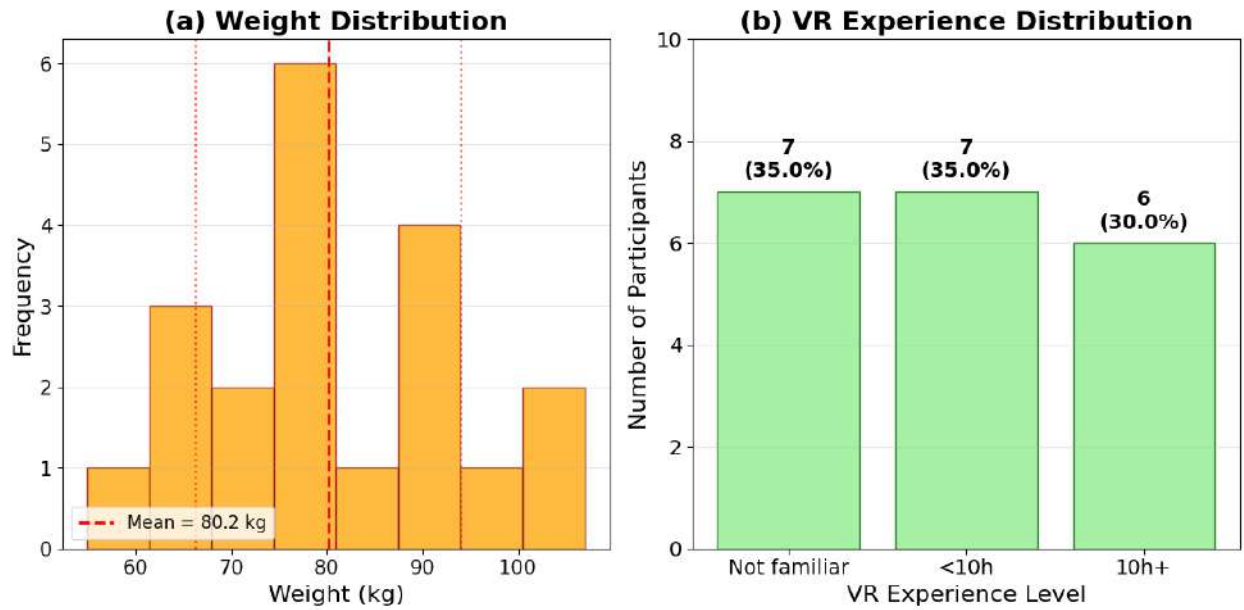


Figure 9.2: Physical characteristics and VR experience showing (a) weight distribution and (b) VR experience levels (n=20)

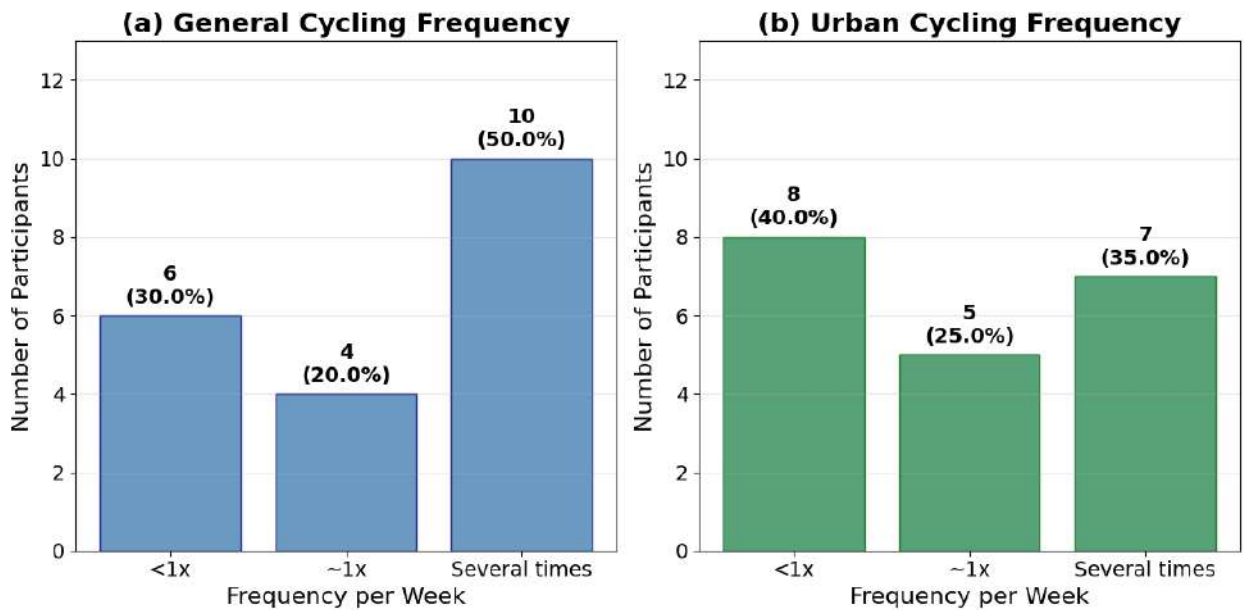


Figure 9.3: Cycling experience characteristics showing (a) general cycling frequency and (b) urban cycling frequency (n=20)

Phase 2 Group Assignment

The assignment of phase 2 groups confirmed the correct implementation of the planned experimental design described in Section 7.4.3. With the achieved sample size of $n = 20$, the group distribution resulted in perfectly balanced assignments as specified in Table 7.3, with each of the four groups (G1, G2, G3, G4) containing exactly 5 participants. All scenario sequences were cor-

rectly implemented according to the experimental protocol, ensuring equal distribution between safe-start participants (G1 + G2, $n = 10$) and danger-start participants (G3 + G4, $n = 10$). This balanced assignment enables robust comparisons between different exposure patterns.

9.1.4 Data Quality Assessment

The response quality analysis identified minor variations in two participants while confirming the overall integrity of the data. A participant (participant 9) showed low response variability (standard deviation = 0.44) and high extreme response use (90.5 %), indicating potential concerns about the response pattern. However, this represents only 5.0 % of the sample, which falls within acceptable limits for behavioral research and does not warrant data exclusion.

The remaining 19 participants showed normal response variability (mean standard deviation = 1.05, range: 0.44-1.53), indicating appropriate use of rating scales and meaningful differentiation between configurations and scenarios. No systematic response biases or data quality issues that would compromise the validity of subsequent analyzes were identified.

9.1.5 Summary

The data exploration and validation procedures confirmed the correct implementation of the experimental design and the collection of high-quality data suitable for hypothesis testing. Key findings include:

- Complete dataset with 100 % participant completion rate ($n = 20$)
- Balanced demographic representation appropriate for VR cycling research
- Correct experimental design implementation with balanced configuration testing and group assignment as planned
- Strong data quality with only minor response pattern variations in 5.0 % of participants
- No missing data or systematic biases that would compromise analytical validity

These results establish a solid foundation for proceeding with the planned Phase 1 configuration analysis and the Phase 2 scenario evaluation.

9.2 Phase 1: Tilt-Steering Dynamics Evaluation

This section presents the results of the Phase 1 evaluation, which examined the effectiveness of different tilt-steering configurations through systematic hypothesis testing. All analyzes were performed using raw and trended data to account for learning effects identified during the preliminary assessment. The results are evaluated against the predefined power thresholds established in Section [7.2.3](#).

9.2.1 Learning Effects Assessment

Before hypothesis testing, an assessment of potential learning or adaptation effects was performed in all trial positions. Linear regression analysis showed significant positive trends for realism ($R^2 = 0.860$, $p = 0.003$) and composite scores ($R^2 = 0.686$, $p = 0.021$), while comfort ($R^2 = 0.493$, $p = 0.079$) and control ($R^2 = 0.395$, $p = 0.131$) showed non-significant trends (Figure 9.4). The significant learning effect for composite scores (slope = 0.097 points per trial) required the creation of detrended data sets for all subsequent analyzes. The detrending procedure successfully removed temporal trends ($p = 1.000$ after detrending), allowing unbiased assessment of configuration effects independent of the position of the test.

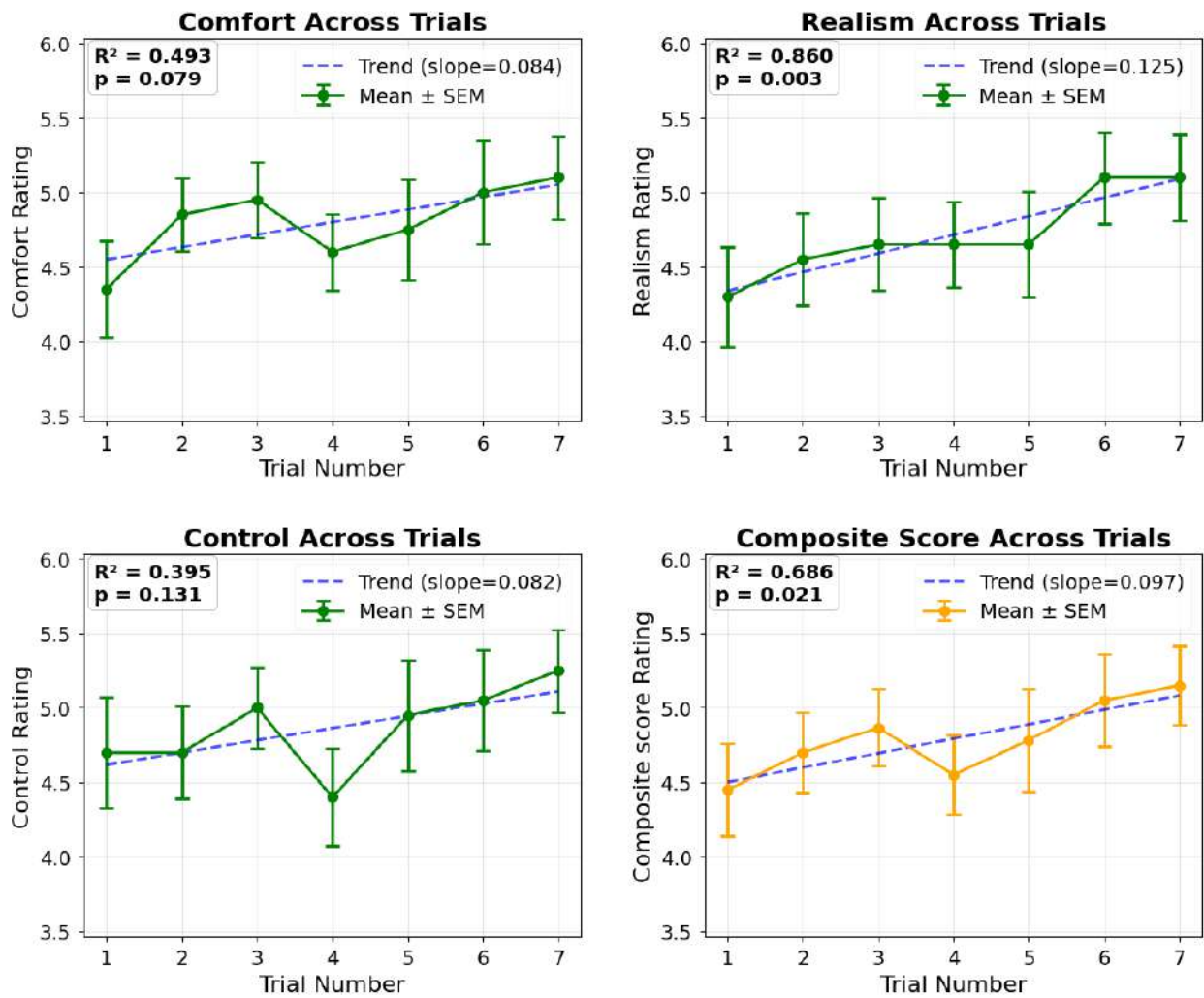


Figure 9.4: Learning effects across trial positions for comfort, realism, control, and composite scores. Error bars represent standard error of the mean. Dashed lines indicate linear trends with corresponding R^2 and p-values shown for each dimension.

9.2.2 Hypothesis 1: Enhanced Configuration Performance

H₁: The best performing enhanced tilt-steering configuration outperforms the baseline configuration (N-B) in composite user experience scores.

The raw data analysis identified M-VB as the best performing configuration with a mean composite score of 5.26 (SD = 0.91), compared to the mean of the baseline N-B configuration of 4.43 (SD = 1.49). The difference of 0.83 points did not reach statistical significance ($t_{(26)} = 1.620$, $p = 0.117$, Cohen's $d = 0.61$). The detrend analysis showed similar results, with L-VB emerging as the best configuration (M = 5.21, SD = 0.86) versus the baseline (M = 4.45, SD = 1.40), resulting in a non-significant difference of 0.76 points ($t_{(26)} = 1.593$, $p = 0.123$, Cohen's $d = 0.60$).

The observed medium effect sizes (raw: $d = 0.61$; detrended: $d = 0.60$) approach the minimum detectable effect size specified in the power analysis ($d \geq 0.5$, Section 7.2.3). However, the non-significant results prevent drawing definitive conclusions. The mean differences (0.83 and 0.76 points) exceed the predefined threshold of 0.5 points for meaningful perceptual changes, but did not reach statistical significance. These inconclusive results could indicate either: (a) a modest real advantage for enhanced configurations that would require a larger sample to detect reliably, or (b) substantial individual variation in preferences, suggesting no single universally optimal configuration exists. Further research with larger sample sizes is needed to distinguish between these possibilities.

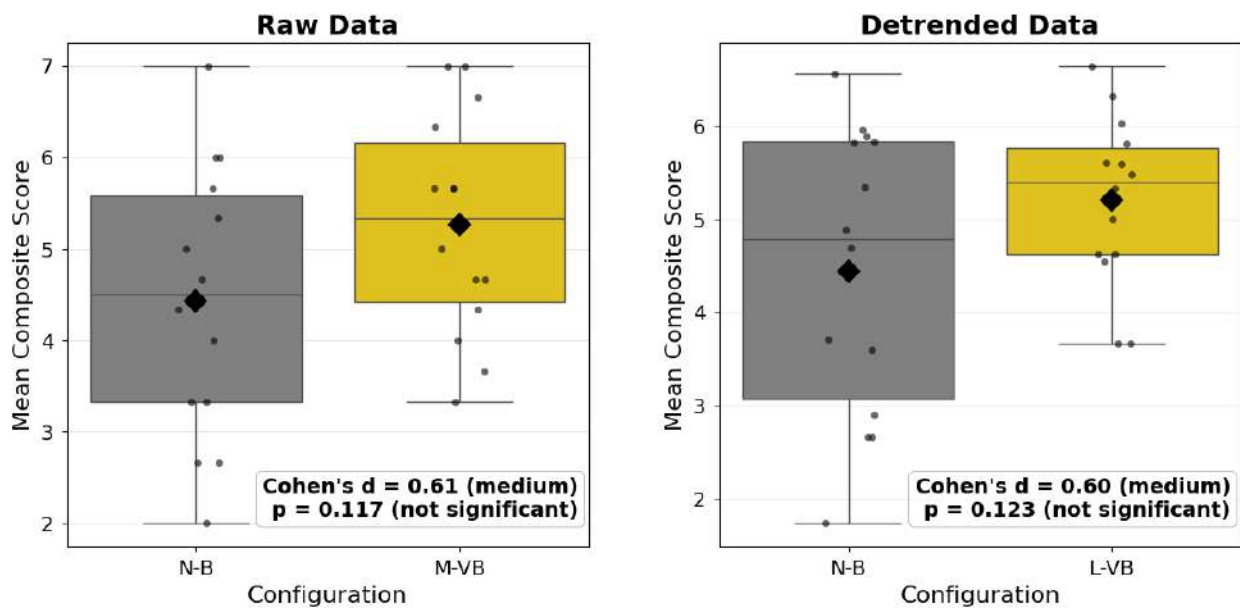


Figure 9.5: Comparison of best performing enhanced configurations versus baseline (N-B) for raw and detrended data. Box plots show median and quartiles with individual data points overlaid. Statistical annotations show Cohen's d effect sizes and p -values.

Result: H₁ is not supported. The null hypothesis ($H_{1,0}$) cannot be rejected. No enhanced configuration showed statistically significant improvements over baseline, despite medium effect sizes and mean differences that exceeded the predefined threshold of 0.5 points. Further investigation

with larger sample sizes is needed.

9.2.3 Hypothesis 2: Personalized Configuration Performance

H₂: Individually selected personalized configurations outperform the baseline configuration.

Within-subject comparisons were made for the 14 participants who provided valid personalized configuration choices. In raw data analysis, personalized configurations ($M = 6.24$, $SD = 0.95$) significantly outperformed baseline ($M = 4.43$, $SD = 1.49$), with a mean difference of 1.81 points ($t_{(13)} = 5.437$, $p = 0.0001$, Cohen's $d = 1.51$). The detrend analysis confirmed these findings with personalized configurations ($M = 5.85$, $SD = 1.10$) that exceeded the baseline ($M = 4.45$, $SD = 1.40$) by 1.40 points ($t_{(13)} = 3.777$, $p = 0.0023$, Cohen $d = 1.05$).

Both analyzes exceed all predefined thresholds: effect sizes surpass $d \geq 0.5$ (raw: $d = 1.51$; detrended: $d = 1.05$), and mean differences exceed 0.5 points (raw: 1.81; detrended: 1.40 points). The reduction from $n = 20$ to $n = 14$ participants with valid personalized choices was anticipated in the study design, as the trial optimization procedure (Section 7.4.2) did not require all participants to make a final configuration selection. Despite this reduced sample, the observed effect sizes substantially exceeded the minimum detectable threshold and the highly significant results ($p < 0.01$) demonstrate robust evidence for the personalization advantage.

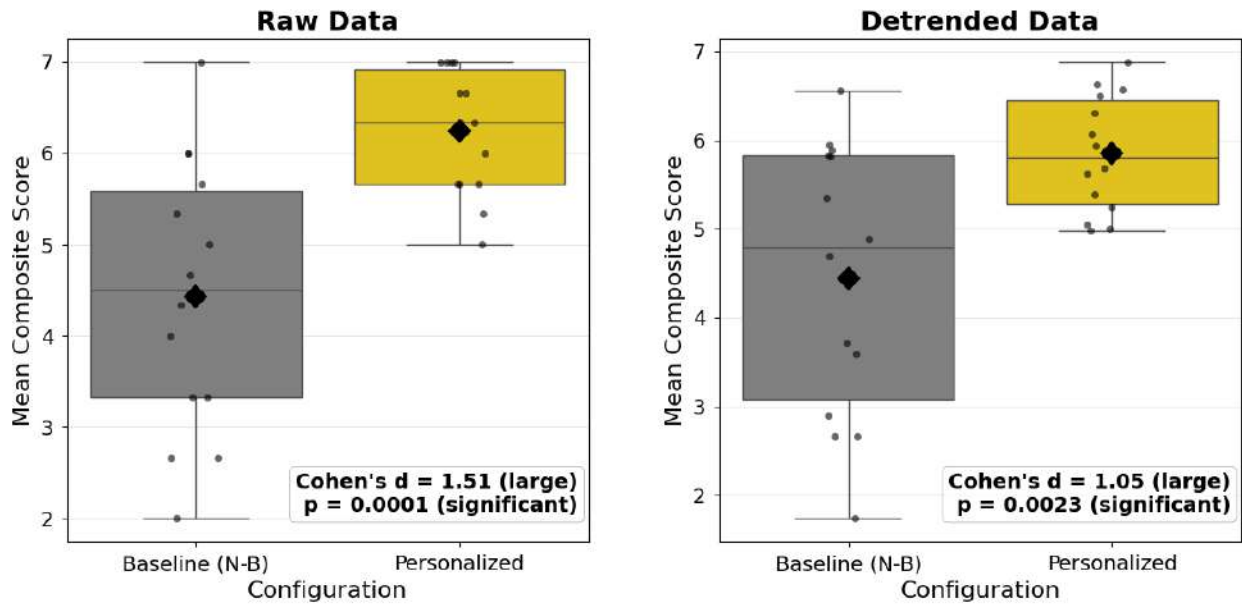


Figure 9.6: Within-subject comparison of personalized configurations versus baseline (N-B) showing paired data for 14 participants. Both raw and detrended analyses show significant improvements with large effect sizes.

Result: H₂ is strongly supported. The alternative hypothesis ($H_{2,1}$) is accepted. Personalized configurations significantly outperform baseline with large effect sizes in both analyses, substantially exceeding all predefined power thresholds.

9.2.4 Hypothesis 3: Weight-Configuration Correlation

H₃: Participants' body weight correlates with their preferred stiffness settings.

Note: The analysis used the best-rated configuration of the participants (highest composite score) to determine the stiffness preference, with stiffness levels coded as None (N) = 0, Low (L) = 1, Medium (M) = 2, High (H) = 3.

The correlation analysis with $n = 20$ participants revealed a strong positive relationship between body weight and preferred stiffness level in both raw ($r = 0.630$, $p = 0.003$) and detrended ($r = 0.598$, $p = 0.005$) analyses. Compared to the predefined detection threshold of $r \geq 0.6$ specified in the power analysis (Section 7.2.3), the correlation of the raw data meets this criterion while the correlation of the detrended declines slightly below it. Both correlations achieve strong statistical significance ($p < 0.01$).

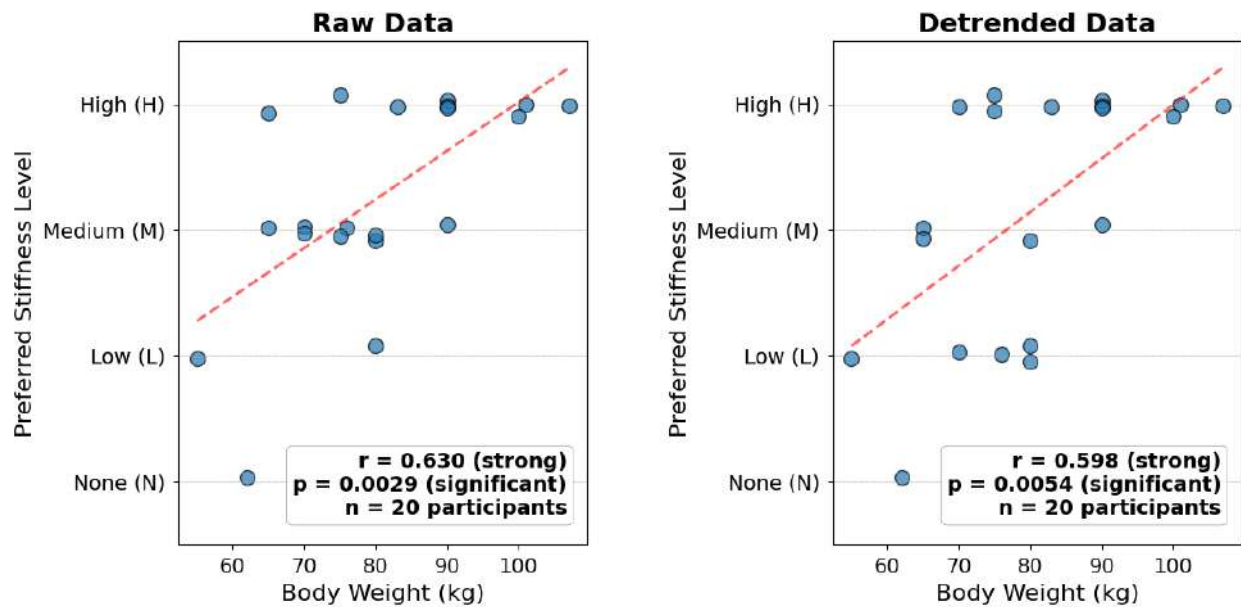


Figure 9.7: Relationship between body weight and stiffness preference based on best-rated configurations. Scatter plots show individual participant choices with linear trend lines.

Result: H₃ is supported based on raw data analysis. The alternative hypothesis ($H_{3,1}$) is accepted based on the raw data, which shows a correlation that meets the predefined power threshold ($r = 0.630 \geq 0.6$). This provides empirical support for weight-based configuration recommendations. The slight reduction in the detrended correlation ($r = 0.598$) suggests a minimal influence of the learning effects on this relationship.

9.2.5 Hypothesis 4: Steering Model Main Effect

H₄: Enhanced steering models (velocity-based and non-linear) outperform the basic steering model.

One-way ANOVA comparing steering models revealed different patterns between raw and detrended analyzes. The raw data showed marginally non-significant differences ($F_{(2,137)} = 2.72$, $p = 0.0696$, $\eta^2 = 0.04$), while the detrended data reached statistical significance ($F_{(2,137)} = 3.25$, $p = 0.0417$, $\eta^2 = 0.05$). However, both effect sizes fell below the predefined medium effect threshold ($\eta^2 \geq 0.06$, Section 7.2.3), indicating only small overall effects among the three steering models.

Performance rankings remained consistent across analyses: velocity-based (VB) performed best (raw: $M = 5.09$; detrended: $M = 5.11$), followed by non-linear (NL; raw: $M = 4.89$; detrended: $M = 4.86$), and basic (B; raw: $M = 4.50$; detrended: $M = 4.51$).

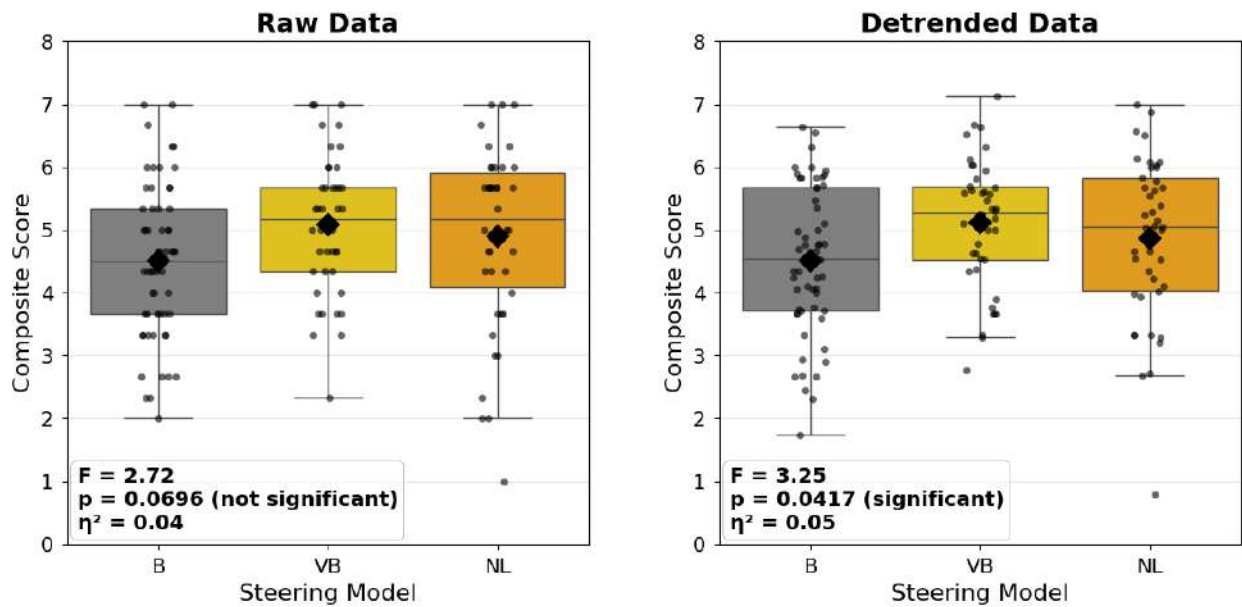


Figure 9.8: Comparison of steering models (Basic, Velocity-Based, and Nonlinear) showing overall performance differences.

Post-hoc comparisons between the best performing model (VB) and baseline (B) showed significant improvements in both raw ($t_{(96)} = 2.419$, $p = 0.0174$, Cohen's $d = 0.49$) and detrended ($t_{(96)} = 2.643$, $p = 0.0096$, Cohen's $d = 0.54$) analyzes. The velocity-based model showed a consistent advantage of approximately 0.60 points over the basic model, exceeding the 0.5-point threshold for meaningful perceptual differences. Only the detrended analysis ($d = 0.54$) meets the predefined Cohen's d threshold ($d \geq 0.5$), while the raw data effect ($d = 0.49$) falls slightly below it.

Overall ANOVA effect sizes (raw: $\eta^2 = 0.038$; detrended: $\eta^2 = 0.045$) fell below the predefined medium effect threshold ($\eta^2 \geq 0.06$), indicating small differences among all three steering models. However, post-hoc comparison revealed that the velocity-based model specifically outperformed

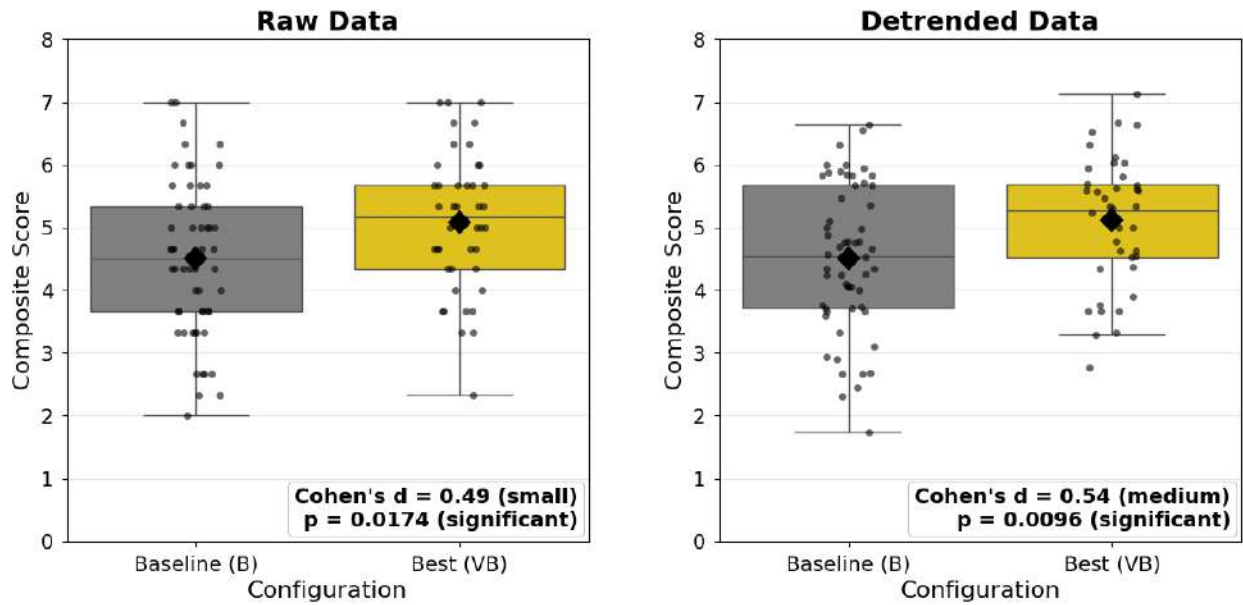


Figure 9.9: Post-hoc comparison of velocity-based (VB) versus basic (B) steering models. Significant improvements are observed in both analyses, with the detrended effect size meeting the predefined threshold.

the basic model with an effect size meeting the predefined threshold in detrended analysis ($d = 0.54 \geq 0.5$), while raw data narrowly missed this criterion ($d = 0.49$).

Result: H_4 is partially supported based on the detrended analysis. The velocity-based model shows significant improvements over the basic model with effect sizes that meet the predefined threshold in the detrended data, highlighting both the importance of controlling for learning effects and the specific advantage of velocity-based steering.

9.2.6 Qualitative Configuration Feedback

The comments of the participants provided additional context for the interpretation of quantitative configuration ratings and revealed implementation-specific factors that affect the user experience. Technical performance issues were consistently reported among multiple participants, with frame rate instabilities observed during trials with enhanced steering models (P3, P9). One participant specifically questioned whether "der extra Berechnungsaufwand für die Lenkung bei NL und VB Settings mit rein[spielt]" (the extra computational load for steering in NL and VB settings plays a role, P9), suggesting that processing demands may have influenced the perceived quality of advanced configurations.

Wind simulation, a component of the velocity-based model, received consistently positive feedback. The participants described the wind effects as "great" (P8) and noted improved realism: "Fahrtwind sehr gut. Realistischer aber auch angenehmer" (headwind very good, more realistic and also more pleasant, P17). This qualitative validation supports the quantitative superiority of the velocity-based configurations observed in the trend analysis.

Steering sensitivity emerged as a cross-cutting concern independent of specific configurations. Multiple participants noted challenges with steering responsiveness, with one stating "Die Lenkung ist in allen Einstellungen ein wenig zu sensibel" (steering is a bit too sensitive in all settings, P6). High stiffness configurations occasionally produced disorienting effects, with one participant describing the experience as feeling "betrunken" (drunk, P19), suggesting that excessive resistance can affect rather than improve control.

9.3 Phase 2: Scenario-Based Training Effects

This section presents the results of Phase 2, which examined the effectiveness of the VR bicycle simulator for training applications using scenario-based exposure patterns. The analysis evaluated two key research questions: whether exposure to the prior danger scenario provides learning advantages (H_5), and if the enhanced simulator shows improvements compared to the previous study by Baumann et al. [3] (H_6). All analyzes used the balanced group assignments verified in Section 9.1.3 and were evaluated against the predefined power thresholds established in Section 7.2.3.

9.3.1 Hypothesis 5: Danger Exposure Benefit

H_5 : Prior danger scenario exposure enhances performance in subsequent danger scenarios compared to safe-to-danger transitions.

To test this hypothesis, the performance of the second scenario was compared between the Safe-Start (safe-to-danger, $n = 10$) and Danger-Start (danger-to-danger, $n = 10$) groups on ten dependent measures. Independent samples t-tests were performed to assess group differences, with results evaluated against the predefined large effect size threshold ($d \geq 0.8$) specified for comparisons between groups in Section 7.2.3.

Motion Sickness and Stress Outcomes

Motion sickness analysis revealed a medium effect size that favors the Danger-Start group (CSQ-VR: $M = 1.67$, $SD = 0.90$) compared to Safe-Start participants ($M = 2.08$, $SD = 0.34$), with Cohen's $d = -0.61$. Although this represents a significant reduction in symptoms of cybersickness, the effect did not meet the predefined threshold of $d \geq 0.8$ and did not achieve statistical significance ($t_{(18)} = -1.368$, $p = 0.188$).

The PASA Stress Index showed a similar pattern, with Danger-Start participants showing lower stress levels ($M = -2.20$, $SD = 1.57$) compared to Safe-Start participants ($M = -1.00$, $SD = 2.03$), resulting in a medium effect size (Cohen's $d = -0.662$). However, this difference neither reached the large effect threshold nor achieved statistical significance ($t_{(18)} = -1.481$, $p = 0.156$). These results suggest potential benefits of exposure to danger for stress management and the reduction of motion sickness, although the evidence is inconclusive and requires further investigation with larger sample sizes. Figure 9.10 illustrates the distribution of both measures between exposure

groups, showing consistent medium-sized trends in the expected direction without achieving the stringent criteria for reliable detection.

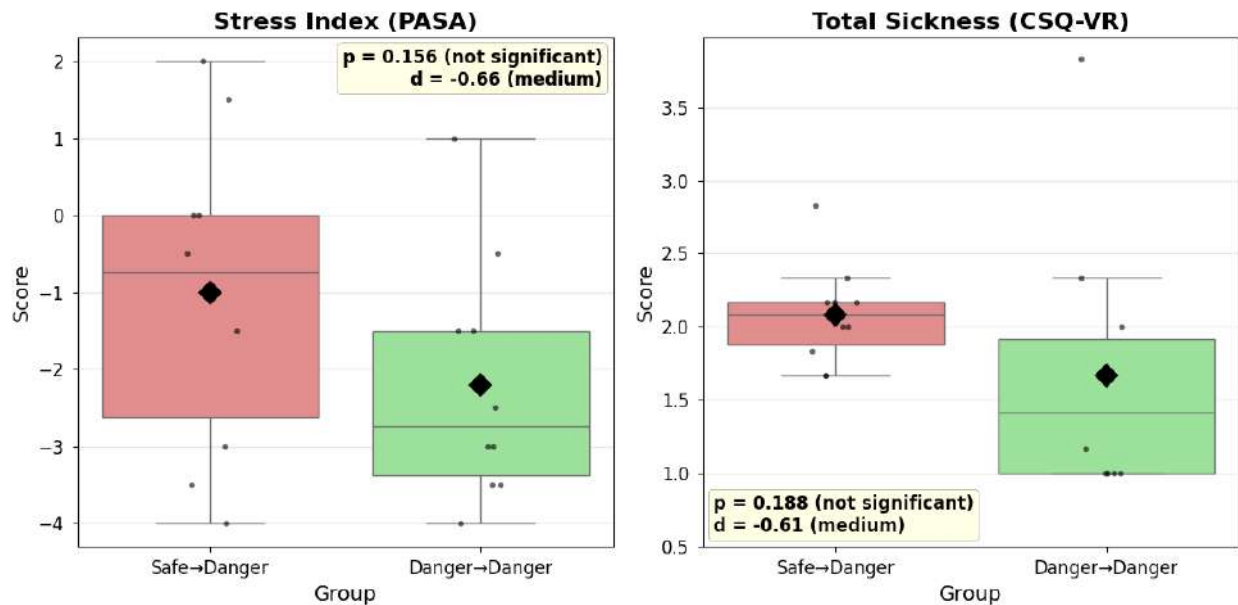


Figure 9.10: Comparison of PASA Stress Index and Total Sickness (CSQ-VR) between Safe-Start and Danger-Start groups in second scenario. Both measures show medium effect sizes in the expected direction but do not reach the predefined threshold or statistical significance.

PASA Subscale Analysis

Examination of the PASA subscales revealed that the Primary Assessment demonstrated a medium effect size (Cohen $d = -0.67$) and Danger-Start participants reported lower perceptions of threat and challenge ($M = 2.60$, $SD = 1.22$) compared to Safe-Start participants ($M = 3.50$, $SD = 1.47$). However, this effect fell below the $d \geq 0.8$ threshold and did not achieve significance ($t_{(18)} = -1.489$, $p = 0.154$).

Secondary Appraisal showed a small effect size (Cohen's $d = 0.382$) with Danger-Start participants demonstrating marginally higher coping resources ($M = 4.80$, $SD = 0.71$) versus Safe-Start participants ($M = 4.50$, $SD = 0.85$), although this difference was neither practically meaningful nor statistically significant ($t_{(18)} = 0.854$, $p = 0.404$).

All three PASA-related measures (Stress Index, Primary Appraisal, and Secondary Appraisal) pointed in the theoretically expected direction, suggesting potential adaptation benefits from exposure to danger. However, none met the stringent criteria for reliable detection in this study, indicating the need for further investigation with greater statistical power. As shown in Figure 9.11, both subscales show the expected directional patterns but with insufficient magnitude to meet the threshold requirements.

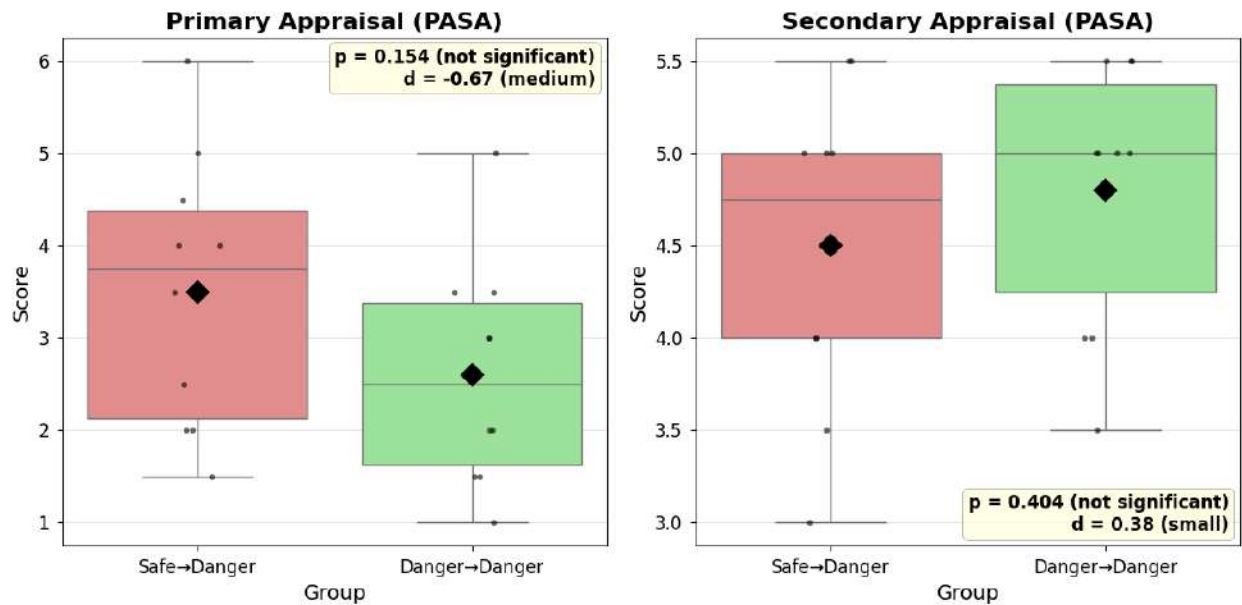


Figure 9.11: PASA subscales comparison showing Primary Appraisal (left) and Secondary Appraisal (right) between exposure groups. Both subscales show effects in the expected direction but remain below threshold values.

Presence and Immersion Outcomes

The four IPQ presence subscales showed mixed and inconsistent patterns between the groups. General Presence (IPQ-G1) showed a small negative effect (Cohen's $d = -0.38$), with Safe-Start participants reporting slightly higher scores ($M = 3.90$, $SD = 0.88$) than Danger-Start participants ($M = 3.60$, $SD = 0.70$), although this difference was not significant ($p = 0.408$).

Spatial Presence (IPQ-SP4) showed a medium effect size (Cohen's $d = -0.70$) with Safe-Start participants reporting higher spatial presence ($M = 4.10$, $SD = 0.57$) compared to Danger-Start participants ($M = 3.60$, $SD = 0.84$). While approaching the threshold, this effect did not reach significance ($p = 0.137$).

Involvement (IPQ-INV1) showed a medium positive effect (Cohen's $d = 0.57$) favoring Danger-Start participants ($M = 3.20$, $SD = 0.79$) over Safe-Start participants ($M = 2.70$, $SD = 0.95$), though significance was not reached ($p = 0.216$). Experienced Realism (IPQ-REAL3) showed a small negative effect (Cohen's $d = -0.38$, $p = 0.408$) that also did not reach significance.

None of the IPQ subscales achieved statistical significance or met the predefined effect size threshold, indicating inconsistent and unreliable presence-related effects across exposure conditions. Figure 9.12 displays the four presence dimensions, illustrating the mixed pattern of effects with no clear advantage for either exposure condition.

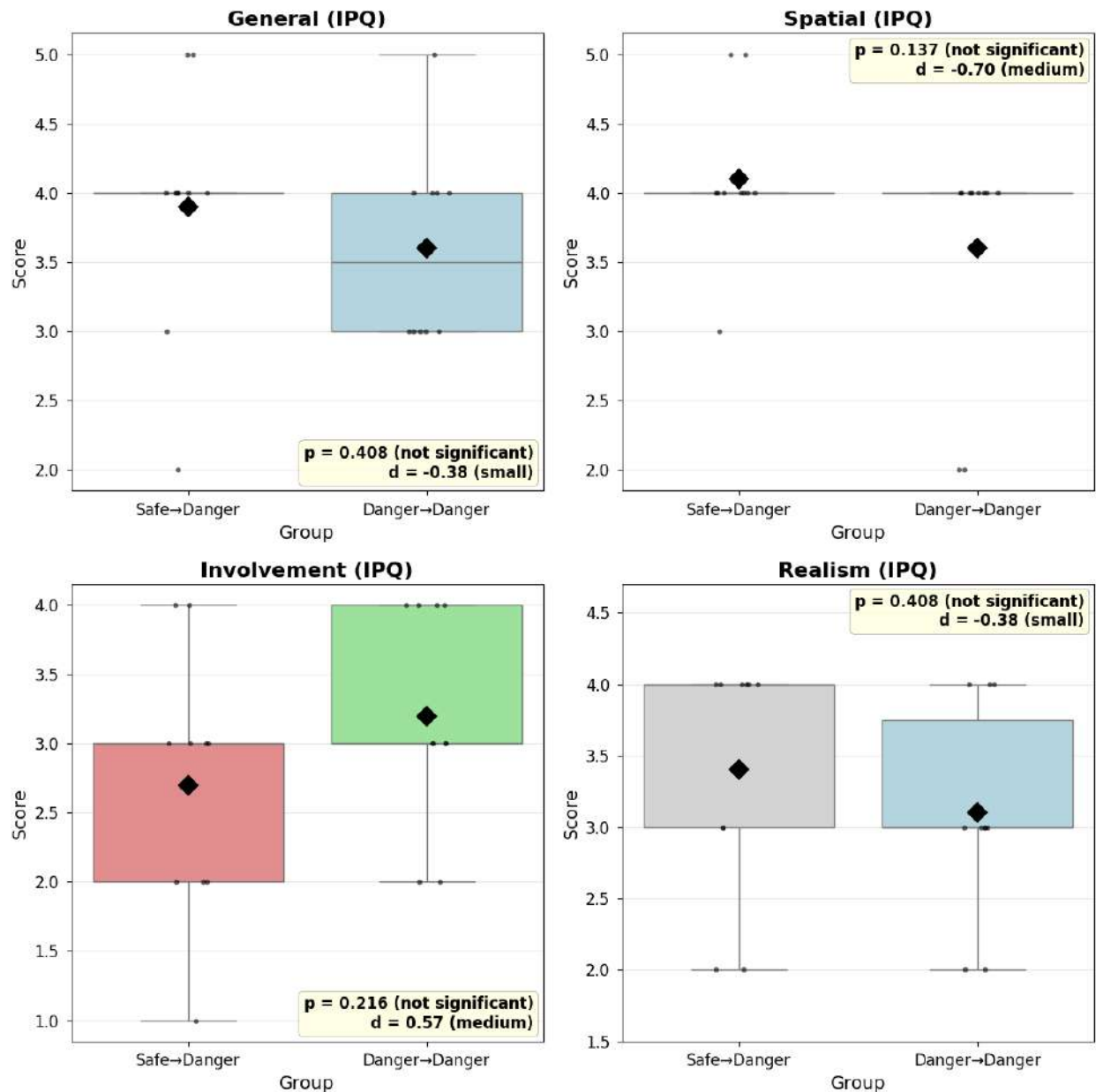


Figure 9.12: IPQ presence subscales showing General (top left), Spatial (top right), Involvement (bottom left), and Experienced Realism (bottom right). Mixed results with no significant differences or threshold-meeting effects.

Perceived Realism and Safety

Analysis of the two custom single item measures revealed different results. Perceived environmental realism showed only a small effect size (Cohen's $d = -0.26$) with substantial variability ($p = 0.572$), indicating an unreliable measurement. This finding aligned with the similarly weak effect observed in the IPQ Experienced Realism subscale, suggesting consistent but negligible perception differences of realism.

In contrast, perceived safety emerged as the only measure that showed both statistical signifi-

cance and a large effect size meeting the predefined threshold. Danger-Start participants reported substantially higher safety perceptions ($M = 5.20$, $SD = 1.32$) compared to Safe-Start participants ($M = 3.80$, $SD = 1.62$), with a difference of 1.40 points ($t_{(18)} = 2.121$, $p = 0.048$, Cohen's $d = 0.95$). This large effect ($d > 0.8$) indicates that prior exposure to danger significantly improved the subjective sense of safety of participants when faced with subsequent challenging scenarios. Figure 9.13 contrasts the two custom measures, clearly showing the substantial safety benefit for the Danger-Start group, while the perceptions of realism remained essentially unchanged.

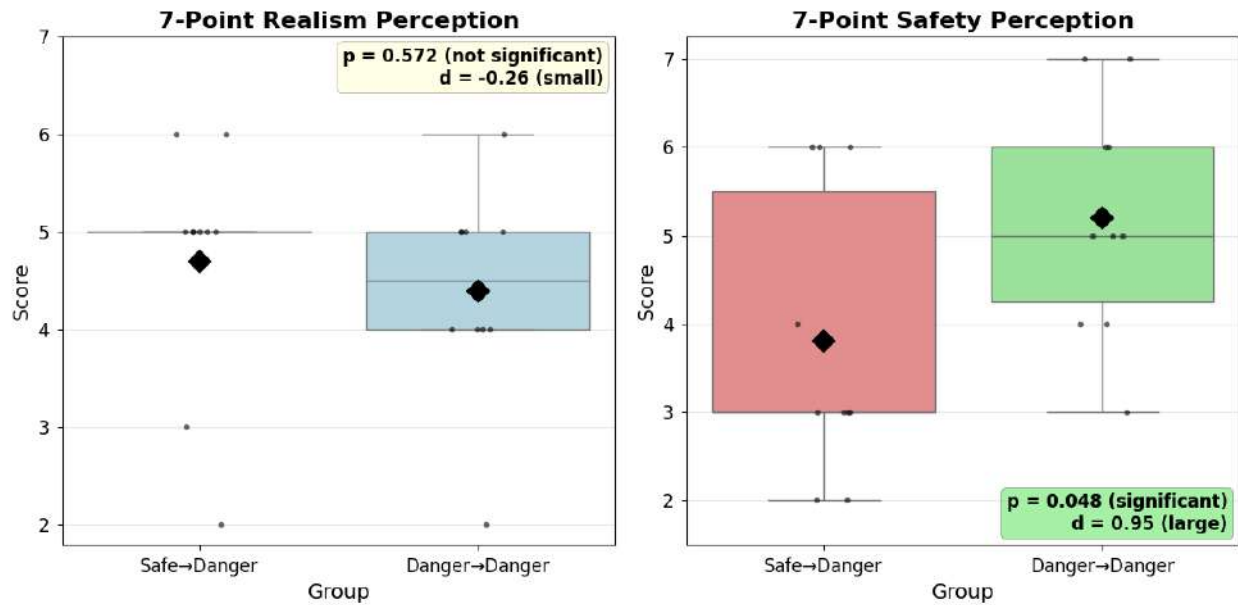


Figure 9.13: Custom single-item measures comparing 7-point environmental realism (left) and perceived safety (right). Only perceived safety showed significant improvement with large effect size for Danger-Start group.

Hypothesis 5 Conclusion

The medium effect sizes for the stress and cybersickness measures suggest potential benefits for the Danger-Start group, although these effects did not reach statistical significance and fell below the predefined large effect size threshold ($d \geq 0.8$). The presence-related measures showed mixed and inconsistent effects, none of which reached statistical significance or the predefined effect size threshold. Perceived safety was the only outcome that showed a statistically significant improvement for the Danger-Start group ($p = 0.048$, $d = 0.95$, large effect).

Result: H_5 is not supported. Although perceived safety demonstrated a clear large effect benefiting the Danger-Start group, most measures (9 of 10) failed to show reliable improvements. The hypothesis predicted significant benefits across multiple domains, but only safety perception met the strict criteria. The promising trends in stress and sickness measures warrant further investigation with larger sample sizes to clarify these potential effects.

9.3.2 Hypothesis 6: System Improvement Validation

H₆: The current enhanced VR bicycle simulator demonstrates improved performance compared to the previous study by Baumann et al. [3].

This hypothesis was evaluated by comparing the results of the current study of the Danger-Start group (Groups G3 + G4, $n = 10$) with participant-level data from the previous study ($n = 24$). The averages per participant across both scenarios were calculated for both studies to provide stable comparison metrics. Independent two-sample t -tests were performed in seven dependent measures, with Cohen's d calculated using pooled standard deviations to properly account for variability in both samples.

PASA Stress Outcomes

Analysis of the PASA Stress Index revealed significant improvement in the enhanced system. The current participants showed substantially lower stress levels ($M = -2.05$, $SD = 1.38$) compared to the previous study ($M = -0.59$, $SD = 1.63$), with a mean difference of -1.46 points. This reduction achieved statistical significance ($t_{(32)} = -2.471$, $p = 0.019$) with a large effect size (Cohen's $d = -0.93$), indicating meaningful improvement in stress management capabilities of the enhanced simulator. Figure 9.14 displays the comparison between the studies, illustrating the substantial reduction in stress index scores achieved with the enhanced tilt-steering system.

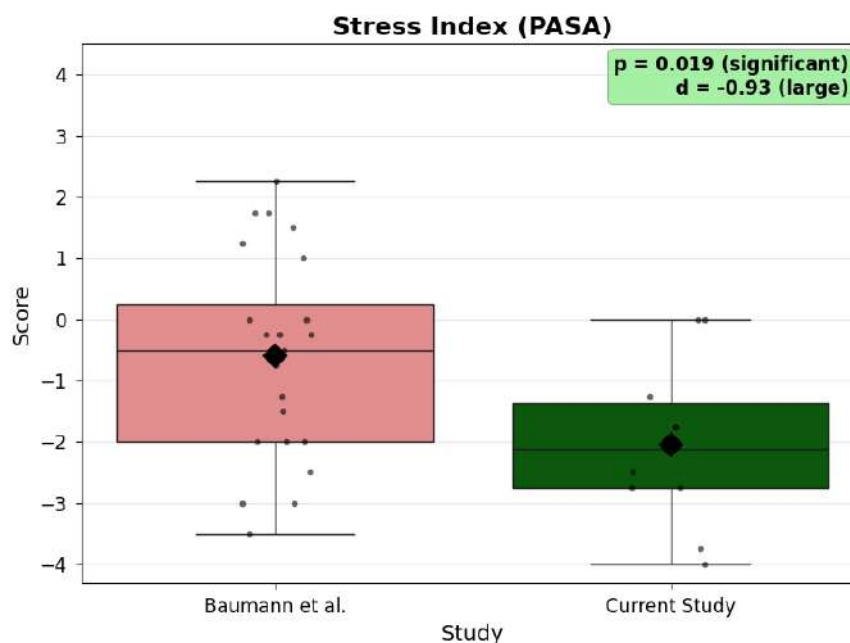


Figure 9.14: PASA Stress Index comparison between current enhanced system and Baumann et al. [3]. Significant reduction in stress shows improved system performance with large effect size.

Examination of the PASA subscales revealed that stress reduction was primarily driven by substantial changes in the Secondary Appraisal rather than the Primary Appraisal. Primary Appraisal

showed virtually no difference between studies (current: $M = 2.92$, $SD = 1.07$; previous: $M = 2.94$, $SD = 1.36$), with a negligible effect size (Cohen's $d = -0.01$, $p = 0.980$). This indicates that the participants perceived similar levels of threat and challenge in both versions of the simulator.

In contrast, Secondary Appraisal showed substantial improvement, and current participants showed markedly higher coping resources ($M = 4.97$, $SD = 0.51$) compared to the previous study ($M = 3.53$, $SD = 0.64$). This produced a very large effect size (Cohen's $d = 2.37$) with strong statistical significance ($t_{(32)} = 6.305$, $p < 0.001$), indicating that the enhanced simulator substantially improved participants' self-perceived abilities to manage challenging scenarios. As shown in Figure 9.15, the large contrast between the negligible Primary Appraisal effect and the very large Secondary Appraisal improvement shows that stress reduction was achieved through enhanced coping resources rather than reduced threat perception.

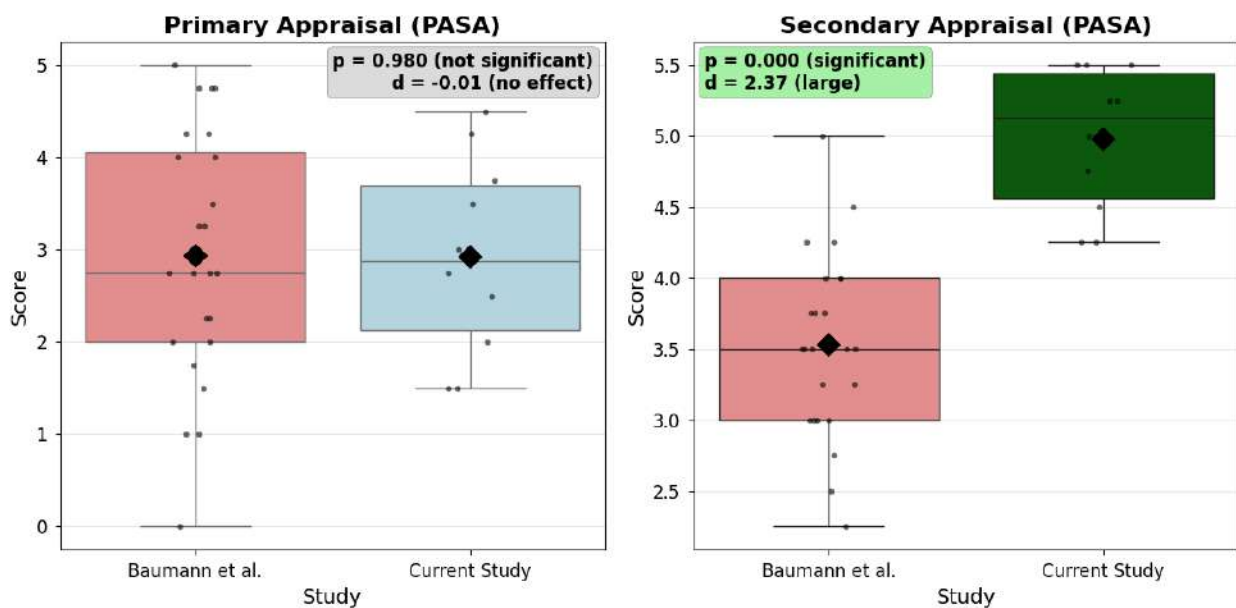


Figure 9.15: PASA subscales comparison showing Primary Appraisal (left) with no effect and Secondary Appraisal (right) with very large significant improvement. The stress index reduction is primarily due to enhanced secondary appraisal.

IPQ Presence Outcomes

Analysis of the four IPQ presence dimensions revealed modest and non-significant differences between studies. General Presence (IPQ-G1) showed no meaningful difference (current: $M = 3.65$, $SD = 0.53$; previous: $M = 3.71$, $SD = 1.09$), with a negligible effect size (Cohen's $d = -0.06$, $p = 0.874$). Similarly, Involvement (IPQ-INV1) showed minimal improvement (current: $M = 3.20$, $SD = 0.71$; previous: $M = 3.10$, $SD = 1.18$) with a negligible effect (Cohen's $d = 0.09$, $p = 0.813$).

Spatial Presence (IPQ-SP4) showed a small positive effect (current: $M = 3.85$, $SD = 0.78$; previous: $M = 3.56$, $SD = 1.16$; Cohen's $d = 0.27$, $p = 0.481$), while Experienced Realism (IPQ-REAL3) showed a small negative effect (current: $M = 3.10$, $SD = 0.81$; previous: $M = 3.42$, $SD =$

1.06; Cohen's $d = -0.32$, $p = 0.404$). Neither achieved statistical significance.

In general, the presence-related measures provided limited evidence for systematic improvements in the enhanced simulator, with the four dimensions showing small or negligible effects that did not reach statistical significance. Figure 9.16 presents all four dimensions of the IPQ, showing the consistently weak effects in all the presence-related outcomes with scores that cluster closely around the Baumann et al. [3] baseline.

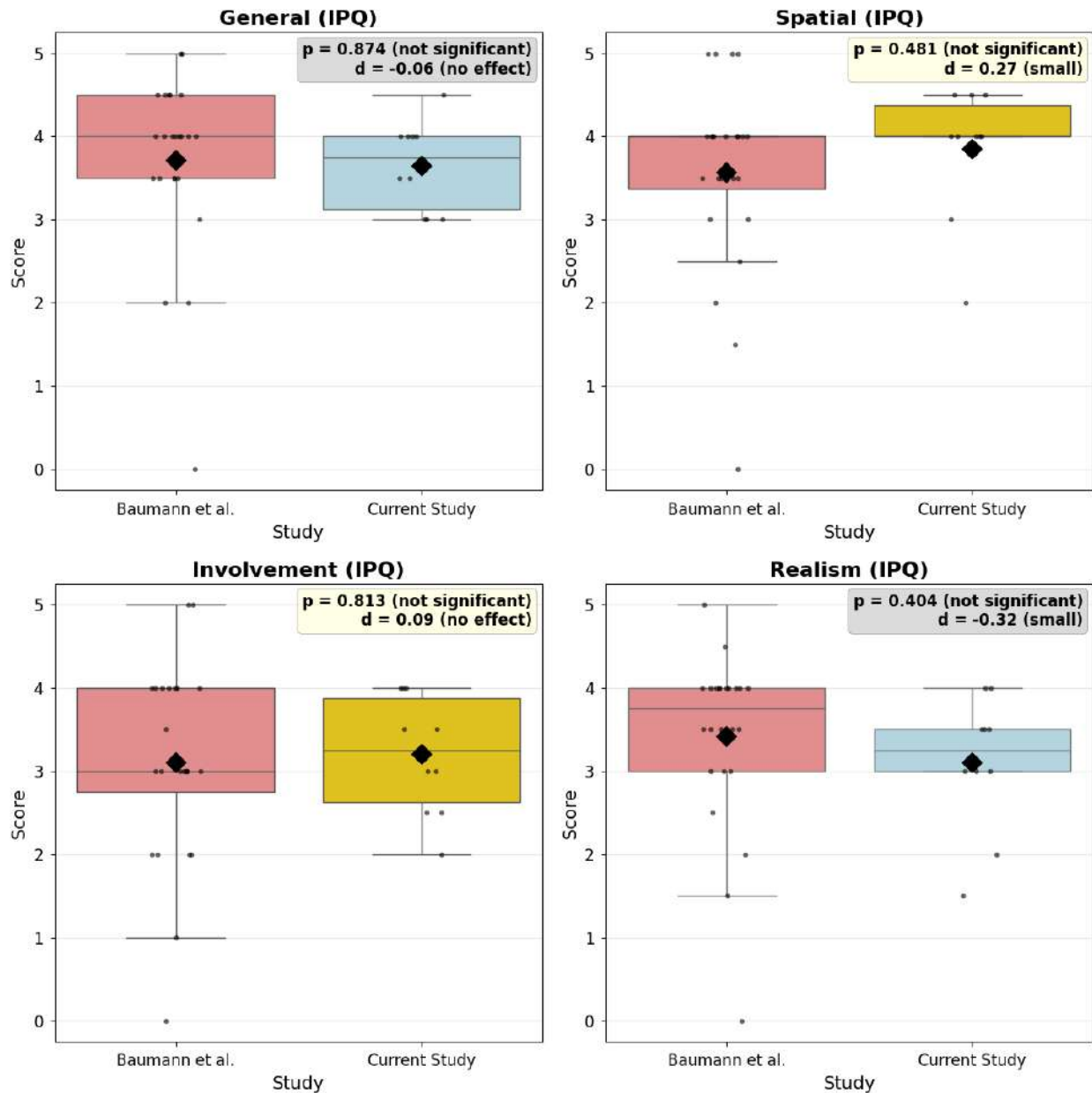


Figure 9.16: IPQ presence dimensions comparison showing General (top left), Spatial (top right), Involvement (bottom left), and Experienced Realism (bottom right). All measures show small or negligible effects without statistical significance.

Hypothesis 6 Conclusion

Comparison with the previous study by Baumann et al. [3] revealed significant improvements in stress management measures, but no changes in presence-related measures.

The stress results showed great improvements. The PASA Stress Index showed a significant reduction (Cohen's $d = -0.93$, $p = 0.019$), exceeding the predefined threshold ($d \geq 0.5$, Section 7.2.3). Secondary Appraisal showed a very large and significant improvement (Cohen's $d = 2.37$, $p < 0.001$), while Primary Appraisal showed no difference (Cohen's $d = -0.01$, $p = 0.980$).

All four IPQ presence dimensions showed small or negligible non-significant effects: General Presence (Cohen's $d = -0.06$, $p = 0.874$), Involvement (Cohen's $d = 0.09$, $p = 0.813$), Spatial Presence (Cohen's $d = 0.27$, $p = 0.481$), and Experienced Realism (Cohen's $d = -0.32$, $p = 0.404$).

Result: H_6 is partially supported. The enhanced simulator showed strong improvements in stress management, particularly through increased secondary evaluation (coping resources) and reduced stress index. These findings validate a key aspect of the hypothesis for better stress management. However, the hypothesis also predicted improved immersion scores, which was not supported by the data. The tilt-steering improvements were effective for stress management, but did not produce the anticipated improvements in presence and immersion.

Scenario Experience Insights

The qualitative feedback from phase 2 provided insight into how environmental effects affected user engagement and perception of realism. The danger scenarios generally received more positive qualitative responses than the quantitative measures alone suggested. One participant noted that the final danger scenario was "bestes von allen" (best of all), explaining that "durch die vielen visuellen Umwelteinflüsse fühlt es sich realistischer an, der Nebel hat den flimmernden Boden weniger präsent gemacht und durch die Autos muss man sich mehr konzentrieren und ist daher mehr in der Welt" (through the many visual environmental influences it feels more realistic, the fog made the flickering ground less noticeable and through the cars you have to concentrate more and are therefore more in the world, P17).

This observation suggests that environmental complexity and visual distraction can mask technical limitations while improving immersion. The comment indicates that challenging scenarios can improve presence through increased cognitive participation, supporting the theoretical framework for progressive difficulty training approaches, although H_5 received limited statistical support.

Technical issues persisted into Phase 2, with participants noting specific control problems such as "steering while at a standstill felt very weird" (P8), indicating that low-speed / standstill dynamics remained problematic despite the implemented enhancements.

10 Discussion

This chapter discusses the results of the evaluation, focusing on the implications of the findings for the improved VR bicycle simulator setup. The discussion addresses both the configuration evaluation results from Phase 1 and the scenario-based training effects from Phase 2, examining what these outcomes mean for the simulator's potential as both a research platform and a training device.

10.1 Phase 1: Configuration Evaluation and Individual Differences

10.1.1 Learning Effects and Analytical Robustness

The results identified substantial learning effects across the seven trial positions, particularly regarding the realism component. Although these effects were not negligible, the dual analytical approach using both raw and detrended data consistently showed alignment in the direction of results. In H_1 , the results were practically identical between approaches. In H_2 , both analyses showed the same direction with different but consistently large effect sizes. Similarly, H_3 showed comparable correlations in both datasets. This consistency suggests that the dual approach increased the robustness of the findings, as the results never contradicted each other.

However, the magnitude of the learning effects was underestimated before conducting the study. The slope of 0.097 points per trial for composite scores represents a cumulative improvement of approximately 0.68 points across all seven trials, which is substantial on a 7-point scale. These large effects must be taken into account in future work, particularly when designing within-subject studies with multiple repeated measurements. The consistent learning trajectory suggests that the participants continued to adapt throughout the entire trial sequence, indicating that even seven trials may not have been sufficient to reach a stable performance plateau.

10.1.2 Enhanced Configurations and Physical Accuracy

The results of H_1 showed clear trends with medium effect sizes indicating that the best performing configurations (M-VB in raw data, L-VB in detrended data) outperformed the baseline configuration. However, these differences did not reach statistical significance. In particular, both top-performing configurations employed the velocity-based (VB) steering model, which most closely approximates realistic bicycle physics through consideration of velocity, handlebar, and tilt inputs.

This finding is particularly interesting when considered in relation to the earlier discussion of objective versus perceived realism (Section 2.1.1). The results suggest that there is probably a balance between physical accuracy and perceived realism. Physical accuracy is likely needed to a

certain degree, but not necessarily to the full extent that would be required for a perfect physical simulation. The velocity-based model, which implements simplified physics principles rather than complete Carvallo–Whipple dynamics, appears to strike an effective balance. However, since the results are not statistically significant, these remain suppositions that require further investigation with larger sample sizes.

10.1.3 Personalization as a Key Factor

The results of H_2 do not undermine the findings of H_1 . Rather, they might offer an explanation for the observed trends. H_2 demonstrated that for both analytical approaches (raw and detrended), personalized configurations significantly outperformed the baseline configuration. The large effect sizes (raw: $d = 1.51$; detrended: $d = 1.05$) with $p < 0.01$ suggest that perceived control, realism, and comfort are not universally applicable to all users, but are instead subject to individual preferences.

These results confirm that the enhancements made to the VR bicycle simulator setup, specifically the adjustable stiffness mechanism and the newly introduced steering models, represent substantial improvements over the previous system. However, the findings also raise important questions about how to determine the best personal configuration of stiffness and steering model for individual users.

This finding has significant implications for simulator design. Rather than pursuing a single “optimal” configuration, future work should focus on developing personalization frameworks that can efficiently identify suitable settings for individual users. This approach aligns with established principles in HCI that emphasize the importance of adaptability and user customization in interactive systems.

10.1.4 Body Weight as a Predictive Factor

H_3 investigated the correlation between body weight and the stiffness settings of the preferred configuration, revealing similar results across both datasets. The raw data identified a correlation of $r = 0.630$, slightly higher than the correlation of the detrended data of $r = 0.598$, which missed the predefined threshold of 0.6. Nevertheless, both results offer a clear directional relationship between the rider’s body weight and the stiffness setting of their preferred configuration. Given the limited scope of this analysis, which examined weight correlations within the context of complete configurations (combining both stiffness and steering models), the topic requires further investigation. A more focused analysis that examines the weight-stiffness relationship independent of steering model variations would provide clearer insights.

Both correlations explain substantial variance (raw: $R^2 = 0.397$; detrended: $R^2 = 0.358$), indicating that body weight accounts for 36–40% of stiffness preference variation. This substantial but incomplete explanatory power suggests that other factors also influence stiffness preferences. Potential additional predictors might include cycling experience, age, balance confidence, or biome-

chanical factors such as center of gravity or torso length. Future research should investigate these additional factors to develop more comprehensive prediction models for initial configuration recommendations.

From a practical point of view, the weight-stiffness correlation provides a valuable starting point for personalization. A simple weight-based initialization could serve as a reasonable first approximation, allowing users to begin with configurations likely to be in the appropriate range, then fine-tune based on subjective experience. This approach would be particularly useful for first-time users who lack the reference experience needed to make initial decisions.

10.1.5 Velocity-Based Steering Model Advantages

H₄ focused exclusively on steering model effects, with ANOVA results showing small overall effects for both data sets. While the detrended data reached statistical significance, neither analysis met the desired medium effect size threshold ($\eta^2 \geq 0.06$). These results align with H₁, which also suggested that settings with velocity-based steering models achieved better results, although without conclusive statistical support.

The post-hoc comparisons between the baseline model (B) and the velocity-based model (VB) revealed larger effects. The raw data set remained within the small effect size range ($d = 0.49$), while the detrended data set achieved a medium effect size ($d = 0.54$) with statistical significance ($p < 0.01$). This shows a clear trend for the superiority of the velocity-based steering model, although the evidence cannot be considered conclusively given the proximity to threshold values.

The theoretical justification for the velocity-based model's advantage lies in its implementation of speed-dependent input blending. At low speeds, where balance control dominates real-world cycling, the model emphasizes tilt input. At higher speeds, where kinematic steering becomes more important, the model transitions to handlebar input. This adaptive weighting mirrors the natural physics of bicycle control and may explain why participants experienced improved realism and control with this model.

The non-linear model, which showed intermediate performance, implements a fixed sinusoidal relationship between tilt and steering that does not adapt to velocity. This static approach may have been less effective because it applies the same input transformation regardless of whether the user is moving slowly through tight corners or traveling straight at higher speeds. The baseline model, which completely ignores the tilt input, showed the lowest performance, confirming that incorporating the tilt information provides measurable benefits.

10.1.6 Summary of Phase 1 Findings

The Phase 1 results provide clear evidence of the benefits of using personalized configurations, with large effect sizes demonstrating substantial improvements over the baseline settings. The analysis also revealed strong indications that riders' body weight correlates with preferred stiffness settings, offering a practical basis for initial configuration recommendations. Furthermore, the

newly introduced velocity-based steering model showed clear signs of superiority over the previous baseline approach.

However, the relatively small sample size ($n = 20$) prevented a conclusive judgment on all research questions. The medium effect sizes observed in H_1 and H_4 , while suggesting potential advantages for enhanced configurations, did not reach statistical significance. Future work with larger sample sizes could help clarify these trends and provide more definitive answers about optimal configuration choices.

The practical implication of these findings is that the enhanced VR bicycle simulator represents a clear improvement over the previous setup, particularly when personalization is employed. The adjustable stiffness mechanism and the velocity-based steering model provide meaningful enhancements to user experience, as evidenced by the large effects observed in personalized configuration comparisons. However, the absence of a universally optimal configuration suggests that the simulator's effectiveness depends significantly on proper individualization of settings.

10.2 Phase 2: Training Applications and System Validation

10.2.1 Learning Effects from Danger Exposure

The evaluation of learning effects from exposure to danger using only two scenarios represents a methodological limitation that must be acknowledged. With such limited training exposure, it is not surprising that not all measures showed clear results. The comparison between subjects required for H_5 , combined with the small group sizes ($n = 10$ per condition, provided limited statistical power to detect effects below the large effect size threshold ($d \geq 0.8$).

The total sickness measure (CSQ-VR) showed a medium effect size ($d = -0.61$) that favors the Danger-Start group, but this did not reach significance. Similarly, the PASA Stress Index showed a medium effect size ($d = -0.66$) without achieving significance. The PASA subscales and the four IPQ presence dimensions showed mixed results, none of which reached significance. The custom realism measure showed effects in the same direction as the IPQ Experienced Realism dimension, again without reaching significance.

The only measure that demonstrated a clear and statistically significant effect was perceived safety, which showed a large effect ($d = 0.95$, $p = 0.048$). This finding indicates that prior exposure to danger scenarios significantly improved the subjective sense of safety of participants when faced with subsequent challenging situations. This result is particularly meaningful because safety perception represents a key training outcome for the VR bicycle simulators intension for traffic safety applications.

The medium effect sizes observed on several measures of stress and sickness, although not significant, suggest potential benefits that warrant further investigation. With only two scenarios and ten participants per group, the study may have been underpowered to detect real but modest learning effects. The consistent directional trends across multiple measures (stress index, primary

appraisal, motion sickness) point toward genuine adaptation processes that require larger samples to confirm reliably.

10.2.2 Training Device Implications

Although H_5 was not fully supported, the results still provide encouraging signals for the potential of the VR bicycle simulator as a training device. The significant improvement in perceived safety demonstrates that the simulator can effectively enhance users' confidence in handling challenging traffic scenarios. This represents a valuable training outcome, particularly for populations who might benefit from safe exposure to traffic situations before encountering them in reality, such as individuals with traffic anxiety.

The theoretical framework of progressive difficulty training, which motivated the danger-to-danger exposure condition, received partial empirical support. Although the comprehensive benefits hypothesized for all ten measures did not materialize, the improvement in safety perception suggests that graduated exposure can produce meaningful benefits. This finding aligns with the principles of exposure therapy from clinical psychology, where repeated exposure to anxiety-provoking situations under safe conditions can reduce fear responses and increase confidence in coping [14].

The absence of significant effects on the presence and immersion measures suggests that the learning benefits may operate through cognitive rather than perceptual mechanisms. Participants may have developed better situation assessment skills and coping strategies rather than experiencing fundamentally different levels of immersion.

Future work should address identified limitations through extended training protocols with more scenarios and longer exposure periods. A longitudinal design with multiple training sessions would allow for a clearer assessment of learning trajectories and skill retention. Furthermore, increasing sample sizes would provide adequate statistical power to detect medium-sized effects observed in several measures that showed promising trends without reaching significance.

10.2.3 System Improvements Compared to Previous Study

H_6 examined whether the enhanced simulator showed improvements compared to the previous study by Baumann et al. [3], revealing mixed results across the seven measured dimensions. The IPQ subscales (General, Spatial, Involvement, Experienced Realism) could not identify any significant improvements, with effect sizes ranging from negligible to small (range d : -0.32 to 0.27). This finding indicates that the tilt-steering enhancements did not produce the anticipated improvements in presence and immersion.

The absence of improvement in presence deserves careful interpretation. It does not necessarily indicate that the enhancements were ineffective, but rather that presence may not be the most sensitive indicator of improved bicycle dynamics. As discussed in Section 2.1.2, presence depends on multiple factors beyond just motion fidelity, including visual quality, natural interaction, and

task participation. The enhanced steering and tilt dynamics may have improved control and physical realism without substantially changing the overall sense of “being there” that the presence questionnaires measure.

In contrast, the PASA stress measures showed very clear improvements. The overall Stress Index across the two danger rides was significantly lower in the current study ($d = -0.93$, $p = 0.019$), indicating that the enhanced setup induces less stress in challenging scenarios. When examining the stress index subscales, the results become even more revealing. Primary Appraisal showed virtually no difference ($d = -0.01$), indicating that participants perceived similar levels of threat and challenge in both simulator versions. However, Secondary Appraisal showed a very large and significant improvement ($d = 2.37$, $p < 0.001$), indicating that the participants felt substantially more capable of handling challenging situations.

This pattern of results demonstrates that the enhanced simulator did not make the scenarios feel less threatening or challenging, but rather increased the confidence of the participants in their ability to manage these situations. The improvement in coping resources rather than threat perception represents a particularly valuable outcome for a training and research platform.

The very large effect size for Secondary Appraisal ($d = 2.37$) is remarkable, especially given the typically modest effect sizes observed in simulator comparison studies. This substantial improvement strongly suggests that the combination of personalized stiffness settings and velocity-based steering provides significantly better control and handling capabilities.

10.2.4 Summary of Phase 2 Findings

The Phase 2 results demonstrate that the VR bicycle simulator shows promise as a training device for increasing perceived safety in challenging scenarios. Significant improvements in stress management compared to the previous study, particularly through enhanced secondary Appraisal (coping resources), validate the effectiveness of the implemented improvements. Participants demonstrated better control and confidence in handling challenging situations, which represents a key outcome for both training and research applications.

However, the limited sample sizes and short training exposure prevented conclusive evaluation of comprehensive learning effects. The medium effect sizes observed in several stress and sickness measures without reaching significance suggest potential benefits that require further investigation with adequate statistical power. The absence of presence improvements indicates that the enhancements primarily affected control and confidence rather than overall immersion, which has important implications for understanding how the simulator produces its effects.

The comparison with the previous study provides strong validation for the technical improvements implemented in this work. The combination of adjustable tilt resistance, velocity-based steering integration, and personalized configuration selection produced measurable benefits in stress management and coping confidence. These findings support the continued development of the simulator as both a training platform and a research tool.

10.3 Tilt as Input: A Middle-Ground Approach

The results support the viability of using the tilt angle (roll) as input rather than output, representing a middle ground between fully actuated systems and handlebar-only input systems. This approach differs from the predominant implementations reviewed in Chapter 2, where most systems calculate the tilt as an output based on the input of speed and steering, then actuate the platforms to match these calculated values.

The passive adjustable tilt mechanism developed in this work, combined with sensor-based tilt measurement and integration into steering commands, provides several advantages. First, it avoids the mechanical complexity, cost, and maintenance requirements of actuated motion platforms. Second, it allows natural rider-induced weight shifts to directly influence bicycle dynamics, potentially increasing the sense of agency and natural control. Third, it provides adjustability without the need for sophisticated control algorithms or real-time platform motion cues.

The observed improvements in Secondary Appraisal suggest that this middle-ground approach successfully enhanced control and confidence. The integration of tilt input into the velocity-based steering model appears to have created more intuitive and manageable bicycle dynamics, particularly for users whose personalized stiffness settings matched their physical characteristics and preferences. This finding validates the conceptual approach of treating tilt as a control input that participants can actively use rather than as a passive motion cue that the system imposes.

However, the absence of presence improvements indicates that passive tilt sensing alone does not provide the full benefits of actively actuated motion platforms. Systems such as those developed by TU Wien [54] or TU Delft [18] may still offer advantages for applications that require maximum immersion or precise motion cues. The choice between passive and active approaches should depend on specific application requirements, balancing cost, complexity, and desired outcomes.

11 Limitations

This chapter acknowledges the key limitations of the present work, focusing on the constraints that most significantly impact the interpretation of the results and directions for future research.

11.1 Sample Limitations

The sample size of $n = 20$ participants, while comparable to related studies [33, 17], limited statistical power to detect small and medium-sized effects. This constraint is particularly evident in H_1 and H_4 , where medium effect sizes ($d \approx 0.5$ – 0.6) suggested potential advantages for enhanced configurations but did not reach statistical significance. Phase 2 comparisons between subjects with only $n = 10$ per group could reliably detect only large effects ($d \geq 0.8$), leaving medium-sized trends in stress and motion sickness measures (H_5) inconclusive.

The sample showed notable demographic concentrations: 85 % aged 18–34 years, 70 % male, and recruitment through university context likely created selection bias toward tech-familiar participants. These characteristics limit generalizability to older adults, broader gender distributions, and populations with limited technology experience or cycling anxiety who might particularly benefit from VR training applications.

11.2 Technical and Implementation Constraints

The passive tilt mechanism, while successfully demonstrating roll-as-input viability, lacks the active vestibular feedback of fully actuated platforms [54, 18]. The adjustable stiffness mechanism implemented only three discrete levels plus a baseline, which means that optimal values for individual users may lie between the tested settings.

Frame-rate instabilities were consistently reported during trials with enhanced steering models, which may have negatively biased their ratings relative to the simpler baseline. Low-speed and standstill dynamics remained problematic, with participants reporting that steering “felt very strange” when stationary, indicating that the velocity-based model’s $v \approx 0$ handling requires refinement.

The parameters of the steering model (Table 6.1) were calibrated by pragmatic tests with only three pilot participants. The non-linear gain ($\alpha = 0.25$) and velocity-based characteristic velocity ($v_0 = 1.5$) were chosen based on subjective impressions rather than systematic optimization. Different sensitivity values between models may have confounded comparisons, though these adjustments were intended to normalize perceived responsiveness.

11.3 Experimental Design Issues

The learning effects of phase 1 were greater than expected and the composite scores improved by 0.68 points in seven trials. Although the dual analytical approach addressed this statistically, the strong learning trajectory suggests that the participants had not reached stable performance by trial 7. The consecutive trial structure without rest periods may have introduced fatigue effects, and each participant testing only 7 of 10 configurations prevented complete within-subject comparisons.

The phase 2 assessment of learning effects suffered from minimal training exposure - only two scenarios per participant in a single session. Real training applications involve multiple sessions over weeks or months.

11.4 Measurement Limitations

The use of questionnaire subsets—PASA reduced from 16 to 4 items, CSQ-VR from 9 to 6 items, IPQ from 14 to 4 items—reduced reliability and construct coverage. Although necessary to prevent participant fatigue, these reductions increased measurement error and may have missed nuanced aspects of stress, motion sickness, and presence.

The exclusive reliance on subjective self-report measures does not provide objective validation of performance differences. No behavioral metrics (collision counts, trajectory smoothness, steering variability) were collected and no physiological data (heart rate, skin conductance). These objective measures would have provided converging evidence for subjective ratings and enabled a assessment of actual versus perceived control quality.

11.5 Comparison and Validation Constraints

Comparison with Baumann et al. [3] implemented multiple improvements simultaneously, adjustable stiffness, enhanced steering, and hardware fixes, preventing isolation of which specific enhancement drove the observed benefits. Only Danger-Start groups ($n = 10$) could be compared, excluding half the current sample. Different sample sizes (current $n = 10$ vs. previous $n = 24$) introduce variability that affects the comparisons.

No real-world validation was performed. The observed improvements in Secondary Appraisal and perceived safety represent simulator-based changes, but whether these transfer to actual outdoor cycling remains unknown. The simplified traffic environment with predictable controlled vehicles lacks many real urban cycling complexities. The safety net effect, which is the awareness of participants that crashes have no real consequences, probably influenced risk-taking and stress responses, leaving the actual transfer of safety skills uncertain.

11.6 Scope Limitations

The steering models, while grounded in principles of bicycle dynamics, were not rigorously derived from complete theoretical frameworks. The velocity-based model uses simplified equations rather than full Carvallo–Whipple dynamics. The non-linear model’s sinusoidal relationship was chosen for perceptual rather than physical reasons.

The weight–stiffness correlation explains 36–40 % of variance, indicating that other unmeasured factors (cycling experience, balance confidence, age, body composition) also influence preferences. The mechanism, which requires stronger riders that require stiffer resistance due to greater inertial forces, remains plausible but not conclusively validated.

The study focused on urban cycling with an emphasis on danger scenarios. The results may not be generalized to rural riding, mountain biking, or other contexts. The scope appropriately limited the investigation to the integration of tilt-steering, not addressing interactions with gradient simulation, wind resistance, or full six-degree-of-freedom motion.

12 Future Work

This chapter outlines key research directions that build on the findings of this thesis, focusing on practical improvements for the VR bicycle simulator as both a research platform and training device.

12.1 Personalization Framework Development

The strong personalization effects (H_2 : $d > 1.0$) and weight–stiffness correlation (H_3 : $r = 0.63$) provide a foundation for developing systematic personalization approaches. Future work should develop prediction models that include body weight, age, cycling experience, and balance confidence. A practical approach could use a short calibration protocol (2–3 trials) to establish initial settings, then adjust them based on observed performance.

Adaptive systems could monitor user performance in real time and suggest configuration changes when control problems are detected. Studies with multiple sessions are needed to determine whether user preferences remain stable over time and when recalibration is helpful.

12.2 Extended Training and Real-World Validation

Limited training exposure (two scenarios, one session) prevented a clear assessment of learning effects despite promising trends. Future work should implement training protocols with at least 40 participants over 2–4 weeks, allowing assessment of learning progress and skill retention. Significant improvement in safety perception (H_5 : $d = 0.95$) suggests that prolonged exposure could produce additional benefits.

Real-world validation is a critical next step. The improvements in Secondary Appraisal ($d = 2.37$ compared to the previous study) indicate better confidence in coping, but it is unknown whether this transfers to actual cycling performance. On-road assessments with behavioral measures (head check, speed adjustment, lane positioning) could validate the effectiveness of training.

Studies should include populations underrepresented in the current sample, particularly older adults and individuals with traffic anxiety who might benefit the most from safe VR exposure.

12.3 Technical Refinements

Steering model optimization through systematic parameter testing with larger pilot samples would identify better values for the velocity-based characteristic velocity (v_0) and non-linear gain (α). The current parameters showed promising results (H_4 : $d = 0.54$) that could be improved through careful optimization.

Continuous stiffness adjustment would replace the three discrete levels with fine-grained control, allowing more precise personalization. **Low-speed dynamics** require dedicated attention, as participants consistently reported steering problems when stationary.

12.4 Objective Measurement Integration

Future work should add objective metrics to subjective measures, including trajectory smoothness, steering variability, speed consistency, and braking response times. Physiological monitoring (heart rate variability, skin conductance) would provide additional evidence for PASA stress ratings. Standardized skill tests with objective scoring would enable benchmarking and progress tracking.

12.5 Comparative Studies and Methodological Improvements

Direct comparisons between passive (current system), actively actuated, and lateral actuation approaches would clarify the trade-offs between cost and benefits. The absence of presence improvements suggests that passive tilt may be sufficient for control, but not for maximum immersion.

A larger validation study would provide enough statistical power to detect the medium effect sizes ($d \approx 0.5 - 0.6$) observed in H_1 and H_4 . Using complete questionnaires (complete PASA, CSQ-VR, and IPQ instruments) would address concerns about the reduced item sets used in this study.

12.6 Summary

This thesis established a foundation for personalized VR bicycle simulation through improved tilt-steering integration. The clear benefits of personalization, practical weight-based prediction, and substantial improvements in coping confidence demonstrate that the implemented enhancements work well. The research directions outlined provide concrete paths to address identified limitations, with immediate priorities focusing on personalization frameworks, extended training validation, and technical optimizations. Future work has a clear direction based on the findings of this study, enabling systematic progress toward more realistic and effective VR bicycle simulation.

13 Conclusion

This thesis investigated the enhancement of a VR bicycle simulator through improved tilt dynamics and steering control. The work focused on three objectives: addressing reliability issues, implementing enhanced tilt-steering integration, and evaluating the system through a user study.

13.1 Technical Achievements

The technical work addressed reliability issues that had limited the potential of the simulator. Hardware improvements, including individual power supplies, protective sensor enclosures, and robust mechanical mounting, eliminated failures and signal instabilities. The improved brake sensor with runtime reset capability resolved persistent "sticking" behavior that had compromised user control.

The main technical contribution is the adjustable tilt resistance system. This mechanism provides variable stiffness through interchangeable pads and carriage positioning. This enables controlled tilt dynamics without the need for an expensive actuated platform. The velocity-based steering model integrates handlebar and tilt inputs through speed-dependent weighting, treating tilt as input rather than calculated output. This represents a departure from most of the simulator implementations reviewed.

13.2 Key Findings: Personalization Matters Most

Phase 1 evaluation with 20 participants revealed that personalization emerged as the dominant factor. Individually selected configurations significantly outperformed the baseline with large effect sizes (raw: Cohen's $d = 1.51$, $p = 0.0001$; detrended: $d = 1.05$, $p = 0.0023$). This has fundamental implications: searching for universally optimal configurations may be less productive than developing approaches that match settings to individual users.

The absence of a universal best configuration among all participants suggests substantial individual variation. This variation appears partially predictable: body weight showed a strong correlation with the stiffness of the preferred configuration ($r = 0.630$, $p = 0.003$) but weight alone explains only 36-40% of variance, indicating that other factors also contribute.

The velocity-based steering model showed consistent advantages over the baseline (detrended: Cohen's $d = 0.54$, $p = 0.0096$). This model emphasizes tilt at low speeds, where balance dominates, and handlebar input at higher speeds, where kinematic steering prevails. However, modest effect sizes suggest steering model selection is only one component of optimal configuration, with stiffness settings and individual characteristics playing equally important roles.

13.3 Training Applications and Validation

Phase 2 demonstrated the simulator’s training potential while revealing methodological limitations. Significant improvement in perceived safety for danger-to-danger exposure (Cohen’s $d = 0.95$, $p = 0.048$) provides evidence that difficulty training can enhance user confidence. However, limited exposure, only two scenarios in one session, prevented a complete assessment across all outcomes.

A comparison with the previous study (Baumann et al. [3]) confirmed the effectiveness of the enhancements through substantial improvements in stress management. The PASA stress index showed a significant reduction (Cohen’s $d = -0.93$, $p = 0.019$), driven primarily by improvements in Secondary Appraisal (Cohen’s $d = 2.37$, $p < 0.001$). Enhanced tilt-steering dynamics did not reduce perceived threat, but instead increased confidence in handling challenging situations.

The absence of improvement in presence (IPQ subscales: $d = -0.32$ to $d = 0.27$, all $p > 0.40$) indicates that the enhancements primarily affected the quality of the control rather than the overall immersion. This suggests that passive tilt sensing with velocity-based steering provides sufficient fidelity for effective control without the immersive motion cues that fully actuated platforms might deliver.

13.4 Practical Implications

The findings support several design principles for VR bicycle simulators. First, personalization frameworks should be prioritized over universal optimal configurations. Large effect sizes for individually selected settings ($d > 1.0$) substantially exceed modest effects for specific components ($d \approx 0.5$).

Second, body weight provides a practical starting point for initial recommendations, explaining approximately 40 % of the variance of the preferred stiffness. Simple weight-based initialization could establish reasonable starting parameters, with subsequent refinement based on experience.

Third, the passive adjustable tilt approach represents a viable middle ground between fixed handlebar-only systems and expensive actuated platforms. User-induced tilt with sensor-based measurement and steering integration provides meaningful control enhancements without the need for complex infrastructure.

Declaration of Authorship

I hereby declare that

1. I have completed this thesis independently and without unauthorized assistance,
2. I have clearly identified and made traceable all passages that have been taken verbatim, approximately verbatim, or conceptually from published and unpublished sources through citations, references, or explanations at the appropriate places within the work,
3. Regarding the use of AI-based tools, I declare that for this thesis, such tools were used in the following manner: for checking spelling, grammar, and linguistic improvement to enhance the readability of my work.

Heilbronn, September 29, 2025



Philipp Baumann

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A Appendix

A.1 Optimized Trial Sequences

For completeness, the full set of trial sequences used in Part 1 of the study is provided here.

Table A.1: Optimized trial sequences for $n = 24$ participants (7 trials per participant)

Trial	Pos1	Pos2	Pos3	Pos4	Pos5	Pos6	Pos7
1	L-B	M-B	L-NL	H-B	H-NL	N-B	M-VB
2	H-NL	M-NL	M-VB	N-B	L-B	H-VB	L-VB
3	L-NL	L-VB	N-B	M-B	M-NL	H-B	H-VB
4	H-B	M-VB	H-NL	L-B	N-B	M-B	M-NL
5	M-B	L-NL	H-VB	L-VB	M-VB	H-NL	N-B
6	N-B	H-VB	L-B	M-NL	L-VB	L-NL	H-B
7	M-VB	N-B	L-VB	H-NL	H-VB	L-B	L-NL
8	M-B	H-B	M-NL	M-VB	L-NL	N-B	L-B
9	M-NL	H-NL	H-VB	M-B	H-B	L-VB	N-B
10	L-VB	L-B	M-B	L-NL	N-B	M-VB	H-VB
11	H-VB	L-B	N-B	H-B	M-B	M-NL	H-NL
12	L-NL	N-B	H-B	H-VB	L-VB	M-NL	M-VB
13	H-NL	L-NL	L-VB	L-B	M-NL	H-B	M-B
14	M-NL	H-VB	M-VB	L-VB	H-B	H-NL	L-NL
15	M-VB	L-VB	M-B	L-NL	H-NL	H-VB	L-B
16	L-VB	M-NL	H-B	H-NL	M-VB	L-B	M-B
17	L-B	H-NL	L-NL	H-VB	M-B	M-VB	M-NL
18	H-VB	H-B	M-NL	N-B	L-B	M-B	L-VB
19	H-B	M-B	L-B	M-VB	L-NL	H-VB	H-NL
20	N-B	H-NL	L-VB	M-NL	L-NL	M-VB	H-B
21	M-NL	M-VB	H-NL	L-NL	H-VB	H-B	N-B
22	N-B	M-B	L-B	L-VB	M-NL	L-NL	H-NL
23	H-VB	M-VB	M-B	L-B	H-B	L-VB	M-NL
24	H-NL	H-VB	N-B	M-B	M-VB	L-NL	L-VB

A.2 Complete Questionnaire Items

This appendix provides a comprehensive documentation of all questionnaire items used in the study, presented in English and German versions as implemented in the study application.

A.2.1 English Questionnaires

Consent and Privacy Statement

Privacy Statement for the Bicycle Simulator Research Study

Thank you for participating in our research study on the bicycle simulator at Hochschule Heilbronn! This Privacy Statement outlines how we collect, use, and protect your personal data during the study.

1. Data Collection and Use: We collect data from participants to assess the effectiveness of the bicycle simulator in safely training urban cycling skills. This includes responses to questionnaires and feedback provided during the test session, as well as any observational data noted during your use of the simulator. Information gathered may include your subjective experiences, any physical reactions (such as nausea or dizziness), and your responses to specific scenarios within the simulation.

2. Purpose of Data Collection: The data collected will help us understand participants' perceptions and experiences within the virtual environment, specifically regarding elements like the brake, steering, and acceleration sensors. By analyzing this information, we aim to improve the realism and effectiveness of the simulator for future users.

3. Data Confidentiality: All information collected will be treated with strict confidentiality. Only authorized members of the research team will have access to the data, and results will be published in a way that prevents identification of individual participants.

4. Data Security: We employ appropriate technical and organizational measures to safeguard your data from unauthorized access, disclosure, or misuse. Collected data will be securely stored and will only be accessible to members of the research team.

5. Voluntary Participation: Your participation in this study is voluntary. You have the right to withdraw from the study at any time without any consequences. By participating in this study, you acknowledge that you have read and understood this Privacy Statement and consent to the collection and use of your data as described above.

Consent Question: "Have you read and understood this Privacy Statement and consent to the collection and use of your data as described above?"

Response options: Yes / No

Demographic Questionnaire

Demographics & Background

1. **Age:** 18-24 / 25-34 / 35-44 / 45-54 / 55-64 / 65+
2. **Gender:** Male / Female / Other / Prefer not to say
3. **VR Experience:** Not familiar / Less than 10 hours / 10 hours or more
4. **Cycling frequency per week:** Less than once / Approximately once / Several times
5. **Urban cycling frequency per week:** Less than once / Approximately once / Several times
6. **Weight (kg):** [Open text field]

Phase 1 Configuration Evaluation Questionnaire

Phase 1 - Trial Configuration Evaluation

After each trial, participants rated their experience with the current configuration using three questions on 7-point Likert scales (1 = "Not at all" to 7 = "Very much"):

1. **How comfortable was this setting?**
2. **How realistic did this feel?**
3. **How much control did you feel?**
4. **Any short notes? (Optional)** [Open text field]

Phase 1 - Final Configuration Selection

1. **Which setting felt best for the next phase?**

Note: The specific configuration options varied by participant according to their assigned trial matrix. An example selection list would include:

- Configuration 1 (L-B)
- Configuration 2 (M-B)
- Configuration 3 (L-NL)
- Configuration 4 (H-B)
- Configuration 5 (H-NL)
- Configuration 6 (N-B)
- Configuration 7 (M-VB)
- Recovery Trial Configuration (if applicable)

2. **How confident are you in your choice?**

7-point scale: 1 = "Not confident" to 7 = "Very confident"

3. **Comments on your preference or feedback on the setups? (Optional)** [Open text field]

Phase 2 Traffic Scenario Evaluation Questionnaire

Phase 2 - Scenario Evaluation

After each traffic scenario, participants completed questionnaires covering four domains:

Stress Appraisal (PASA Subset)

Four statements rated on 6-point scales (1 = "Completely False" to 6 = "Completely True"):

1. This situation is very uncomfortable for me.

-
2. This situation challenges me.
 3. I can best protect myself from failure in this situation through my own actions.
 4. I have no idea what I should do in this situation.

Motion Sickness (CSQ-VR Subset)

Six symptom items rated on 7-point scales (1 = "Absent" to 7 = "Extreme"):

1. Do you experience nausea (e.g., stomach pain, acid reflux, or tension to vomit)?
2. Do you experience dizziness (e.g., light-headedness or spinning feeling)?
3. Do you experience disorientation (e.g., spatial confusion or vertigo)?
4. Do you experience postural instability (i.e., imbalance)?
5. Do you experience visually induced fatigue (e.g., feeling of tiredness or sleepiness)?
6. Do you experience visually induced discomfort (e.g., eyestrain, blurred vision, or headache)?

Immersion / Presence (IPQ Subset)

Four items with question-specific 5-point scales:

1. To what extent did you have a sense of "being there" in the computer-generated world?
Scale: 1 = "Not at all" to 5 = "Very much"
2. To what extent do you agree with the following statement: I had a sense of acting in the virtual space, rather than operating something from outside.
Scale: 1 = "Strongly disagree" to 5 = "Strongly agree"
3. How aware were you of the real-world surroundings while navigating in the virtual world (e.g., sounds, room temperature, other people)?
Scale: 1 = "Not at all aware" to 5 = "Extremely aware"
4. How real did the virtual world seem to you?
Scale: 1 = "Not at all real" to 5 = "Completely real"

Perceived Realism & Safety

Two items on 7-point scales:

1. How realistic did the environment feel?
Scale: 1 = "Not realistic at all" to 7 = "Completely realistic"
2. How safe or threatened did you feel?
Scale: 1 = "Very threatened" to 7 = "Very safe"

Comments

Any thoughts or feelings about this scenario? (Optional) [Open text field]

Final Feedback Questionnaire

Final Feedback

Do you have any other thoughts, ideas, or concerns you'd like to share? (Optional) [Open text field]

A.2.2 German Questionnaires

Einverständniserklärung und Datenschutzerklärung

Datenschutzerklärung für die Fahrrad-Simulator-Forschungsstudie

Vielen Dank für deine Teilnahme an unserer Forschungsstudie zum Fahrrad-Simulator an der Hochschule Heilbronn! Diese Datenschutzerklärung erläutert, wie wir deine personenbezogenen Daten während der Studie erheben, verwenden und schützen.

1. Datenerhebung und -verwendung: Wir erheben Daten von Teilnehmern, um die Wirksamkeit des Fahrrad-Simulators beim sicheren Training städtischer Radfahrfähigkeiten zu bewerten. Dazu gehören Antworten auf Fragebögen und Feedback während der Testsitzung sowie alle Beobachtungsdaten, die während deiner Nutzung des Simulators notiert werden. Die gesammelten Informationen können deine subjektiven Erfahrungen, körperliche Reaktionen (wie Übelkeit oder Schwindel) und deine Reaktionen auf bestimmte Szenarien in der Simulation umfassen.

2. Zweck der Datenerhebung: Die gesammelten Daten helfen uns, die Wahrnehmungen und Erfahrungen der Teilnehmer in der virtuellen Umgebung zu verstehen, insbesondere in Bezug auf Elemente wie Brems-, Lenk- und Beschleunigungssensoren. Durch die Analyse dieser Informationen wollen wir den Realismus und die Wirksamkeit des Simulators für zukünftige Benutzer verbessern.

3. Datenvertraulichkeit: Alle gesammelten Informationen werden streng vertraulich behandelt. Nur autorisierte Mitglieder des Forschungsteams haben Zugang zu den Daten, und die Ergebnisse werden so veröffentlicht, dass die Identifikation einzelner Teilnehmer verhindert wird.

4. Datensicherheit: Wir setzen angemessene technische und organisatorische Maßnahmen ein, um deine Daten vor unbefugtem Zugriff, Offenlegung oder Missbrauch zu schützen. Die gesammelten Daten werden sicher gespeichert und sind nur für Mitglieder des Forschungsteams zugänglich.

5. Freiwillige Teilnahme: Deine Teilnahme an dieser Studie ist freiwillig. Du hast das Recht, jederzeit ohne Konsequenzen von der Studie zurückzutreten. Durch die Teilnahme an dieser Studie bestätigst du, dass du diese Datenschutzerklärung gelesen und verstanden hast und der Erhebung und Verwendung deiner Daten wie oben beschrieben zustimmst.

Einverständnisfrage: „Hast du diese Datenschutzerklärung gelesen und verstanden und stimmst du der Erhebung und Verwendung deiner Daten wie oben beschrieben zu?“

Antwortoptionen: Ja / Nein

Demografischer Fragebogen

Demografie & Hintergrund

1. **Alter:** 18-24 / 25-34 / 35-44 / 45-54 / 55-64 / 65+

2. **Geschlecht:** Männlich / Weiblich / Anderes / Keine Angabe

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3. **VR Erfahrung:** Nicht vertraut / Weniger als 10 Stunden / 10 Stunden oder mehr
 4. **Fahrrad-Häufigkeit pro Woche:** Weniger als einmal / Ungefähr einmal / Mehrmals
 5. **Stadtradfahren-Häufigkeit pro Woche:** Weniger als einmal / Ungefähr einmal / Mehrmals
 6. **Gewicht (kg):** [Offenes Textfeld]

Phase 1 Konfigurationsbewertung Fragebogen

Phase 1 - Versuchskonfiguration Bewertung

Nach jedem Versuch bewerteten die Teilnehmer ihre Erfahrung mit der aktuellen Konfiguration anhand von drei Fragen auf 7-Punkt-Likert-Skalen (1 = „Gar nicht“ bis 7 = „Sehr viel“):

1. **Wie komfortabel war diese Fahrt für dich?**
2. **Wie realistisch hat sich das für dich angefühlt?**
3. **Wie viel Kontrolle hast du gespürt?**
4. **Kurze Notizen? (Optional)** [Offenes Textfeld]

Phase 1 - Finale Konfigurationsauswahl

1. **Welche Einstellung hat sich für dich am besten für die nächste Phase angefühlt?**

Hinweis: Die spezifischen Konfigurationsoptionen variierten je nach Teilnehmer entsprechend ihrer zugewiesenen Versuchsmatrix. Eine beispielhafte Auswahlliste würde umfassen:

- Konfiguration 1 (L-B)
- Konfiguration 2 (M-B)
- Konfiguration 3 (L-NL)
- Konfiguration 4 (H-B)
- Konfiguration 5 (H-NL)
- Konfiguration 6 (N-B)
- Konfiguration 7 (M-VB)
- Wiederherstellungsversuch-Konfiguration (falls zutreffend)

2. **Wie sicher bist du bei deiner Wahl?**

7-Punkt-Skala: 1 = „Nicht sicher“ bis 7 = „Sehr sicher“

3. **Kommentare zu deiner Präferenz oder Feedback zu den Einstellungen? (Optional)**
[Offenes Textfeld]

Phase 2 Verkehrsszenario Bewertung Fragebogen

Phase 2 - Szenario Bewertung

Nach jedem Verkehrsszenario füllten die Teilnehmer Fragebögen aus, die vier Bereiche abdeckten:

Stress-Bewertung (PASA Teilmenge)

Vier Aussagen bewertet auf 6-Punkt-Skalen (1 = „Völlig falsch“ bis 6 = „Völlig wahr“):

1. Diese Situation ist sehr unangenehm für mich.
2. Diese Situation fordert mich heraus.
3. Ich kann mich am besten vor Misserfolg in dieser Situation durch meine eigenen Handlungen schützen.
4. Ich habe keine Ahnung, was ich in dieser Situation tun soll.

Bewegungskrankheit (CSQ-VR Teilmenge)

Sechs Symptom-Items bewertet auf 7-Punkt-Skalen (1 = „Abwesend“ bis 7 = „Extrem“):

1. Erleben Sie Übelkeit (z. B. Magenschmerzen, Sodbrennen oder das Gefühl, sich übergeben zu müssen)?
2. Erleben Sie Schwindel (z. B. Benommenheit oder das Gefühl, dass sich alles dreht)?
3. Erleben Sie Orientierungslosigkeit (z.B. Räumliche Verwirrung oder Schwindel)?
4. Erleben Sie eine Haltungsinstabilität (d. h. Gleichgewichtsstörungen)?
5. Erleben Sie eine visuell bedingte Müdigkeit (z. B. Gefühl von Erschöpfung oder Schläfrigkeit)?
6. Erleben Sie ein visuell bedingtes Unwohlsein (z. B. Augenbelastung, verschwommene Sicht oder Kopfschmerzen)?

Immersion / Präsenz (IPQ Teilmenge)

Vier Items mit fragenspezifischen 5-Punkt-Skalen:

1. In welchem Ausmaß hatten Sie in der computergenerierten Welt das Gefühl, „dort zu sein“?
Skala: 1 = „Überhaupt nicht“ bis 5 = „Sehr viel“
2. Inwieweit stimmen Sie der folgenden Aussage zu: Ich hatte das Gefühl, im virtuellen Raum zu handeln, statt etwas von außen zu bedienen.
Skala: 1 = „Stimme überhaupt nicht zu“ bis 5 = „Stimme voll und ganz zu“

-
3. Wie bewusst war Ihnen die reale Umgebung, während Sie sich in der virtuellen Welt bewegten (z. B. Geräusche, Raumtemperatur, andere Personen)?

Skala: 1 = „Gar nicht bewusst" bis 5 = „Äußerst bewusst"

4. Wie realistisch erschien Ihnen die virtuelle Welt?

Skala: 1 = „Überhaupt nicht realistisch" bis 5 = „Vollständig realistisch"

Wahrgenommener Realismus & Sicherheit

Zwei Items auf 7-Punkt-Skalen:

1. Wie realistisch fühlte sich die Umgebung an?

Skala: 1 = „Überhaupt nicht realistisch" bis 7 = „Vollständig realistisch"

2. Wie sicher oder bedroht fühlten Sie sich?

Skala: 1 = „Sehr bedroht" bis 7 = „Sehr sicher"

Kommentare

Gedanken oder Gefühle zu diesem Szenario? (Optional) [Offenes Textfeld]

Abschließender Feedback Fragebogen

Abschließendes Feedback

Hast du andere Gedanken, Ideen oder Bedenken, die du teilen möchtest? (Optional) [Offenes Textfeld]